



# TRIGONOMETRY

## QUESTIONS

### FROM 2010 TO 2020 MAINS

#### SSC CGL MAINS 2011

{04 Sept. 2011}

1. If  $\sin^2\alpha + \sin^2\beta = 2$ , then the value of  $\left(\cos \frac{\alpha+\beta}{2}\right)$  is-

यदि  $\sin^2\alpha + \sin^2\beta = 2$  तो  $\left(\cos \frac{\alpha+\beta}{2}\right)$  का मान है-

- (a) 1 (b) -1  
(c) 0 (d) 0.5

2. The value of/का मान है-

$$\cot \frac{\pi}{20} \cot \frac{3\pi}{20} \cot \frac{5\pi}{20} \cot \frac{7\pi}{20} \cot \frac{9\pi}{20}$$

- (a) -1 (b) 1/2  
(c) 0 (d) 1

3. If  $\sin\theta + \cos\theta = 17/13$ ,  $0 < \theta < 90^\circ$ , then the value of  $\sin\theta - \cos\theta$  is

यदि  $\sin\theta + \cos\theta = 17/13$ ,  $0 < \theta < 90^\circ$  तो  $\sin\theta - \cos\theta$  का मान है-

- (a) 5/17 (b) 3/19  
(c) 7/10 (d) 7/13

4. If  $\tan\theta \cdot \tan 2\theta = 1$ , then the value of  $\sin^2 2\theta + \tan^2 2\theta$  is equal to

यदि  $\tan\theta \cdot \tan 2\theta = 1$  तो  $\sin^2 2\theta + \tan^2 2\theta$  का मान है-

- (a) 3/4 (b) 10/3  
(c)  $3\frac{3}{4}$  (d) 3

#### SSC CGL MAINS 2012

{16 Sept. 2012}

5. If  $\theta$  is a positive acute angle and  $\tan 2\theta \cdot \tan 3\theta = 1$ , then the value of  $\left(2\cos^2 \frac{5\theta}{2} - 1\right)$  is

अगर  $\theta$  धनात्मक न्यूनकोण हो तथा  $\tan 2\theta \cdot \tan 3\theta = 1$  हो, तब  $\left(2\cos^2 \frac{5\theta}{2} - 1\right)$  है

(a) 0 (b)  $\frac{1}{2}$   
(c)  $-\frac{1}{2}$  (d) 1

6. If  $\sin 17^\circ = \frac{x}{y}$ , then the value of  $(\sec 17^\circ - \sin 73^\circ)$  is -

यदि  $\sin 17^\circ = \frac{x}{y}$  है, तब  $(\sec 17^\circ - \sin 73^\circ)$  है

(a)  $\frac{x^2}{y\sqrt{x^2-y^2}}$  (b)  $\frac{y^2}{x\sqrt{x^2-y^2}}$

(c)  $\frac{y^2}{x\sqrt{y^2-x^2}}$  (d)  $\frac{x^2}{y\sqrt{y^2-x^2}}$

7. If  $0^\circ < \theta < 90^\circ$ , then the value of  $\sin\theta + \cos\theta$  is-

यदि  $0^\circ < \theta < 90^\circ$ , तब  $\sin\theta + \cos\theta$  होगा-

- (a) less than 1/1 से कम  
(b) equal to 2/2 के बराबर  
(c) equal to 1/1 के बराबर  
(d) greater than 1/1 से बड़ा

8. The minimum value of  $\sin^2\theta + \cos^2\theta + \sec^2\theta + \csc^2\theta + \tan^2\theta + \cot^2\theta$ .

$\sin^2\theta + \cos^2\theta + \sec^2\theta + \csc^2\theta + \tan^2\theta + \cot^2\theta$  का न्यूनतम मान ज्ञात कीजिए?

- (a) 5 (b) 7  
(c) 1 (d) 3

9. If  $2 \sin \left(\frac{\pi x}{2}\right) = x^2 + \frac{1}{x^2}$ , then the value of  $\left(x - \frac{1}{x}\right)$  is -

यदि  $2 \sin \left(\frac{\pi x}{2}\right) = x^2 + \frac{1}{x^2}$  है, तो

$\left(x - \frac{1}{x}\right)$  का मान ज्ञात करें?

- (a) 1 (b) 0  
(c) -1 (d) 2

10. The expression  $\frac{\tan 57^\circ + \cot 37^\circ}{\tan 33^\circ + \cot 53^\circ}$  is equal to-

व्यंजक  $\frac{\tan 57^\circ + \cot 37^\circ}{\tan 33^\circ + \cot 53^\circ}$  किसके बराबर है।

- (a)  $\tan 33^\circ \cdot \cot 53^\circ$   
(b)  $\tan 53^\circ \cdot \cot 37^\circ$   
(c)  $\tan 33^\circ \cdot \cot 57^\circ$   
(d)  $\tan 57^\circ \cdot \cot 37^\circ$

#### SSC CGL MAINS 2013

{29 Sept. 2013}

11. If  $\sin\theta + \cos\theta = \sqrt{2} \cos\theta$ , then the value of  $(\cos\theta - \sin\theta)$  is:

यदि  $\sin\theta + \cos\theta = \sqrt{2} \cos\theta$  हो, तो  $(\cos\theta - \sin\theta)$  का मान कितना होगा?

- (a)  $\sqrt{3} \cos\theta$  (b)  $\sqrt{3} \sin\theta$   
(c)  $\sqrt{2} \cos\theta$  (d)  $\sqrt{2} \sin\theta$

12. If  $x \sin 45^\circ = y \operatorname{cosec} 30^\circ$ , then  $\frac{x^4}{y^4}$  is equal to-  
यदि  $x \sin 45^\circ = y \operatorname{cosec} 30^\circ$  हो, तो  $\frac{x^4}{y^4}$  किसके बराबर होगा?

- (a)  $4^3$  (b)  $6^3$   
(c)  $2^3$  (d)  $8^3$

13.  $\frac{\sin A}{1 + \cos A} + \frac{\sin A}{1 - \cos A}$  is  $(0^\circ < A < 90^\circ)$ ?

$(0^\circ < A < 90^\circ)$  की स्थिति में  $\frac{\sin A}{1 + \cos A}$

$+ \frac{\sin A}{1 - \cos A}$  कितना होगा?

- (a)  $2 \operatorname{cosec} A$  (b)  $2 \sec A$   
(c)  $2 \sin A$  (d)  $2 \cos A$

14. If  $r \sin\theta = 1$ ,  $r \cos\theta = \sqrt{3}$  then the value of  $(\sqrt{3} \tan\theta + 1)$  is-

यदि  $r \sin\theta = 1$ ,  $r \cos\theta = \sqrt{3}$  हो, तो  $(\sqrt{3} \tan\theta + 1)$  का मान क्या होगा-

- (a)  $\sqrt{3}$  (b)  $\frac{1}{\sqrt{3}}$   
(c) 1 (d) 2

## SSC CGL MAINS 2013

{21 Sept. 2014}

15. If  $\tan x = \sin 45^\circ \cdot \cos 45^\circ + \sin 30^\circ$ , then value of  $x$  is -

यदि  $\tan x = \sin 45^\circ \cdot \cos 45^\circ + \sin 30^\circ$  हो, तो  $x$  का मान क्या होगा?

- (a)  $30^\circ$  (b)  $45^\circ$   
(c)  $60^\circ$  (d)  $90^\circ$

16. For any real values of  $\theta$ ,

$$\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} = ?$$

$\theta$  के किसी वास्तविक मान के लिए

$$\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} = ?$$

- (a)  $\cot \theta - \operatorname{cosec} \theta$   
(b)  $\sec \theta - \tan \theta$   
(c)  $\operatorname{cosec} \theta - \cot \theta$   
(d)  $\tan \theta - \sec \theta$

17. If  $\sin 3A = \cos (A - 26^\circ)$ , where  $3A$  is an acute angle, then the value of  $A$  is -

यदि  $\sin 3A = \cos (A - 26^\circ)$  जहाँ  $3A$  एक न्यून कोण है, तो  $A$  का मान क्या होगा?

- (a)  $29^\circ$  (b)  $26^\circ$   
(c)  $23^\circ$  (d)  $28^\circ$

18.  $\sec^2 \theta - \frac{\sin^2 \theta - 2 \sin^4 \theta}{2 \cos^4 \theta - \cos^2 \theta} = ?$

- (a) 1 (b) 2  
(c) -1 (d) 0

19. If  $x = a(\sin \theta + \cos \theta)$  and  $y = b(\sin \theta - \cos \theta)$ , then the value of

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \text{ is -}$$

यदि  $x = a(\sin \theta + \cos \theta)$  and  $y = b(\sin$

$\theta - \cos \theta)$ , तो  $\frac{x^2}{a^2} + \frac{y^2}{b^2}$  का मान क्या होगा?

- (a) 0 (b) 1  
(c) 2 (d) -2

20. If  $\sin 5\theta = \cos 2\theta$  ( $0^\circ < \theta < 90^\circ$ ), then the value of  $\theta$  is -

यदि  $\sin 5\theta = \cos 2\theta$  ( $0^\circ < \theta < 90^\circ$ ) हो, तो  $\theta$  का मान क्या होगा?

- (a)  $4^\circ$  (b)  $22^\circ$   
(c)  $10^\circ$  (d)  $14^\circ$

## SSC CGL MAINS 2014

{12 Apr. 2015} PAPER - I

21. The value of  $\theta$  ( $0^\circ \leq \theta \leq 90^\circ$ ) satisfying  $2 \sin^2 \theta = 3 \cos \theta$  is

$\theta$  ( $0^\circ \leq \theta \leq 90^\circ$ ) का कौन सा मान,  $2 \sin^2 \theta = 3 \cos \theta$  को संतुष्ट करेगा?

- (a)  $30^\circ$  (b)  $45^\circ$   
(c)  $60^\circ$  (d)  $90^\circ$

22. If  $a \sin \theta + b \cos \theta = c$ , then  $a \cos \theta - b \sin \theta$  is equal to -

यदि  $a \sin \theta + b \cos \theta = c$  है, तो  $a \cos \theta - b \sin \theta$  का मान किसके बराबर है?

- (a)  $\pm \sqrt{a^2 + b^2 + c^2}$   
(b)  $\pm \sqrt{c^2 + a^2 - b^2}$   
(c)  $\pm \sqrt{a + b - c}$   
(d)  $\pm \sqrt{a^2 + b^2 - c^2}$

23. If  $\sin \theta + \cos \theta = \sqrt{2} \sin \theta (90^\circ - \theta)$  then the value of  $\cot \theta$  is -

यदि  $\sin \theta + \cos \theta = \sqrt{2} \sin \theta (90^\circ - \theta)$  है, तो  $\cot \theta$  का मान ज्ञात करें।

- (a) -1 (b)  $-\sqrt{2} + 1$   
(c)  $-\sqrt{2} - 1$  (d)  $\sqrt{2} + 1$

24. If  $\theta$  is a positive acute angle and  $3(\sec^2 \theta + \tan^2 \theta) = 5$ , then the value of  $\cos^2 \theta$  is -

यदि  $\theta$  एक धनात्मक न्यून कोण है तथा  $3(\sec^2 \theta + \tan^2 \theta) = 5$  है, तो  $\cos^2 \theta$  का मान ज्ञात करें।

- (a)  $\frac{1}{\sqrt{2}}$  (b) 1  
(c)  $\frac{1}{2}$  (d)  $\frac{\sqrt{3}}{2}$

25. If  $x \cos^2 30^\circ \cdot \sin 60^\circ = \frac{\tan^2 45^\circ \cdot \sec 60^\circ}{\operatorname{cosec} 60^\circ}$

then the value of  $x$  is -

$$\text{यदि } x \cos^2 30^\circ \cdot \sin 60^\circ = \frac{\tan^2 45^\circ \cdot \sec 60^\circ}{\operatorname{cosec} 60^\circ}$$

है, तो  $x$  का मान ज्ञात करें।

- (a)  $\frac{1}{\sqrt{2}}$  (b)  $\frac{1}{2}$   
(c)  $\frac{1}{\sqrt{3}}$  (d)  $2 \frac{2}{3}$

26. If  $\tan \alpha = 2$ , then the value of

$$\frac{\operatorname{cosec}^2 \alpha - \sec^2 \alpha}{\operatorname{cosec}^2 \alpha + \sec^2 \alpha} \text{ is -}$$

यदि  $\tan \alpha = 2$  है, तो  $\frac{\operatorname{cosec}^2 \alpha - \sec^2 \alpha}{\operatorname{cosec}^2 \alpha + \sec^2 \alpha}$  का मान ज्ञात करें।

- (a)  $-\frac{3}{5}$  (b)  $\frac{1}{2}$   
(c)  $-\frac{15}{9}$  (d)  $\frac{3}{5}$

27. If  $\sin(\theta + 30^\circ) = \frac{3}{\sqrt{12}}$  then the value of  $\cos^2 \theta$  is -

यदि  $\sin(\theta + 30^\circ) = \frac{3}{\sqrt{12}}$  है, तो  $\cos^2 \theta$  का मान ज्ञात करें।

- (a)  $\frac{\sqrt{3}}{2}$  (b)  $\frac{1}{2}$   
(c)  $\frac{1}{4}$  (d)  $\frac{3}{4}$

28. If  $0^\circ \leq \theta \leq 90^\circ$  and  $4 \cos^2 \theta - 4\sqrt{3} \cos \theta + 3 = 0$ , then the value of  $\theta$  is -

यदि  $0^\circ \leq \theta \leq 90^\circ$  और  $4 \cos^2 \theta - 4\sqrt{3} \cos \theta + 3 = 0$  है, तो  $\theta$  का मान क्या है?

- (a)  $45^\circ$  (b)  $60^\circ$   
(c)  $30^\circ$  (d)  $90^\circ$

## SSC CGL MAINS 2014

{12 Apr. 2015} PAPER - II

29. If  $\sec \theta - \cos \theta = \frac{3}{2}$  where  $\theta$  is a positive acute angle. then the value of  $\sec \theta$  is:

यदि  $\sec \theta - \cos \theta = \frac{3}{2}$  यहाँ  $\theta$  धनात्मक न्यूनकोण है, तो  $\sec \theta$  का मान क्या है?

- (a) 2 (b) 0  
(c)  $-1/2$  (d) 1

30. If  $\tan(5x - 10^\circ) = \cot(5y + 20^\circ)$ , then value of  $(x + y)$  is:-

यदि  $\tan(5x - 10^\circ) = \cot(5y + 20^\circ)$  तो  $(x + y)$  का मान क्या है?

- (a)  $24^\circ$  (b)  $20^\circ$   
(c)  $15^\circ$  (d)  $16^\circ$

31. If  $\sin \theta + \sin^2 \theta = 1$ , then the value of  $\cos^{12} \theta + 3 \cos^{10} \theta + 3 \cos^8 \theta + \cos^6 \theta - 1$  is -

यदि  $\sin \theta + \sin^2 \theta = 1$  तो  $\cos^{12} \theta + 3 \cos^{10} \theta + 3 \cos^8 \theta + \cos^6 \theta - 1$  का मान क्या होगा?

- (a) 3 (b) 0  
(c) 1 (d) 2

32. If  $a(\tan\theta + \cot\theta) = 1$ ,  $\sin\theta + \cos\theta = b$  with  $0^\circ < \theta < 90^\circ$ , then the relation between 'a' and 'b' is-

यदि  $a(\tan\theta + \cot\theta) = 1$ ,  $\sin\theta + \cos\theta = b$  तथा  $0^\circ < \theta < 90^\circ$  तो a और b के बीच क्या संबंध है?

- (a)  $2a = b^2 - 1$  (b)  $2a = b^2 + 1$   
(c)  $b^2 = 2(a+1)$  (d)  $b^2 = 2(a-1)$

33. The value of  $\tan 11^\circ \tan 17^\circ \tan 79^\circ \tan 73^\circ$  is-

$\tan 11^\circ \tan 17^\circ \tan 79^\circ \tan 73^\circ$  का मान है-

- (a) 1 (b)  $1/\sqrt{2}$   
(c)  $1/2$  (d) 0

34. If for any acute angle A,  $\sin A + \sin^2 A = 1$ , then the value of  $\cos^2 A + \cos^4 A$  is-

यदि किसी न्यूनकोण A, के लिए  $\sin A + \sin^2 A = 1$  तो  $\cos^2 A + \cos^4 A$  का मान क्या होगा?

- (a) 2 (b) 0  
(c) -1 (d) 1

35. The value of  $(1 + \sec 20^\circ + \cot 70^\circ)(1 - \operatorname{cosec} 20^\circ + \tan 70^\circ)$  is equal to-

$(1 + \sec 20^\circ + \cot 70^\circ)(1 - \operatorname{cosec} 20^\circ + \tan 70^\circ)$  का मान कितना है?

- (a) 2 (b) -1  
(c) 0 (d) 1

36. If  $0^\circ < A < 90^\circ$ , then the value of  $\frac{\tan A - \sec A - 1}{\tan A + \sec A + 1}$

यदि  $0^\circ < A < 90^\circ$  तो का  $\frac{\tan A - \sec A - 1}{\tan A + \sec A + 1}$  का मान क्या है?

- (a)  $\frac{1 - \cos A}{\sin A}$  (b)  $\frac{\sin A + 1}{\cos A}$   
(c)  $\frac{\sin A - 1}{\cos A}$  (d)  $\frac{1 + \sin A}{\cos A}$

### SSC CGL MAINS 2015

{25 Aug. 2015}

37. If  $\sec\theta - \tan\theta = \frac{1}{\sqrt{3}}$  the value of  $\sec\theta \cdot \tan\theta$  is -

यदि  $\sec\theta - \tan\theta = \frac{1}{\sqrt{3}}$  है, तो  $\sec\theta \cdot \tan\theta$  का मान है-

- (a)  $2/3$  (b)  $4/\sqrt{3}$   
(c)  $2/\sqrt{3}$  (d)  $1/\sqrt{3}$

38. If  $x = a \sin\theta - b \cos\theta$ ,  $y = a \cos\theta + b \sin\theta$ , then which of the following is true?

यदि  $x = a \sin\theta - b \cos\theta$ ,  $y = a \cos\theta + b \sin\theta$  है, तो निम्नलिखित में से कौन-सा सही है?

- (a)  $x^2 + y^2 = a^2 - b^2$   
(b)  $x^2 + y^2 = a^2 + b^2$

(c)  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(d)  $\frac{x^2}{y^2} + \frac{a^2}{b^2} = 1$

39. If  $\tan A = n \tan B$  and  $\sin A = m \sin B$ , then the value of  $\cos^2 A$  is यदि  $\tan A = n \tan B$  और  $\sin A = m \sin B$ , तो  $\cos^2 A$  का मान है

- (a)  $\frac{m^2 - 1}{n^2 + 1}$  (b)  $\frac{m^2 + 1}{n^2 + 1}$   
(c)  $\frac{m^2 + 1}{n^2 - 1}$  (d)  $\frac{m^2 - 1}{n^2 - 1}$

40. If  $\tan\theta - \cot\theta = 0$  and  $\theta$  is positive acute angle, then the value of

$\frac{\tan(\theta + 15^\circ)}{\tan(\theta - 15^\circ)}$  is-

यदि  $\tan\theta - \cot\theta = 0$  और  $\theta$  धनात्मक न्यून

कोण है, तो  $\frac{\tan(\theta + 15^\circ)}{\tan(\theta - 15^\circ)}$  का मान क्या होगा?

- (a)  $1/\sqrt{3}$  (b) 3  
(c)  $1/3$  (d)  $\sqrt{3}$

41. If  $\sin A + \sin^2 A = 1$ , then the value of  $\cos^2 A + \cos^4 A$  is-

यदि  $\sin A + \sin^2 A = 1$ , तो  $\cos^2 A + \cos^4 A$  का मान है-

- (a) 2 (b) 1

(c)  $1\frac{2}{3}$  (d)  $1\frac{1}{2}$

42. If  $5\cos\theta + 12\sin\theta = 13$ ,  $0^\circ < \theta < 90^\circ$ , then the value of  $\sin\theta$  is-

यदि  $5\cos\theta + 12\sin\theta = 13$ ,  $0^\circ < \theta < 90^\circ$ , तो  $\sin\theta$  का मान क्या है?

- (a)  $\frac{12}{13}$  (b)  $-\frac{12}{13}$   
(c)  $\frac{5}{13}$  (d)  $\frac{6}{13}$

43. Find the value of the following

निम्न का मान ज्ञात करें।

$\cot 41^\circ \cdot \cot 42^\circ \cdot \cot 43^\circ \cdot \cot 44^\circ \cdot \cot 45^\circ \cdot \cot 46^\circ \cdot \cot 47^\circ \cdot \cot 48^\circ \cdot \cot 49^\circ$

- (a)  $1/\sqrt{2}$  (b) 1  
(c) 0 (d)  $\sqrt{3}/2$

44. If  $7\sin^2\theta + 3\cos^2\theta = 4$ , then the value of  $\tan\theta$  is ( $\theta$  is acute angle)

यदि  $7\sin^2\theta + 3\cos^2\theta = 4$ , तो  $\tan\theta$  का मान क्या है ( $\theta$  न्यून कोण है)?

- (a)  $1/\sqrt{2}$  (b) 1  
(c)  $\sqrt{3}$  (d)  $1/\sqrt{3}$

45. The value of  $(\operatorname{cosec} \alpha - \sin \alpha)(\sec \alpha - \cos \alpha)(\tan \alpha + \cot \alpha) -$

$(\operatorname{cosec} \alpha - \sin \alpha)(\sec \alpha - \cos \alpha)(\tan \alpha + \cot \alpha)$  का मान क्या है?

- (a) 1 (b) 6  
(c) 2 (d) 4

### SSC CGL MAINS 2016

{30 Nov. 2016}

46. If  $x \tan 60^\circ + \cos 45^\circ = \sec 45^\circ$  then the value of  $x^2 + 1$  is-

यदि  $x \tan 60^\circ + \cos 45^\circ = \sec 45^\circ$  है, तो  $x^2 + 1$  का मान है-

- (a)  $6/7$  (b)  $7/6$   
(c)  $5/6$  (d)  $6/5$

47.  $x, y$  be two acute angles,  $x + y < 90^\circ$  and  $\sin(2x - 20^\circ) = \cos(2y + 20^\circ)$ , the value of  $\tan(x + y)$  is

$x, y$  दो न्यून कोण हैं,  $x + y < 90^\circ$  और तो  $\sin(2x - 20^\circ) = \cos(2y + 20^\circ)$  है, तो  $\tan(x + y)$  का मान बताइए?

- (a)  $\sqrt{3}$  (b)  $1/\sqrt{3}$   
(c) 1 (d)  $2 + \sqrt{2}$

48. If  $a^2 \sec^2 x - b^2 \tan^2 x = c^2$  then the value of  $\sec^2 x + \tan^2 x$  is equal to (assume  $b^2 \neq a^2$ )

यदि  $a^2 \sec^2 x - b^2 \tan^2 x = c^2$  है, तो  $\sec^2 x + \tan^2 x$  का मान बताइए (यह मानते हुए कि  $b^2 \neq a^2$ )

(a)  $\frac{b^2 - a^2 + 2c^2}{b^2 + a^2}$

(b)  $\frac{b^2 + a^2 - 2c^2}{b^2 - a^2}$

(c)  $\frac{b^2 - a^2 - 2c^2}{b^2 + a^2}$

(d)  $\frac{b^2 - a^2}{b^2 + a^2 + 2c^2}$

49.  $(1 + \sec 20^\circ + \cot 70^\circ)(1 - \operatorname{cosec} 20^\circ + \tan 70^\circ)$  is equal to  $(1 + \sec 20^\circ + \cot 70^\circ)(1 - \operatorname{cosec} 20^\circ + \tan 70^\circ)$  किसके बराबर होगा?
- (a) 0 (b) 1  
(c) 2 (d) 3

50. If  $\tan^4 \theta + \tan^2 \theta = 1$  then the value of  $\cos^4 \theta + \cos^2 \theta$  is  
यदि  $\tan^4 \theta + \tan^2 \theta = 1$  हो, तो  $\cos^4 \theta + \cos^2 \theta$  का मान क्या होगा?

- (a) 2 (b) 0  
(c) 1 (d) -1

51. The value of  $8(\sin^6 \theta + \cos^6 \theta) - 12(\sin^4 \theta + \cos^4 \theta)$  is equal to  $8(\sin^6 \theta + \cos^6 \theta) - 12(\sin^4 \theta + \cos^4 \theta)$  का मान क्या होगा?

- (a) 20 (b) -20  
(c) -4 (d) 4

### SSC CGL MAINS 2016

{01 Dec. 2016}

52. The value of the expression  $(1 + \sec 22^\circ + \cot 68^\circ)(1 - \operatorname{cosec} 22^\circ + \tan 68^\circ)$  is-  
 $(1 + \sec 22^\circ + \cot 68^\circ)(1 - \operatorname{cosec} 22^\circ + \tan 68^\circ)$  व्यंजक का मान बताइए?

- (a) 0 (b) 1  
(c) -1 (d) 2

53. If  $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$  and  $x \sin \theta - y \cos \theta = 0$ , then the value of  $x^2 + y^2$  equals-

यदि  $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$  हो और  $x \sin \theta - y \cos \theta = 0$  हो तो  $x^2 + y^2$  का मान बताइए?

- (a) 1 (b)  $1/2$   
(c)  $3/2$  (d) 2

54. If  $\sec \theta + \tan \theta = m$  ( $>1$ ), then the value of  $\sin \theta$  is ( $0^\circ < \theta < 90^\circ$ )

यदि  $\sec \theta + \tan \theta = m$  ( $>1$ ) हो, तो  $\sin \theta$  का मान ज्ञात कीजिए ( $0^\circ < \theta < 90^\circ$ )

- (a)  $\frac{1-m^2}{1+m^2}$  (b)  $\frac{m^2-1}{m^2+1}$   
(c)  $\frac{m^2+1}{m^2-1}$  (d)  $\frac{1+m^2}{1-m^2}$

55. If  $(a^2 - b^2) \sin \theta + 2ab \cos \theta = a^2 + b^2$ , then  $\tan \theta =$

यदि  $(a^2 - b^2) \sin \theta + 2ab \cos \theta = a^2 + b^2$  हो, तो  $\tan \theta$  का मान बताइए?

- (a)  $\frac{2ab}{a^2 - b^2}$  (b)  $\frac{a^2 - b^2}{2ab}$   
(c)  $\frac{ab}{a^2 - b^2}$  (d)  $\frac{a^2 - b^2}{ab}$

56. If  $2y \cos \theta = x \sin \theta$  and  $2x \sec \theta - y \operatorname{cosec} \theta = 3$ , then the value of  $x^2 + 4y^2$  is

यदि  $2y \cos \theta = x \sin \theta$  and  $2x \sec \theta - y \operatorname{cosec} \theta = 3$  हो, तो  $x^2 + 4y^2$  का मान क्या होगा?

- (a) 1 (b) 2  
(c) 3 (d) 4

57.  $\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta + \operatorname{cosec} \theta + 1} = ?$

- (a)  $\frac{1 + \cos \theta}{\sin \theta}$  (b)  $\frac{1 - \cos \theta}{\sin \theta}$   
(c)  $\frac{\cot \theta - 1}{\operatorname{cosec} \theta}$  (d)  $\frac{\cot \theta - 1}{\operatorname{cosec} \theta}$

### SSC CGL MAINS 2016

{12 Jan. 2017}

58. If  $\sec A + \tan A = a$ , then the value of  $\cos A$  is-

यदि  $\sec A + \tan A = a$  हो, तो  $\cos A$  का मान क्या होगा?

- (a)  $\frac{a^2+1}{2a}$  (b)  $\frac{2a}{a^2+1}$   
(c)  $\frac{a^2-1}{2a}$  (d)  $\frac{2a}{a^2-1}$

59. If  $\sin P + \operatorname{cosec} P = 2$ , then the value of  $\sin^7 P + \operatorname{cosec}^7 P$  is

यदि  $\sin P + \operatorname{cosec} P = 2$  हो, तो  $\sin^7 P + \operatorname{cosec}^7 P$  का मान क्या होगा?

- (a) 1 (b) 2  
(c) 3 (d) 0

60. If  $\cos x \cdot \cos y + \sin x \cdot \sin y = -1$  then  $\cos x + \cos y$  is-

यदि  $\cos x \cdot \cos y + \sin x \cdot \sin y = -1$  हो, तो  $\cos x + \cos y$  का मान क्या होगा?

- (a) -2 (b) 1  
(c) 0 (d) 2

61. The value of the expression  $2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) + 1$  is-

व्यंजक  $2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) + 1$  का मान क्या होगा?

- (a) -1 (b) 0  
(c) 1 (d) 2

62. If  $\cos \theta = \frac{x^2 - y^2}{x^2 + y^2}$  then the value of  $\cot \theta$  is equal to [If  $0 \leq \theta \leq 90^\circ$ ]

यदि  $\cos \theta = \frac{x^2 - y^2}{x^2 + y^2}$  हो, तो  $\cot \theta$  का मान किसके बराबर होगा? (यदि  $0 \leq \theta \leq 90^\circ$ )

- (a)  $\frac{2xy}{x^2 - y^2}$  (b)  $\frac{2xy}{x^2 + y^2}$

- (c)  $\frac{x^2 + y^2}{2xy}$  (d)  $\frac{x^2 - y^2}{2xy}$

63. If  $x = \operatorname{cosec} \theta - \sin \theta$  and  $y = \sec \theta - \cos \theta$ , then the relation between  $x$  and  $y$  is -

यदि  $x = \operatorname{cosec} \theta - \sin \theta$  और  $y = \sec \theta - \cos \theta$  हो, तो  $x$  और  $y$  के बीच संबंध होगा?

- (a)  $x^2 + y^2 + 3 = 1$   
(b)  $x^2 y^2 (x^2 + y^2 + 3) = 1$   
(c)  $x^2 y^2 (x^2 + y^2 - 5) = 1$   
(d)  $y^2 (x^2 + y^2 - 5) = 1$

### SSC CGL MAINS 2017

{17 Feb. 2018}

64. What is the value of  $[(\sin x + \sin y)(\sin x - \sin y)] / [(\cos x + \cos y)(\cos y - \cos x)]?$

$[(\sin x + \sin y)(\sin x - \sin y)] / [(\cos x + \cos y)(\cos y - \cos x)]$  का मान क्या है?

- (a) 0 (b) 1  
(c) -1 (d) 2

65. What is the value of  $[(\tan 5\theta + \tan 3\theta) / 4 \cos 4\theta (\tan 5\theta - \tan 3\theta)]?$

$[(\tan 5\theta + \tan 3\theta) / 4 \cos 4\theta (\tan 5\theta - \tan 3\theta)]$  का मान क्या है?

- (a)  $\sin 2\theta$  (b)  $\cos 2\theta$   
(c)  $\tan 4\theta$  (d)  $\cot 2\theta$

66. What is the value of  $(4/3) \cot^2(\pi/6) + 3 \cos^2(150^\circ) - 4 \operatorname{cosec}^2 45^\circ + 8 \sin(\pi/2)?$

$(4/3) \cot^2(\pi/6) + 3 \cos^2(150^\circ) - 4 \operatorname{cosec}^2 45^\circ + 8 \sin(\pi/2)$  का मान क्या है?

- (a)  $25/4$  (b) 1  
(c)  $-7/2$  (d)  $13/2$

67. What is the value of  $\sin(B - C) \cos(A - D) + \sin(A - B) \cos(C - D) + \sin(C - A) \cos(B - D)?$

$\sin(B - C) \cos(A - D) + \sin(A - B) \cos(C - D) + \sin(C - A) \cos(B - D)$  का मान क्या है?

- (a)  $3/2$  (b) -3  
(c) 1 (d) 0

68. What is the value of/का मान क्या है?

$$\frac{\left[ \begin{aligned} &4 \cos(90 - A) \sin^3(90 + A) \\ &4 \sin(90 + A) \cos^3(90 - A) \end{aligned} \right] -}{\cos\left(\frac{180 + 8A}{2}\right)}?$$

- (a) 1 (b) -1  
(c) 0 (d) 2

69. What is the value of  $\cos[(180 - \theta)/2] \cdot \cos[(180 - 90)/2] + \sin[(180 - 30)/2] \cdot \sin[(180 - 130)/2]$ ?  
 $\cos[(180 - \theta)/2] \cdot \cos[(180 - 90)/2] + \sin[(180 - 30)/2] \cdot \sin[(180 - 130)/2]$  का मान क्या है?

(a)  $\sin 2\theta \cdot \sin 4\theta$   
 (b)  $\cos 2\theta \cdot \cos 6\theta$   
 (c)  $\sin 2\theta \cdot \sin 6\theta$   
 (d)  $\cos 2\theta \cdot \cos 4\theta$

70. What is the value of  $[\tan^2(90 - \theta) - \sin^2(90 - \theta)] \operatorname{cosec}^2(90 - \theta) \cot^2(90 - \theta)$ ?

$[\tan^2(90 - \theta) - \sin^2(90 - \theta)] \operatorname{cosec}^2(90 - \theta) \cot^2(90 - \theta)$  का मान क्या है?

(a) 0 (b) 1  
 (c) -1 (d) 2

### SSC CGL MAINS 2017

{18 Feb. 2018}

71. What is the value of  $[(\cos^3 2\theta + 3 \cos 2\theta) \div (\cos^6 \theta - \sin^6 \theta)]$ ?

$[(\cos^3 2\theta + 3 \cos 2\theta) \div (\cos^6 \theta - \sin^6 \theta)]$  का मान क्या है?

(a) 0 (b) 1  
 (c) 4 (d) 2

72. What is the value of  $\tan\left(\frac{\pi}{4} + A\right) \times \tan\left(\frac{3\pi}{4} + A\right)$ ?

$\tan\left(\frac{\pi}{4} + A\right) \times \tan\left(\frac{3\pi}{4} + A\right)$

(a) 1 (b) 0  
 (c)  $\cot A/2$  (d) -1

73. What is the value of  $\frac{1}{2}(\sec 2\theta + 1) \sqrt{\sec^2 \theta - 1} \times (\cot \theta - \tan \theta)$ ?

$\frac{1}{2}(\sec 2\theta + 1) \sqrt{\sec^2 \theta - 1} \times (\cot \theta - \tan \theta)$

(a) 0 (b) 1  
 (c)  $\operatorname{cosec} \theta$  (d)  $\sec \theta$

74. What is the value of  $\sin(630^\circ + A) + \cos A$ ?

$\sin(630^\circ + A) + \cos A$  का मान क्या है?

(a)  $\sqrt{3}/2$  (b)  $1/2$   
 (c) 0 (d)  $2/\sqrt{3}$

75. What is the value of  $[(\sin 59^\circ \cdot \cos 31^\circ + \cos 59^\circ \cdot \sin 31^\circ) \div (\cos 20^\circ \cos 25^\circ - \sin 20^\circ \sin 25^\circ)]$ ?

$[(\sin 59^\circ \cdot \cos 31^\circ + \cos 59^\circ \cdot \sin 31^\circ) \div (\cos 20^\circ \cos 25^\circ - \sin 20^\circ \sin 25^\circ)]$

(a)  $1/\sqrt{2}$  (b)  $2\sqrt{2}$   
 (c)  $\sqrt{3}$  (d)  $\sqrt{2}$

76. What is the value of  $\cos(90 - B) \cdot \sin(C - A) + \sin(90 + A) \cdot \cos(B + C) - \sin(90 - C) \cdot \cos(A + B)$ ?

$\cos(90 - B) \cdot \sin(C - A) + \sin(90 + A) \cdot \cos(B + C) - \sin(90 - C) \cdot \cos(A + B)$  का मान क्या है?

(a) 1  
 (b)  $\sin(A + B - C)$   
 (c)  $\cos(B + C - A)$   
 (d) 0

### SSC CGL MAINS 2017

{19 Feb. 2018}

77. What is the value of  $[(\cos 7A + \cos 5A) \div (\sin 7A - \sin 5A)]$ ?

$[(\cos 7A + \cos 5A) \div (\sin 7A - \sin 5A)]$  का मान क्या है?

(a)  $\tan A$  (b)  $\tan 4A$   
 (c)  $\cot 4A$  (d)  $\cot A$

78. What is the value of  $[1 - \sin(90 - 2A)]/[1 + \sin(90 + 2A)]$ ?

$[1 - \sin(90 - 2A)]/[1 + \sin(90 + 2A)]$  का मान क्या है?

(a)  $\sin A \cdot \cos A$  (b)  $\cot^2 A$   
 (c)  $\tan^2 A$  (d)  $\sin^2 A \cdot \cos A$

79. What is the value of  $\sin 75^\circ + \sin 15^\circ$ ?

$\sin 75^\circ + \sin 15^\circ$  का मान क्या है?

(a)  $\sqrt{3}$  (b)  $2\sqrt{3}$   
 (c)  $\sqrt{3/2}$  (d)  $3/\sqrt{2}$

80. What is the value of  $[(\cos 3\theta + 2 \cos 5\theta + \cos 7\theta) \div (\cos \theta + 2 \cos 3\theta + \cos 5\theta)] + \sin 2\theta \cdot \tan 3\theta$ ?

$[(\cos 3\theta + 2 \cos 5\theta + \cos 7\theta) \div (\cos \theta + 2 \cos 3\theta + \cos 5\theta)] + \sin 2\theta \cdot \tan 3\theta$  का मान क्या है?

(a)  $\cos 2\theta$  (b)  $\sin 2\theta$   
 (c)  $\tan 2\theta$  (d)  $\cot \theta \cdot \sin 2\theta$

81. What is the value of  $[2 \sin(45 + \theta) \cdot \sin(45 - \theta)]/\cos 2\theta$ ?

$[2 \sin(45 + \theta) \cdot \sin(45 - \theta)]/\cos 2\theta$  का मान क्या है?

(a) 0 (b)  $\tan 2\theta$   
 (c)  $\cot 2\theta$  (d) 1

82. What is the value of  $\sin(90^\circ + 2A)[4 - \cos^2(90^\circ - 2A)]$ ?

$\sin(90^\circ + 2A)[4 - \cos^2(90^\circ - 2A)]$  का मान क्या है?

(a)  $2(\cos^2 A - \sin^3 A)$   
 (b)  $2(\cos^3 A + \sin^3 A)$   
 (c)  $4(\cos^6 A + \sin^6 A)$   
 (d)  $4(\cos^6 A - \sin^6 A)$

83. What is the value of  $[\cos(90 + A) \div \sec(270 - A)] + [\sin(270 + A) \div \operatorname{cosec}(630 - A)]$ ?

$[\cos(90 + A) \div \sec(270 - A)] + [\sin(270 + A) \div \operatorname{cosec}(630 - A)]$  का मान क्या है?

(a)  $3 \sec A$  (b)  $\tan A \cdot \sec A$   
 (c) 0 (d) 1

### SSC CGL MAINS 2017

{20 Feb. 2018}

84. What is the value of  $[\sin(y - z) + \sin(y + z) + 2 \sin y]/[\sin(x - z) + \sin(x + z) + 2 \sin x]$ ?

$[\sin(y - z) + \sin(y + z) + 2 \sin y]/[\sin(x - z) + \sin(x + z) + 2 \sin x]$  का मान क्या है?

(a)  $\cos x \cdot \sin y$  (b)  $(\sin y)/(\sin x)$   
 (c)  $\sin z$  (d)  $\sin x \cdot \tan y$

85. What is the value of  $\{[\sin(x + y) - 2 \sin x + \sin(x - y)]/[\cos(x - y) + \cos(x + y) - 2 \cos x]\} \times \{[\sin 10x - \sin 8x]/[\cos 10x + \cos 8x]\}$ ?

$\{[\sin(x + y) - 2 \sin x + \sin(x - y)]/[\cos(x - y) + \cos(x + y) - 2 \cos x]\} \times \{[\sin 10x - \sin 8x]/[\cos 10x + \cos 8x]\}$  का मान क्या है?

(a) 0 (b)  $\tan^2 x$   
 (c) 1 (d)  $2 \tan x$

86. What is the value of  $[\sin(90^\circ - 10\theta) - \cos(\pi - 6\theta)]/[\cos(\pi/2 - 10\theta) - \sin(\pi - 6\theta)]$ ?

$[\sin(90^\circ - 10\theta) - \cos(\pi - 6\theta)]/[\cos(\pi/2 - 10\theta) - \sin(\pi - 6\theta)]$  का मान क्या है?

(a)  $\tan 2\theta$  (b)  $\cot 2\theta$   
 (c)  $\cot \theta$  (d)  $\cot 3\theta$

87. If  $\sec \theta (\cos \theta + \sin \theta) = \sqrt{2}$ , then what is the value of  $(2 \sin \theta)/(\cos \theta - \sin \theta)$ ?

यदि  $\sec \theta (\cos \theta + \sin \theta) = \sqrt{2}$  है, तो  $(2 \sin \theta)/(\cos \theta - \sin \theta)$  का मान क्या है?

(a)  $3\sqrt{2}$  (b)  $3/\sqrt{2}$   
 (c)  $1/\sqrt{2}$  (d)  $\sqrt{2}$

88. What is the value of  $\frac{1}{\sin^4(90 - \theta)} + \frac{1}{[\cos^2(90 - \theta)] - 1}$ ?

$\frac{1}{\sin^4(90 - \theta)} + \frac{1}{[\cos^2(90 - \theta)] - 1}$

(a)  $\tan^2 \theta \cdot \sec^2 \theta$   
 (b)  $\sec^4 \theta$   
 (c)  $\tan^4 \theta$   
 (d)  $\tan^2 \theta \cdot \sin^2 \theta$

89. What is the value of  $[\tan(90 - A) + \cot(90 - A)]^2 / [2 \sec^2(90 - 2A)]$ ?  
 $[\tan(90 - A) + \cot(90 - A)]^2 / [2 \sec^2(90 - 2A)]$  का मान क्या है?

(a) 0 (b) 1  
(c) 2 (d) -1

90. What is the value of  $\{\sin(90 - x) \cdot \cos[\pi - (x - y)]\} + \{\cos(90 - x) \cdot \sin[\pi - (y - x)]\}$ ?

$\{\sin(90 - x) \cdot \cos[\pi - (x - y)]\} + \{\cos(90 - x) \cdot \sin[\pi - (y - x)]\}$  का मान क्या है?

(a)  $-\cos y$  (b)  $-\sin y$   
(c)  $\cos x$  (d)  $\tan y$

### SSC CGL MAINS 2017

{21 Feb. 2018}

91. What is the value of  $\{(\sin 4x + \sin 4y) \cdot [(\tan 2x - 2y)] / (\sin 4x - \sin 4y)\}$ ?

$\{(\sin 4x + \sin 4y) \cdot [(\tan 2x - 2y)] / (\sin 4x - \sin 4y)\}$  का मान क्या है?

(a)  $\tan 2(2x + 2y)$   
(b)  $\tan^2$   
(c)  $\cot^2(x - y)$   
(d)  $\tan(2x + 2y)$

92. What is the value of  $(32 \cos^6 x - 48 \cos^4 x + 18 \cos^2 x - 1) / [4 \sin x \cdot \cos x \cdot \sin(60 - x) \cdot \cos(60 - x) \cdot \sin(60 + x) \cdot \cos(60 + x)]$ ?

$(32 \cos^6 x - 48 \cos^4 x + 18 \cos^2 x - 1) / [4 \sin x \cdot \cos x \cdot \sin(60 - x) \cdot \cos(60 - x) \cdot \sin(60 + x) \cdot \cos(60 + x)]$

का मान क्या है?

(a)  $4 \tan 6x$  (b)  $4 \cot 6x$   
(c)  $8 \cot 6x$  (d)  $8 \tan 6x$

93. What is the value of  $[2 \cot x (\pi - A) / 2] / [1 + \tan^2 x (2\pi - A) / 2]$ ?

$[2 \cot x (\pi - A) / 2] / [1 + \tan^2 x (2\pi - A) / 2]$  का मान क्या है?

(a)  $2 \sin^2 A / 2$  (b)  $\cos A$   
(c)  $\sin A$  (d)  $2 \cos^2 A / 2$

94. If  $\tan \theta + \sec \theta = (x - 2) / (x + 2)$ , then what is the value of  $\cos \theta$ ?

यदि  $\tan \theta + \sec \theta = (x - 2) / (x + 2)$  है, तो  $\cos \theta$  का मान क्या है?

(a)  $(x^2 - 1) / (x^2 + 1)$   
(b)  $(2x^2 - 4) / (2x^2 + 4)$   
(c)  $(x^2 - 4) / (x^2 + 4)$   
(d)  $(x^2 - 2) / (x^2 + 2)$

95. What is the value of  $(\cos 40^\circ - \cos 140^\circ) / (\sin 80^\circ + \sin 20^\circ)$ ?

$(\cos 40^\circ - \cos 140^\circ) / (\sin 80^\circ + \sin 20^\circ)$  का मान क्या है?

(a)  $2\sqrt{3}$  (b)  $2/\sqrt{3}$   
(c)  $1/\sqrt{3}$  (d)  $\sqrt{3}$

96. What is the value of  $[1 - \tan(90 - \theta) + \sec(90 - \theta)] / [\tan(90 - \theta) + \sec(90 - \theta) + 1]$ ?

$[1 - \tan(90 - \theta) + \sec(90 - \theta)] / [\tan(90 - \theta) + \sec(90 - \theta) + 1]$  का मान क्या है?

(a)  $\cos(\theta/2)$  (b)  $\tan(\theta/2)$   
(c)  $\sin \theta$  (d)  $\cos \theta$

97. What is the value of  $[\sin(90 - A) + \cos(180 - 2A)] / [\cos(90 - 2A) + \sin(180 - A)]$ ?

$[\sin(90 - A) + \cos(180 - 2A)] / [\cos(90 - 2A) + \sin(180 - A)]$  का मान क्या है?

(a)  $\sin(A/2) \cdot \cos A$   
(b)  $\cot(A/2)$   
(c)  $\tan(A/2)$   
(d)  $\sin A \cdot \cos(A/2)$

### SSC CGL MAINS 2017

{09 Mar. 2018} Paper - I

98. What is the value of  $\frac{2(1 - \sin^2 \theta) \operatorname{cosec}^2 \theta}{\cot^2 \theta (1 + \tan^2 \theta)} - 1$ ?

$\frac{2(1 - \sin^2 \theta) \operatorname{cosec}^2 \theta}{\cot^2 \theta (1 + \tan^2 \theta)} - 1$  का मान क्या है?

(a)  $\sin 2\theta$  (b)  $\sin^2 \theta$   
(c)  $\cos^2 \theta$  (d)  $\cos 2\theta$

99. What is the value of  $\frac{\cos 2A + 2 \cos^2 A - 2 \cos 2A \cos A}{\sin 2A - 2 \sin^2 A \sin 2A}$ ?

$\frac{\cos 2A + 2 \cos^2 A - 2 \cos 2A \cos A}{\sin 2A - 2 \sin^2 A \sin 2A}$

का मान क्या है?

(a)  $2 \cot A$  (b)  $2 \tan A$   
(c)  $\cot A$  (d)  $\tan A$

100. What is the value of  $\cos 15^\circ - \cos 165^\circ$ ?

$\cos 15^\circ - \cos 165^\circ$  का मान क्या है?

(a)  $\sqrt{3} / \sqrt{2}$  (b)  $2 / (\sqrt{3} - 1)$   
(c)  $(\sqrt{3} + 1) / \sqrt{2}$  (d)  $(\sqrt{3} + 1) / 2$

101. If  $P + Q + R = 60^\circ$ , then what is the value of  $\cos Q \cdot \cos R (\cos P - \sin P) + \sin Q \cdot \sin R (\sin P - \cos P)$ ?

यदि  $P + Q + R = 60^\circ$  है, तो  $\cos Q \cdot \cos R (\cos P - \sin P) + \sin Q \cdot \sin R (\sin P - \cos P)$  का मान क्या है?

(a)  $1/2$  (b)  $\sqrt{3} / \sqrt{2}$   
(c)  $1/\sqrt{2}$  (d)  $\sqrt{2}$

102. What is the value of  $[1 - \tan(90 - \theta)]^2 / [\cos^2(90 - \theta)] - 1$ ?

$[1 - \tan(90 - \theta)]^2 / [\cos^2(90 - \theta)] - 1$  का मान क्या है?

(a)  $-\sin 2\theta$  (b)  $-\cos 2\theta$   
(c)  $\cos 2\theta$  (d)  $\sin 2\theta$

103. What is the value of  $1 + 2 \cot^2(90 - x) - 2 \operatorname{cosec}(90 - x) \cdot \cot(90 - x) / [\operatorname{cosec}(90 - x) - \cot(90 - x)]$ ?

$1 + 2 \cot^2(90 - x) - 2 \operatorname{cosec}(90 - x) \cdot \cot(90 - x) / [\operatorname{cosec}(90 - x) - \cot(90 - x)]$  का मान क्या है?

(a)  $\cos x + \sin x$  (b)  $\sin x - \cos x$   
(c)  $\sec x + \tan x$  (d)  $\sec x - \tan x$

104. What is the value of  $\sin(180 - \theta) \sin(90 - \theta) + [\cot(90 - \theta) / (1 + \tan^2 \theta)]$ ?

$\sin(180 - \theta) \sin(90 - \theta) + [\cot(90 - \theta) / (1 + \tan^2 \theta)]$  का मान क्या है?

(a)  $\cos^2 \theta \cdot \sin \theta$   
(b)  $\cot \theta / (1 + \cot^2 \theta)^2$   
(c)  $\tan \theta / (1 + \tan^2 \theta)^2$   
(d)  $2 \cos \theta \cdot \sin \theta$

### SSC CGL MAINS 2017

{09 Mar. 2018} Paper - II

105.  $\cos 15^\circ + \cos 105^\circ = ?$

(a)  $\sqrt{3}$  (b)  $\frac{1}{\sqrt{2}}$   
(c)  $\frac{\sqrt{3}}{2}$  (d)  $\frac{1}{\sqrt{3}}$

106. If  $(A + B + C) = 90^\circ$ , then what is the value of  $\sin\left(\frac{A}{2}\right) \cdot \sin\left(\frac{180 - B - C}{2}\right) + \cos\left(\frac{A}{2}\right) \cdot \sin(B + C) / 2$ ?

$\sin\left(\frac{A}{2}\right) \cdot \sin\left(\frac{180 - B - C}{2}\right) + \cos\left(\frac{A}{2}\right) \cdot \sin(B + C) / 2$  का मान क्या है?

यदि  $(A + B + C) = 90^\circ$  है, तो  $\sin\left(\frac{A}{2}\right) \cdot \sin\left(\frac{180 - B - C}{2}\right) + \cos\left(\frac{A}{2}\right) \cdot \sin(B + C) / 2$  का मान क्या है?

$\sin\left(\frac{A}{2}\right) \cdot \sin\left(\frac{180 - B - C}{2}\right) + \cos\left(\frac{A}{2}\right) \cdot \sin(B + C) / 2$  का मान क्या है?

(a)  $\frac{1}{2}$  (b) 1  
(c)  $\frac{1}{\sqrt{2}}$  (d)  $\frac{\sqrt{3}}{2}$

107. What is the value of  $\cot(90-x)$   
 $\sin^4(90-x) + \cot(180-x) \sin^4(180-x)$ ?  
 $\cot(90-x) \sin^4(90-x) + \cot(180-x)$   
 $\sin^4(180-x)$  का मान क्या है?

- (a)  $\frac{(\cos^4 x)}{4}$  (b)  $\frac{(\sin^2 2x)}{2}$   
 (c)  $\frac{(\cos^2 2x)}{2}$  (d)  $\frac{(\sin 4x)}{4}$

108. If  $\sin x = 1/2$  and  $\sin y = 2/3$ , then what is the value of  $[(6\cos^2 x - 4\cos^4 x)/(18\cos^2 y - 27\cos^4 y)]$ ?

यदि  $\sin x = 1/2$  तथा  $\sin y = 2/3$  है, तो  $[(6\cos^2 x - 4\cos^4 x)/(18\cos^2 y - 27\cos^4 y)]$  का मान क्या है?

- (a) 27/20 (b) 15/14  
 (c) 25/21 (d) 17/14

109. If  $\sin(A-B) = 1/2$  and  $\cos(A+B) = 1/2$ , then what is the value of  $\sin A \cos A + \sin^2 A \sin B \cos B + \cos^3 A \cos B \tan A$ ?

यदि  $\sin(A-B) = 1/2$  तथा  $\cos(A+B) = 1/2$  का तो मान  $\sin A \cos A + \sin^2 A \sin B \cos B + \cos^3 A \cos B \tan A$  क्या है?

- (a)  $\frac{1}{2}$  (b)  $\frac{6+\sqrt{3}}{8}$   
 (c)  $\frac{1}{4}$  (d)  $\frac{3}{2}$

110. What is the value of  $\cot(90^\circ+75^\circ)$ ?  
 $\cot(90^\circ+75^\circ)$  का मान क्या है?

- (a)  $2+\sqrt{3}$  (b)  $-(2+\sqrt{3})$   
 (c)  $\sqrt{3}+1$  (d)  $-\sqrt{3}-1$

111. What is the value of  $2\sin 15^\circ \cos 15^\circ - 4\sin^3 15^\circ \cos 15^\circ$ ?  
 $2\sin 15^\circ \cos 15^\circ - 4\sin^3 15^\circ \cos 15^\circ$  का मान क्या है?

- (a)  $\frac{3}{\sqrt{2}}$  (b)  $\frac{\sqrt{3}}{2}$   
 (c)  $\frac{\sqrt{3}}{4}$  (d)  $\frac{1}{2}$

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112.  $\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\sin 56^\circ \sec 34^\circ + \cos 25^\circ \operatorname{cosec} 65^\circ} = ?$

- (a) 0.5 (b) 4  
 (c) 2 (d) 0.25

113.  $\operatorname{Cosec}(65^\circ + \theta) - \sec(25^\circ - \theta) +$

$\tan^2 20^\circ - \operatorname{cosec}^2 70^\circ = ?$

- (a) 2 (b) -1  
 (c) 1 (d) 0

114.  $\sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$  is equal to:

$\sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$  निम्न में से किसके बराबर है?

- (a)  $1 + \sec \theta \tan \theta$   
 (b)  $\sec \theta - \tan \theta$   
 (c)  $\sec \theta + \tan \theta$   
 (d)  $1 - \sec \theta - \tan \theta$

115. If  $\theta$  lies in the first quadrant and

$\cos^2 \theta - \sin^2 \theta = \frac{1}{2}$ , then the value

of  $\tan^2 2\theta + \sin^2 3\theta$  is

यदि  $\theta$  प्रथम चतुर्थांश में है तथा

$\cos^2 \theta - \sin^2 \theta = \frac{1}{2}$  है, तो

$\tan^2 2\theta + \sin^2 3\theta$  का मान है

- (a) 4 (b) 4/3

- (c)  $\frac{7}{2}$  (d) 3

116. If  $5\sin \theta - 4\cos \theta = 0$ ,

$0^\circ < \theta < 90^\circ$  then the value of

$\frac{5\sin \theta - 2\cos \theta}{5\sin \theta + 3\cos \theta}$  is:

- (a) 2/7 (b) 5/8  
 (c) 3/7 (d) 3/8

117.  $\frac{(1+\cos \theta)^2 + \sin^2 \theta}{(\operatorname{cosec}^2 \theta - 1)\sin^2 \theta} = ?$

- (a)  $\cos \theta(1+\sin \theta)$   
 (b)  $2\sec \theta(1+\sec \theta)$   
 (c)  $\sec \theta(1+\sin \theta)$   
 (d)  $2\cos \theta(1+\sec \theta)$

118. If  $\sin \theta = \sqrt{3} \cos \theta$ ,  $0^\circ < \theta < 90^\circ$ , then the value of  $2\sin^2 \theta + \sec^2 \theta + \sin \theta + \operatorname{cosec} \theta$  is

यदि  $\sin \theta = \sqrt{3} \cos \theta$ ,  $0^\circ < \theta < 90^\circ$  है, तो  $2\sin^2 \theta + \sec^2 \theta + \sin \theta + \operatorname{cosec} \theta$  का मान है?

- (a)  $\frac{33+10\sqrt{3}}{3}$  (b)  $\frac{19+10\sqrt{3}}{3}$   
 (c)  $\frac{19+10\sqrt{3}}{6}$  (d)  $\frac{33+10\sqrt{3}}{6}$

119.  $\left(\frac{1-\tan \theta}{1-\cot \theta}\right)^2 + 1 = ?$

- (a)  $\sec^2 \theta$  (b)  $\cos^2 \theta$   
 (c)  $\sin^2 \theta$  (d)  $\operatorname{cosec}^2 \theta$

120.  $\frac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$  is equal to-

$\frac{(2\sin A)(1+\sin A)}{1+\sin A+\cos A}$  का मान निम्न में से किसके बराबर है?

- (a)  $1 + \sin A \cos A$   
 (b)  $1 - \sin A \cos A$   
 (c)  $1 + \cos A - \sin A$   
 (d)  $1 + \sin A - \cos A$

121. The value of the expression

$(\cos^6 \theta + \sin^6 \theta - 1)(\tan^2 \theta + \cot^2 \theta + 2)$  is -

व्यंजक  $(\cos^6 \theta + \sin^6 \theta - 1)(\tan^2 \theta + \cot^2 \theta + 2)$  निम्न में से किसके बराबर है?

- (a) -3 (b) 0  
 (c) 1 (d) -1

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122. The value of

$\left(\frac{\sin A}{1-\cos A} + \frac{1-\cos A}{\sin A}\right) \div \left(\frac{\cot^2 A}{1+\operatorname{cosec} A} + 1\right)$

is: / का मान है:

- (a) 3/2 (b) 1/2  
 (c) 1 (d) 2

123. If  $\frac{1+\sin \theta}{1-\sin \theta} = \frac{p^2}{q^2}$ ,

then  $\sec \theta$  is equal to:

यदि  $\frac{1+\sin \theta}{1-\sin \theta} = \frac{p^2}{q^2}$  है, तो  $\sec \theta$  निम्नलिखित में से किसके बराबर है:

- (a)  $\frac{2p^2 q^2}{p^2 + q^2}$  (b)  $\frac{1}{2} \left(\frac{q}{p} + \frac{p}{q}\right)$   
 (c)  $\frac{1}{p^2} + \frac{1}{q^2}$  (d)  $\frac{p^2 q^2}{p^2 + q^2}$

124. The value of

$\frac{\sec \theta(1-\sin \theta)(\sin \theta + \cos \theta)(\sec \theta + \tan \theta)}{\sin \theta(1+\tan \theta) + \cos \theta(1+\cot \theta)}$

is equal to: / का मान है:

- (a)  $2\cos \theta$  (b)  $\operatorname{cosec} \theta \sec \theta$   
 (c)  $2\sin \theta$  (d)  $\sin \theta \cos \theta$

125. The value of  $\tan^2 \theta + \cot^2 \theta - \sec^2 \theta \operatorname{cosec}^2 \theta$  is equal to:

$\tan^2 \theta + \cot^2 \theta - \sec^2 \theta \operatorname{cosec}^2 \theta$  का मान निम्नलिखित में से किसके बराबर होगा?

- (a) -2 (b) 1  
 (c) 0 (d) -1

126.  $(\sec\theta - \tan\theta)^2(1 + \sin\theta)^2 \div \sin^2\theta = ?$

- (a)  $\cos\theta$  (b)  $\cot^2\theta$   
(c)  $\sec\theta$  (d)  $\cos^2\theta$

127.  $3(\cot^2\theta - \cos^2\theta) = \cos^2\theta$ ,  $0^\circ < \theta < 90^\circ$ , then the value of  $(\tan^2\theta + \operatorname{cosec}^2\theta + \sin^2\theta)$  is-

यदि  $3(\cot^2\theta - \cos^2\theta) = \cos^2\theta$ ,  $0^\circ < \theta < 90^\circ$  है, तो  $(\tan^2\theta + \operatorname{cosec}^2\theta + \sin^2\theta)$  का मान है:

- (a)  $\frac{13}{3}$  (b)  $\frac{61}{12}$   
(c)  $\frac{25}{12}$  (d)  $\frac{15}{4}$

128. If  $\frac{\sin^2\theta - 3\sin\theta + 2}{\cos^2\theta} = 1$ , where  $0^\circ < \theta < 90^\circ$ , then what is the value of  $(\cos 2\theta + \sin 3\theta + \operatorname{cosec} 2\theta)$ ?

यदि  $\frac{\sin^2\theta - 3\sin\theta + 2}{\cos^2\theta} = 1$  है, तो जिसमें  $0^\circ < \theta < 90^\circ$  है, तो  $(\cos 2\theta + \sin 3\theta + \operatorname{cosec} 2\theta)$  का मान क्या है?

- (a)  $\frac{2 + \sqrt{3}}{3}$  (b)  $\frac{3 + 4\sqrt{3}}{6}$   
(c)  $\frac{9 + 4\sqrt{3}}{6}$  (d)  $\frac{3 + 2\sqrt{3}}{3}$

129. The value of

$$\frac{\sin(78^\circ + \theta) - \cos(12^\circ - \theta) + (\tan^2 70^\circ - \operatorname{cosec}^2 20^\circ)}{\sin 25^\circ \cos 65^\circ + \cos 25^\circ \sin 65^\circ}$$

is: / का मान है:

- (a) 2 (b) -1  
(c) -2 (d) 0

130. The value of

$$\frac{\sqrt{\operatorname{cosec}\theta - \cot\theta}}{\sqrt{\operatorname{cosec}\theta + \cot\theta}} \div \frac{\sin\theta}{1 + \cos\theta} \text{ is equal to:}$$

$\frac{\sqrt{\operatorname{cosec}\theta - \cot\theta}}{\sqrt{\operatorname{cosec}\theta + \cot\theta}} \div \frac{\sin\theta}{1 + \cos\theta}$  का मान निम्नलिखित में से किसके बराबर है?

- (a)  $\operatorname{cosec}\theta$  (b)  $\frac{1}{2}$   
(c)  $\sec\theta$  (d) 1

131. The value of  $(\tan 29^\circ \cot 61^\circ - \operatorname{cosec}^2 61^\circ) + \cot^2 54^\circ - \sec^2 36^\circ + (\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ)$  is / का मान है:

- (a)  $20\frac{1}{2}$  (b) 21  
(c)  $22\frac{1}{2}$  (d) 22

## SSC CGL MAINS 2018

{13 Sept. 2019}

132. If  $\sec\theta + \tan\theta = p$ , ( $p > 1$ ) then

$$\frac{\operatorname{cosec}\theta + 1}{\operatorname{cosec}\theta - 1} = ?$$

- (a)  $\frac{p+1}{p-1}$  (b)  $p^2$   
(c)  $\frac{p-1}{p+1}$  (d)  $2p^2$

133.  $(67^\circ + \theta) - \sec(23^\circ - \theta) + \cos 15^\circ \cos 35^\circ \operatorname{cosec} 55^\circ \cdot \cos 60^\circ \cdot \operatorname{cosec} 75^\circ = ?$

- (a) 2 (b) 0  
(c) 1 (d) 0.5

134. If  $2\cos^2\theta + 3\sin\theta = 3$ , where  $0^\circ < \theta < 90^\circ$ , then what is the value of  $\sin^2 2\theta + \cos^2\theta + \tan^2 2\theta + \operatorname{cosec}^2 2\theta$ ?

यदि  $2\cos^2\theta + 3\sin\theta = 3$ , जहाँ  $0^\circ < \theta < 90^\circ$ , तो  $\sin^2 2\theta + \cos^2\theta + \tan^2 2\theta + \operatorname{cosec}^2 2\theta$  का मान ज्ञात करें।

- (a)  $\frac{35}{12}$  (b)  $\frac{29}{3}$   
(c)  $\frac{35}{6}$  (d)  $\frac{29}{6}$

135.  $(1 + \cot\theta - \operatorname{cosec}\theta)(1 + \cos\theta + \sin\theta) \cdot \sec\theta = ?$

- (a) -2 (b) 2  
(c)  $\sec\theta \cdot \operatorname{cosec}\theta$  (d)  $\sin\theta \cdot \cos\theta$

136.  $\frac{\sec^2\theta}{\operatorname{cosec}^2\theta} + \frac{\operatorname{cosec}^2\theta}{\sec^2\theta} - (\sec^2\theta + \operatorname{cosec}^2\theta) = ?$

- (a) 0 (b) -2  
(c) 2 (d) 1

137.  $\frac{2(\sin^6\theta + \cos^6\theta) - 3(\sin^4\theta + \cos^4\theta)}{\cos^4\theta - \sin^4\theta - 2\cos^2\theta} = ?$

- (a) -1 (b) -2  
(c) 2 (d) 1

138.  $\sin^2 64^\circ + \cos 64^\circ \sin 26^\circ \cdot 2\cos 43^\circ \operatorname{cosec} 47^\circ = ?$

- (a) 4 (b) 1  
(c) 2 (d) 3

139.  $\frac{(\sin\theta - \cos\theta)(1 + \tan\theta + \cot\theta)}{1 + \sin\theta \cos\theta} = ?$

- (a)  $\sec\theta - \operatorname{cosec}\theta$   
(b)  $\operatorname{cosec}\theta - \sec\theta$   
(c)  $\sin\theta + \cos\theta$   
(d)  $\tan\theta - \cot\theta$

140.  $\frac{\sin\theta + \cos\theta - 1}{\sin\theta - \cos\theta + 1} \times \frac{\tan^2\theta(\operatorname{cosec}^2\theta - 1)}{\sec\theta - \tan\theta}$

- (a) 0 (b) -1  
(c) 1 (d)  $1/2$

141. If  $\frac{\sin\theta}{1 + \cos\theta} + \frac{1 + \cos\theta}{\sin\theta} = \frac{4}{\sqrt{3}}$ ,  $0^\circ < \theta < 90^\circ$ , then the value of  $(\tan\theta + \sec\theta)^{-1}$  is:

- (a)  $2 - \sqrt{3}$  (b)  $3 - \sqrt{2}$   
(c)  $2 + \sqrt{3}$  (d)  $3 + \sqrt{2}$

## SSC CGL MAINS 2019

{15 Nov. 2020}

142.  $\frac{\operatorname{cosec}^2 30^\circ \sin^2 45^\circ + \sec^2 60^\circ}{\tan 60^\circ \operatorname{cosec}^2 45^\circ - \sec^2 60^\circ \tan 45^\circ} = ?$

- (a)  $3(2 + \sqrt{3})$  (b)  $2(\sqrt{3} - 2)$   
(c)  $-2\sqrt{3} - 2$  (d)  $-3(2 + \sqrt{3})$

143.  $\frac{\sin\theta(1 - \tan\theta)\tan\theta + \sec^2\theta}{(1 - \sin\theta)\tan\theta(1 + \tan\theta)(\sec\theta + \tan\theta)} = ?$

- (a) 1 (b)  $\operatorname{cosec}\theta \cdot \sec\theta$   
(c)  $\sin\theta \cdot \cos\theta$  (d) -1

144.  $\cos A(\sec A - \cos A) \cdot (\cot A + \tan A) = ?$

- (a)  $\sec A$  (b)  $\cot A$   
(c)  $\sin A$  (d)  $\tan A$

145. If  $\sin 3A = \cos(A + 10^\circ)$ , where  $3A$  is an acute angle, then what is the value of  $2\operatorname{cosec} \frac{3A}{2} + 6\sin^2 3A - \frac{3}{2} \tan^2 3A$ ?

यदि  $\sin 3A = \cos(A + 10^\circ)$  है, जहाँ  $3A$  न्यून कोण है, तो  $2\operatorname{cosec} \frac{3A}{2} + 6\sin^2 3A - \frac{3}{2} \tan^2 3A$  का मान ज्ञात कीजिए।

$2\operatorname{cosec} \frac{3A}{2} + 6\sin^2 3A - \frac{3}{2} \tan^2 3A$

का मान ज्ञात कीजिए।

- (a)  $\frac{7}{4}$  (b) 4  
(c)  $\frac{17}{2}$  (d) 5

146.  $\left(\frac{1}{\cos\theta} - \frac{1}{\sin\theta}\right) + \frac{1}{\operatorname{cosec}\theta - \cot\theta} - \frac{1}{\sec\theta + \tan\theta} = ?$

- (a)  $\sec\theta \cdot \operatorname{cosec}\theta$   
(b)  $\sin\theta \cdot \tan\theta$   
(c)  $\operatorname{cosec}\theta \cdot \cot\theta$   
(d)  $\sin\theta \cdot \cos\theta$

## SSC CGL MAINS 2019

{16 Nov. 2020}

147. In  $\Delta PQR$ ,  $\angle Q = 90^\circ$ . If  $\cot R = \frac{1}{3}$ ,

then what is the value of

$$\frac{\sec P(\cos R + \sin P)}{\operatorname{cosec} R(\sin R - \operatorname{cosec} P)} ?$$

$\Delta PQR$  में,  $\angle Q = 90^\circ$  है। यदि  $\cot R =$

$$\frac{1}{3} \text{ है, तो } \frac{\sec P(\cos R + \sin P)}{\operatorname{cosec} R(\sin R - \operatorname{cosec} P)} \text{ का}$$

मान ज्ञात कीजिए।

(a)  $\frac{2}{7}$  (b)  $-\frac{2}{7}$

(c)  $\frac{2}{3}$  (d)  $-\frac{2}{3}$

148. If  $\sec \theta = \frac{a}{b}$ ,  $b \neq 0$ , then

$$\frac{1 - \tan^2 \theta}{2 - \sin^2 \theta} = ?$$

यदि  $\sec \theta = \frac{a}{b}$ ,  $b \neq 0$  है, तो

$$\frac{1 - \tan^2 \theta}{2 - \sin^2 \theta} = ?$$

(a)  $\frac{a^2(2b^2 + a^2)}{b^2(a^2 - b^2)}$  (b)  $\frac{a^2(2b^2 + a^2)}{b^2(a^2 + b^2)}$

(c)  $\frac{a^2(2b^2 - a^2)}{b^2(a^2 + b^2)}$  (d)  $\frac{b^2(2b^2 - a^2)}{a^2(a^2 + b^2)}$

149.  $\frac{\sec^2 \theta (2 + \tan^2 \theta + \cot^2 \theta) + (\sin^2 \theta - \tan^2 \theta)}{(\operatorname{cosec}^2 \theta + \sec^2 \theta)(1 + \cot^2 \theta)^2}$

(a) -1 (b) 1

(c) -2 (d) 2

150. The value of

$$\frac{2 \sin^2 38^\circ \sec^2 52^\circ + \cos 64^\circ \sin 26^\circ + \sin^2 64^\circ}{\tan^2 23^\circ + \cot^2 23^\circ - \sec^2 67^\circ - \operatorname{cosec}^2 67^\circ}$$

(a) -2 (b)  $\frac{3}{2}$

(c) 2 (d)  $-\frac{3}{2}$

151. The value of

$$\frac{\cos^6 \theta + \sin^6 \theta + 3 \sin^2 \theta \cos^2 \theta}{\operatorname{cosec} \theta \sec \theta (\sin \theta + \cos \theta - 1)(\sin \theta + \cos \theta + 1)}$$
 is:

(a) 1 (b) 2

(c)  $\frac{1}{2}$  (d) 3

152. If  $x = \sec 57^\circ$ , then

$$\cot^2 33^\circ + \sin^2 57^\circ + \sin^2 33^\circ + \operatorname{cosec}^2 57^\circ \cos^2 33^\circ + \sec^2 33^\circ \sin^2 57^\circ \text{ is equal to :}$$

यदि  $x = \sec 57^\circ$  है, तो  $\cot^2 33^\circ + \sin^2 57^\circ + \sin^2 33^\circ + \operatorname{cosec}^2 57^\circ \cos^2 33^\circ + \sec^2 33^\circ \sin^2 57^\circ$  के बराबर है।

(a)  $x^2 + 2$  (b)  $2x^2 + 1$

(c)  $x^2 + 1$  (d)  $\frac{1}{x^2 + 1}$

153. The value of  $(\tan^2 A + \cot^2 A - 2) - \sec^2 A \operatorname{cosec}^2 A$  is:

$(\tan^2 A + \cot^2 A - 2) - \sec^2 A \operatorname{cosec}^2 A$  का मान ज्ञात कीजिए।

(a) -4 (b) -1

(c) 1 (d) 4

154.  $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\tan^2 60^\circ - \sin^2 30^\circ - \cos^2 45^\circ} = ?$

(a)  $\frac{19}{9}$  (b)  $\frac{67}{27}$

(c)  $\frac{22}{9}$  (d)  $\frac{67}{24}$

155. If  $\cos(2\theta + 54^\circ) = \sin \theta$ ,  $0^\circ < (2\theta + 54^\circ) < 90^\circ$ , then what is

the value of  $\frac{1}{\tan 5\theta + \operatorname{cosec} \frac{5\theta}{2}}$  ?

यदि  $\cos(2\theta + 54^\circ) = \sin \theta$ ,  $0^\circ < (2\theta + 54^\circ) < 90^\circ$  है, तो

$\frac{1}{\tan 5\theta + \operatorname{cosec} \frac{5\theta}{2}}$  का मान ज्ञात कीजिए।

(a)  $2 + \sqrt{3}$  (b)  $3\sqrt{2}$

(c)  $2\sqrt{3}$  (d)  $2 - \sqrt{3}$

156. If  $\operatorname{cosec} \theta = \frac{b}{a}$ , then

$\frac{\sqrt{3} \cot \theta + 1}{\tan \theta + \sqrt{3}}$  is equal to :

यदि  $\operatorname{cosec} \theta = \frac{b}{a}$  है, तो  $\frac{\sqrt{3} \cot \theta + 1}{\tan \theta + \sqrt{3}}$

.....के बराबर है।

(a)  $\frac{\sqrt{b^2 - a^2}}{a}$  (b)  $\frac{\sqrt{a^2 + b^2}}{a}$

(c)  $\frac{\sqrt{a^2 + b^2}}{b}$  (d)  $\frac{\sqrt{b^2 - a^2}}{b}$

157.  $\frac{\sin^2 \theta}{\cos \theta(1 + \cos \theta)} + \frac{1 + \cos \theta}{\cos \theta} = ?$

(a)  $2 \cos \theta$  (b)  $\sec \theta$

(c)  $2 \sec \theta$  (d)  $\operatorname{cosec} \theta$

158. In  $\Delta ABC$ , right angled at B, if  $\tan A = \frac{1}{2}$ , then the value of

$\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$  is:

$\Delta ABC$  में, जो B पर समकोण है, यदि  $\tan A$

$= \frac{1}{2}$  है, तो  $\frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$  का

मान ज्ञात कीजिए।

(a)  $2\sqrt{5}$  (b) 3

(c) 2 (d) 1

159.  $\frac{\sin \theta + \cos \theta - 1}{\sin \theta - \cos \theta + 1} \times \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} = ?$

(a) 1 (b) -1

(c) -2 (d) 2

160.  $\frac{(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta)}{(\sec \theta + \tan \theta)(1 - \sin \theta)}$

(a)  $2 \sec \theta$  (b)  $2 \sec \theta$

(c)  $2 \operatorname{cosec} \theta$  (d)  $\operatorname{cosec} \theta$

161.  $\frac{\sec A(\sec A + \tan A)(1 - \sin A)}{(\operatorname{cosec}^2 A - 1) \sin^2 A} = ?$

(a)  $\sec^2 A$  (b)  $\cos^2 A$

(c)  $\cot A$  (d)  $\cos A$

## SSC CGL MAINS 2019

{18 Nov. 2020}

162. If  $\sec \theta + \tan \theta = 3$ , then the value of  $\sec \theta$  is:

यदि  $\sec \theta + \tan \theta = 3$  है, तो  $\sec \theta$  का मान ज्ञात कीजिए।

(a)  $\frac{4}{3}$  (b)  $\frac{3}{4}$

(c)  $\frac{5}{3}$  (d)  $\frac{3}{5}$

163. If  $\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} = 2 \frac{51}{79}$ , then the value of  $\sin \theta$  is equal to:

यदि  $\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} = 2 \frac{51}{79}$  है, तो  $\sin \theta$

का मान ज्ञात कीजिए।

(a)  $\frac{91}{144}$  (b)  $\frac{39}{72}$

(c)  $\frac{65}{144}$  (d)  $\frac{35}{72}$

164. If  $3 \sin x + 4 \cos x = 2$ , then the value of  $3 \cos x - 4 \sin x$  is equal to:  
यदि  $3 \sin x + 4 \cos x = 2$  है, तो  $3 \cos x - 4 \sin x$  का मान ज्ञात कीजिए।

- (a)  $\sqrt{23}$  (b)  $\sqrt{29}$   
(c) 21 (d)  $\sqrt{21}$

165. If  $\sin \theta + \sin^2 \theta = 1$ , then the value of  $\cos^2 \theta + \cos^4 \theta$  is equal to:

- यदि  $\sin \theta + \sin^2 \theta = 1$  है, तो  $\cos^2 \theta + \cos^4 \theta$  का मान ज्ञात कीजिए।  
(a) 5 (b) 2  
(c) 1 (d)  $1/2$

166. If  $\frac{\cos^2 \theta}{\cot^2 \theta - \cos^2 \theta} = 3$ , where  $0^\circ < \theta < 90^\circ$  then the value of  $\theta$  is:

- यदि  $\frac{\cos^2 \theta}{\cot^2 \theta - \cos^2 \theta}$  जहाँ  $0^\circ < \theta < 90^\circ$  है, तो  $\theta$  का मान ज्ञात कीजिए।  
(a)  $60^\circ$  (b)  $45^\circ$   
(c)  $50^\circ$  (d)  $30^\circ$

167. If  $\cos \theta = \frac{5}{13}$ , then the value of  $\tan^2 \theta + \sec^2 \theta$  is equal to:

यदि  $\cos \theta = \frac{5}{13}$  है, तो  $\tan^2 \theta + \sec^2 \theta$  का मान ज्ञात कीजिए।

- (a)  $\frac{233}{25}$  (b)  $\frac{303}{25}$   
(c)  $\frac{313}{25}$  (d)  $\frac{323}{25}$

168. If  $\alpha + \beta = 90^\circ$  and  $\alpha = 2\beta$ , then the value of  $3\cos^2 \alpha - 2\sin^2 \beta$  is equal to:

यदि  $\alpha + \beta = 90^\circ$  और  $\alpha = 2\beta$  है, तो  $3\cos^2 \alpha - 2\sin^2 \beta$  का मान ज्ञात कीजिए।

- (a)  $\frac{4}{3}$  (b)  $\frac{3}{4}$   
(c)  $\frac{3}{2}$  (d)  $\frac{1}{4}$

169. If  $\sin(x+y) = \cos(x-y)$ , then the value of  $\cos^2 x$  is:

यदि  $\sin(x+y) = \cos(x-y)$  है, तो  $\cos^2 x$  का मान ज्ञात कीजिए।

- (a)  $\frac{1}{2}$  (b) 3  
(c)  $\frac{1}{4}$  (d) 5

170. If  $\operatorname{cosec} 39^\circ = x$ , then the value of

$$\frac{1}{\operatorname{cosec}^2 51^\circ} + \sin^2 39^\circ + \tan^2 51^\circ - \frac{1}{\sin^2 51^\circ \sec^2 39^\circ} \text{ is:}$$

यदि  $\operatorname{cosec} 39^\circ = x$  है, तो  $\frac{1}{\operatorname{cosec}^2 51^\circ} + \sin^2 39^\circ + \tan^2 51^\circ -$

$$\frac{1}{\sin^2 51^\circ \sec^2 39^\circ} \text{ का मान ज्ञात कीजिए।}$$

- (a)  $1 - x^2$  (b)  $\sqrt{x^2 - 1}$   
(c)  $x^2 - 1$  (d)  $\sqrt{1 - x^2}$

171. If  $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = k + \tan^2 \theta + \cot^2 \theta$ , then the value of  $k$  is equal to:

यदि  $(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = k + \tan^2 \theta + \cot^2 \theta$  है, तो  $k$  का मान ज्ञात कीजिए।

- (a) 7 (b) 5  
(c) 9 (d) 2

### SSC CGL MAINS 2020

{29 Jan. 2022}

172. If  $3 \tan \theta = 2\sqrt{3} \sin \theta$ ,  $0^\circ < \theta < 90^\circ$  then the value of

$$\frac{\cos^2 2\theta + \cot^2 2\theta}{\sin^2 \theta + \tan^2 \theta} \text{ is:}$$

यदि  $3 \tan \theta = 2\sqrt{3} \sin \theta$ ,  $0^\circ < \theta < 90^\circ$  है, तो  $\frac{\cos^2 2\theta + \cot^2 2\theta}{\sin^2 \theta + \tan^2 \theta}$  का मान क्या होगा?

- (a)  $\frac{4}{13}$  (b)  $\frac{20}{39}$   
(c)  $\frac{4}{3}$  (d)  $\frac{20}{27}$

173. The value of

$$3(\cot^2 47^\circ - \sec^2 43^\circ) - 2(\tan^2 23^\circ - \operatorname{cosec}^2 67^\circ) \\ \operatorname{cosec}^2(68^\circ + \theta) - \tan(\theta + 61^\circ) \\ - \tan^2(22^\circ - \theta) + \cot(29^\circ - \theta)$$

- (a) 1 (b) 0  
(c) 5 (d) -1

174. If  $1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$ ,  $0^\circ$

$< \theta < 90^\circ$ , then  $\frac{a+b}{a-b} = ?$

यदि  $1 + 2 \tan^2 \theta + 2 \sin \theta \sec^2 \theta = \frac{a}{b}$ ,

$0^\circ < \theta < 90^\circ$  तो  $\frac{a+b}{a-b} = ?$

- (a)  $\sin \theta$  (b)  $\cos \theta$   
(c)  $\operatorname{cosec} \theta$  (d)  $\sec \theta$

175.  $\left( \frac{\tan^3 \theta}{\sec^2 \theta} + \frac{\cot^3 \theta}{\operatorname{cosec}^2 \theta} 2 \sin \theta \cos \theta \right) \div (1 + \operatorname{cosec}^2 \theta + \tan^2 \theta)$ ,  $0^\circ < \theta < 90^\circ$

$\left( \frac{\tan^3 \theta}{\sec^2 \theta} + \frac{\cot^3 \theta}{\operatorname{cosec}^2 \theta} 2 \sin \theta \cos \theta \right) \div (1 + \operatorname{cosec}^2 \theta + \tan^2 \theta)$ ,  $0^\circ < \theta < 90^\circ$  का मान इनमें से किसके बराबर होगा?

- (a)  $\sin \theta \cos \theta$  (b)  $\operatorname{cosec} \theta$   
(c)  $\sec \theta$  (d)  $\operatorname{cosec} \theta \sec \theta$

176. If  $7 \sin^2 \theta + 4 \cos^2 \theta = 5$  and  $\theta$  lies in the first quadrant, then what

is the value of  $\frac{\sqrt{3} \sec \theta + \tan \theta}{\sqrt{2} \cot \theta - \sqrt{3} \cos \theta}$ ?

यदि  $7 \sin^2 \theta + 4 \cos^2 \theta = 5$  और  $\theta$  पहले चतुर्थांश में स्थित है, तो  $\frac{\sqrt{3} \sec \theta + \tan \theta}{\sqrt{2} \cot \theta - \sqrt{3} \cos \theta}$

का मान क्या है?

- (a)  $2(1 + \sqrt{2})$  (b)  $3\sqrt{2}$   
(c)  $2(\sqrt{2} - 1)$  (d)  $4\sqrt{2}$

177. Let,  $0^\circ < \theta < 90^\circ$ ,  $(1 + \cot^2 \theta)(1 + \tan^2 \theta) \times (\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)$  is equal to-

मान लें कि  $0^\circ < \theta < 90^\circ$ , है तो  $(1 + \cot^2 \theta)(1 + \tan^2 \theta) \times (\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)$  का मान इनमें से किसके बराबर होगा?

- (a)  $\sec \theta \operatorname{cosec} \theta$  (b)  $\sec \theta + \operatorname{cosec} \theta$   
(c)  $\sin \theta + \cos \theta$  (d)  $\sin \theta \cos \theta$

178. If  $0^\circ < \theta < 90^\circ$  then find the value of

$$\frac{(1 + \sec \theta \operatorname{cosec} \theta)^2 (\sec \theta - \tan \theta)^2 (1 + \sin \theta)}{(\sin \theta + \sec \theta)^2 + (\cos \theta + \operatorname{cosec} \theta)^2}$$

- (a)  $1 - \cos \theta$  (b)  $1 - \sin \theta$   
(c)  $\cos \theta$  (d)  $\sin \theta$

179. If  $0^\circ < \theta < 90^\circ$  then find the value of

$$\frac{1 + \cos \theta - \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}{\tan \theta + \cot \theta}$$

- (a)  $\tan \theta$  (b)  $\sec \theta$   
(c)  $\operatorname{cosec} \theta$  (d)  $\cot \theta$

180. If  $\sin A = \frac{5}{13}$  and  $7 \cot B = 24$ , then the value of  $(\sec A \cos B)(\operatorname{cosec} B \tan A)$  is:

यदि  $\sin A = \frac{5}{13}$  और  $7 \cot B = 24$  है, तो  $(\sec A \cos B)(\operatorname{cosec} B \tan A)$  का मान क्या होगा?

- (a)  $\frac{13}{7}$  (b)  $\frac{65}{42}$   
(c)  $\frac{15}{13}$  (d)  $\frac{13}{14}$

$$4 \tan^2 30^\circ + \sin^2 30^\circ \cos^2 45^\circ + \sec^2 48^\circ - \cot^2 42^\circ$$

181.  $\cos 37^\circ \sin 53^\circ + \sin 37^\circ \cos 53^\circ + \tan 18^\circ \tan 72^\circ$

- (a)  $\frac{35}{48}$  (b)  $\frac{59}{48}$   
(c)  $\frac{49}{24}$  (d)  $\frac{35}{24}$

### SSC CGL MAINS 2020

{03 Feb. 2022}

182. The value of/का मान है:

$$\frac{3(\cos^2 26^\circ - \tan^2 64^\circ) + (\cot^2 42^\circ - \sec^2 48^\circ)}{\cot(22^\circ - \theta) - \operatorname{cosec}(62^\circ + \theta) - \tan(\theta + 68^\circ) + \tan^2(28^\circ - \theta)}$$

- (a) 3 (b) 4  
(c) -1 (d) -2

183. If  $\cos \theta = \frac{12}{13}$  then the value of

$$\frac{\sin \theta(1 - \tan \theta)}{\tan \theta(1 + \operatorname{cosec} \theta)}$$
 is:

यदि  $\cos \theta = \frac{12}{13}$  है तो  $\frac{\sin \theta(1 - \tan \theta)}{\tan \theta(1 + \operatorname{cosec} \theta)}$  का मान क्या होगा?

- (a)  $\frac{25}{78}$  (b)  $\frac{35}{234}$   
(c)  $\frac{35}{108}$  (d)  $\frac{25}{156}$

184. If  $\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} = \frac{1}{7}$ ,  $\theta$  lies in first quadrant, then the value of

$$\frac{\operatorname{cosec} \theta + \cot^2 \theta}{\operatorname{cosec} \theta - \cot^2 \theta}$$
 is:

यदि  $\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} = \frac{1}{7}$ ,  $\theta$  प्रथम चतुर्भांश में स्थित है, तो  $\frac{\operatorname{cosec} \theta + \cot^2 \theta}{\operatorname{cosec} \theta - \cot^2 \theta} = ?$

- (a)  $\frac{19}{5}$  (b)  $\frac{22}{3}$

- (c)  $\frac{37}{12}$  (d)  $\frac{37}{19}$

185. The expression  $(\tan \theta + \cot \theta)(\sec \theta + \tan \theta)(1 - \sin \theta)$ ,  $0^\circ < A < 90^\circ$ , is equal to:

व्यंजक  $(\tan \theta + \cot \theta)(\sec \theta + \tan \theta)(1 - \sin \theta)$ ,  $0^\circ < A < 90^\circ$ , का मान है:

- (a)  $\sec \theta$  (b)  $\operatorname{cosec} \theta$   
(c)  $\cot \theta$  (d)  $\sin \theta$

186. The expression  $\frac{\tan^6 \theta - \sec^6 \theta + 3 \sec^2 \theta \tan^2 \theta}{\tan^2 \theta + \cot^2 \theta + 1}$ ,  $0^\circ < \theta < 90^\circ$ , is equal to:

$$\frac{\tan^6 \theta - \sec^6 \theta + 3 \sec^2 \theta \tan^2 \theta}{\tan^2 \theta + \cot^2 \theta + 1}$$

का मान बताइए जहाँ  $0^\circ < \theta < 90^\circ$  है।

- (a)  $\sec^2 \theta \operatorname{cosec}^2 \theta$   
(b)  $-\sec^2 \theta \operatorname{cosec}^2 \theta$   
(c)  $\cos^2 \theta \sin^2 \theta$   
(d)  $-\cos^2 \theta \sin^2 \theta$

187. The expression

$$\frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2}, 0^\circ < \theta < 90^\circ, \text{ is equal to}$$

व्यंजक

$$\frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2}, \text{ का मान बताइए जहाँ } 0^\circ < \theta < 90^\circ \text{ है।}$$

- (a)  $\frac{1}{2} \sin \theta$  (b)  $2 \sin \theta$   
(c)  $\sec \theta$  (d)  $2 \operatorname{cosec} \theta$

188. The expression

$$\frac{(1 - \sin \theta + \cos \theta)^2 (1 - \cos \theta) \sec^3 \theta \operatorname{cosec}^2 \theta}{(\sec \theta - \tan \theta)(\tan \theta + \cot \theta)},$$

$0^\circ < \theta < 90^\circ$  is equal to

व्यंजक

$$\frac{(1 - \sin \theta + \cos \theta)^2 (1 - \cos \theta) \sec^3 \theta \operatorname{cosec}^2 \theta}{(\sec \theta - \tan \theta)(\tan \theta + \cot \theta)},$$

, का मान बताइए, जहाँ  $0^\circ < \theta < 90^\circ$  है।

- (a)  $\sin \theta$  (b)  $2 \cos \theta$   
(c)  $\cot \theta$  (d)  $2 \tan \theta$

189. The value of/का मान है?

$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\operatorname{cosec}^2 71^\circ + \tan^2(90^\circ - 71^\circ) + 1}$$

- (a) 1 (b) 4  
(c) -3 (d) 2

190. If  $\frac{\sin^2 \theta}{\cos^2 \theta - 3 \cos \theta + 2} = 1$ ,  $\theta$  lies in the first quadrant, then the

$$\text{value of } \frac{\tan^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2}}{\tan \theta + \sin \theta} \text{ is.}$$

यदि  $\frac{\sin^2 \theta}{\cos^2 \theta - 3 \cos \theta + 2} = 1$ ,  $\theta$  चतुर्भांश में है, तो  $\frac{\tan^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2}}{\tan \theta + \sin \theta}$  का मान है?

$$\frac{\tan^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2}}{\tan \theta + \sin \theta}$$

- (a)  $\frac{2\sqrt{3}}{27}$  (b)  $\frac{5\sqrt{3}}{27}$

- (c)  $\frac{2\sqrt{3}}{9}$  (d)  $\frac{7\sqrt{3}}{54}$

191. The expression

$$\frac{(1 - 2 \sin^2 \theta \cos \theta)(\cot \theta + 1) \cos \theta}{(\sin^4 \theta + \cos^4 \theta)(1 + \tan \theta) \operatorname{cosec} \theta} - 1,$$

$0^\circ < \theta < 90^\circ$ , is equal to:

व्यंजक

$$\frac{(1 - 2 \sin^2 \theta \cos \theta)(\cot \theta + 1) \cos \theta}{(\sin^4 \theta + \cos^4 \theta)(1 + \tan \theta) \operatorname{cosec} \theta} - 1$$

का मान बताइए जहाँ  $0^\circ < \theta < 90^\circ$  है।

- (a)  $\cos^2 \theta$  (b)  $-\sin^2 \theta$   
(c)  $-\cos 2\theta$  (d)  $\sin^2 \theta$

**ANSWER KEY - TRIGONOMETRY**

- |          |          |          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1. (c)   | 2. (d)   | 3. (d)   | 4. (c)   | 5. (a)   | 6. (d)   | 7. (d)   | 8. (b)   | 9. (b)   | 10. (d)  |
| 11. (d)  | 12. (a)  | 13. (a)  | 14. (d)  | 15. (b)  | 16. (c)  | 17. (a)  | 18. (a)  | 19. (c)  | 20. (d)  |
| 21. (c)  | 22. (d)  | 23. (d)  | 24. (c)  | 25. (d)  | 26. (a)  | 27. (d)  | 28. (c)  | 29. (a)  | 30. (d)  |
| 31. (b)  | 32. (a)  | 33. (a)  | 34. (d)  | 35. (a)  | 36. (c)  | 37. (a)  | 38. (b)  | 39. (d)  | 40. (b)  |
| 41. (b)  | 42. (a)  | 43. (b)  | 44. (d)  | 45. (a)  | 46. (b)  | 47. (c)  | 48. (b)  | 49. (c)  | 50. (c)  |
| 51. (c)  | 52. (d)  | 53. (a)  | 54. (b)  | 55. (b)  | 56. (d)  | 57. (b)  | 58. (b)  | 59. (b)  | 60. (c)  |
| 61. (b)  | 62. (d)  | 63. (b)  | 64. (b)  | 65. (b)  | 66. (a)  | 67. (d)  | 68. (b)  | 69. (b)  | 70. (b)  |
| 71. (c)  | 72. (d)  | 73. (b)  | 74. (c)  | 75. (d)  | 76. (d)  | 77. (d)  | 78. (c)  | 79. (c)  | 80. (a)  |
| 81. (d)  | 82. (d)  | 83. (d)  | 84. (b)  | 85. (b)  | 86. (b)  | 87. (d)  | 88. (a)  | 89. (c)  | 90. (a)  |
| 91. (d)  | 92. (c)  | 93. (c)  | 94. (c)  | 95. (b)  | 96. (b)  | 97. (c)  | 98. (d)  | 99. (*)  | 100. (c) |
| 101. (a) | 102. (*) | 103. (d) | 104. (d) | 105. (b) | 106. (c) | 107. (d) | 108. (a) | 109. (b) | 110. (b) |
| 111. (c) | 112. (c) | 113. (b) | 114. (c) | 115. (a) | 116. (a) | 117. (b) | 118. (d) | 119. (a) | 120. (d) |
| 121. (a) | 122. (d) | 123. (b) | 124. (d) | 125. (a) | 126. (b) | 127. (b) | 128. (c) | 129. (b) | 130. (d) |
| 131. (a) | 132. (b) | 133. (d) | 134. (c) | 135. (b) | 136. (b) | 137. (d) | 138. (d) | 139. (a) | 140. (c) |
| 141. (a) | 142. (d) | 143. (a) | 144. (d) | 145. (b) | 146. (a) | 147. (b) | 148. (c) | 149. (a) | 150. (d) |
| 151. (c) | 152. (a) | 153. (a) | 154. (b) | 155. (d) | 156. (a) | 157. (c) | 158. (b) | 159. (a) | 160. (a) |
| 161. (a) | 162. (c) | 163. (c) | 164. (d) | 165. (c) | 166. (a) | 167. (c) | 168. (d) | 169. (a) | 170. (c) |
| 171. (a) | 172. (b) | 173. (d) | 174. (c) | 175. (a) | 176. (a) | 177. (a) | 178. (b) | 179. (d) | 180. (b) |
| 181. (b) | 182. (d) | 183. (b) | 184. (a) | 185. (b) | 186. (d) | 187. (b) | 188. (d) | 189. (d) | 190. (d) |
| 191. (b) |          |          |          |          |          |          |          |          |          |

Gagan

## Hints &amp; Solutions {TRIGONOMETRY}

## SSC CGL MAINS 2011

{04 Sept. 2011}

1. (c)  $\sin^2\alpha + \sin^2\beta = 2$

(b)  $\sin^2\alpha = 1 \quad \sin^2\beta = 1$

(d)  $\sin\alpha = 1 \quad \sin\beta = 1$

(c)  $\alpha = 90^\circ \quad \beta = 90^\circ$

(c)  $\cos \frac{\alpha + \beta}{2} = \cos \frac{90^\circ + 90^\circ}{2}$

(d)  $= \cos 90^\circ = 0$

2. (d)

(b)  $\cot \frac{\pi}{20} \cot \frac{3\pi}{20} \cot \frac{5\pi}{20} \cot \frac{7\pi}{20} \cot \frac{9\pi}{20}$

(d)  $= \left( \cot \frac{\pi}{20} \cdot \cot \frac{9\pi}{20} \right)$

(b)  $\left( \cot \frac{3\pi}{20} \cdot \cot \frac{7\pi}{20} \right) \cot \frac{5\pi}{20}$

(c)  $= 1 \times 1 \times 1 = 1$

3. (d)  $\sin\theta + \cos\theta = \frac{17}{13}$

(c)  $\frac{P}{H} + \frac{B}{H} = \frac{P+B}{H} = \frac{17}{13} \Rightarrow \frac{5+12}{13}$

(d)  $\therefore \sin\theta - \cos\theta = \frac{12-5}{13} = \frac{7}{13}$

Alternatively:-

$$\sin\theta + \cos\theta = \frac{17}{13}$$

$$(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta \cdot \cos\theta$$

$$\left(\frac{17}{13}\right)^2 = 1 + 2\sin\theta \cdot \cos\theta$$

$$\frac{289}{169} - 1 = 2\sin\theta \cdot \cos\theta$$

$$\sin\theta \cdot \cos\theta = \frac{60}{169}$$

$$(\sin\theta - \cos\theta)^2 = \sin^2\theta + \cos^2\theta - 2\sin\theta \cdot \cos\theta$$

$$\therefore (\sin\theta - \cos\theta)^2 = 1 - 2 \times \frac{60}{169}$$

$$\sin\theta - \cos\theta = \sqrt{\frac{49}{169}} = \frac{7}{13}$$

4. (c)  $\tan\theta \cdot \tan 2\theta = 1$

$$\tan 2\theta = \cot\theta$$

$$\tan 2\theta = \tan (90^\circ - \theta)$$

$$2\theta = 90^\circ - \theta$$

$$3\theta = 90^\circ \Rightarrow \theta = 30^\circ$$

$$\sin^2 2\theta + \tan^2 2\theta = \sin^2 60^\circ + \tan^2 60^\circ$$

$$= \left(\frac{\sqrt{3}}{2}\right)^2 + (\sqrt{3})^2$$

$$= 3 + \frac{3}{4} = 3\frac{3}{4}$$

## SSC CGL MAINS 2012

{16 Sept. 2012}

5. (a)  $\tan 2\theta \cdot \tan 3\theta = 1$

$$\Rightarrow \tan 2\theta = \frac{1}{\tan 3\theta}$$

$$\Rightarrow \tan 2\theta = \cot 3\theta$$

$$\therefore 2\theta + 3\theta = 90^\circ$$

$$5\theta = 90^\circ$$

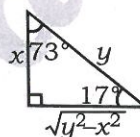
$$\theta = 18^\circ$$

$$\therefore 2\cos^2 \frac{5\theta}{2} - 1 = 2\cos^2 \frac{90^\circ}{2} - 1$$

$$= 2\cos^2 45^\circ - 1$$

$$= 2 \times \frac{1}{2} - 1 = 0$$

6. (d)  $\sin 17^\circ = \frac{x}{y}$



$$= \sec 17^\circ - \sin 73^\circ$$

$$= \frac{y}{\sqrt{y^2 - x^2}} - \frac{\sqrt{y^2 - x^2}}{y}$$

$$= \frac{y^2 - y^2 + x^2}{y\sqrt{y^2 - x^2}}$$

$$= \frac{x^2}{y\sqrt{y^2 - x^2}}$$

7. (d) Take any value of  $\theta$  between  $0^\circ$  and  $90^\circ$

$$\text{take } \theta = 45^\circ$$

$$\sin\theta + \cos\theta$$

$$\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \Rightarrow \frac{2}{\sqrt{2}} = \sqrt{2} = 1.732$$

means it is **Greater than 1**

8. (b)  $\sin^2\theta + \cos^2\theta + \sec^2\theta + \csc^2\theta$

$$+ \tan^2\theta + \cot^2\theta$$

$$= 1 + \sec^2\theta + \csc^2\theta + \tan^2\theta + \cot^2\theta$$

$$= 1 + \tan^2\theta + 1 + \cot^2\theta + 1 + \tan^2\theta$$

$$+ \cot^2\theta$$

$$= 3 + 2(\tan^2\theta + \cot^2\theta)$$

$$\text{Put, } \theta = 45^\circ$$

$$= 3 + 2(1 + 1) = 7$$

9. (b)  $2 \sin\left(\frac{\pi x}{2}\right) = x^2 + \frac{1}{x^2}$

make perfect square of  $(x - \frac{1}{x})$  by adding  $(+2 - 2)$ 

$$2 \sin\left(\frac{\pi x}{2}\right) = \left(x - \frac{1}{x}\right)^2 + 2$$

$$x - \frac{1}{x} = \sqrt{2 \sin\left(\frac{\pi x}{2}\right) - 2}$$

Only  $x = 1$  satisfy the equation

$$x - \frac{1}{x} = 0$$

10. (d)  $\frac{\tan 57^\circ + \cot 37^\circ}{\tan 33^\circ + \cot 53^\circ}$

$$\frac{\cot 33^\circ + \tan 53^\circ}{\tan 33^\circ + \frac{1}{\tan 53^\circ}}$$

$$= \frac{1 + \tan 53^\circ \tan 33^\circ}{\tan 33^\circ \tan 53^\circ + 1} \times \frac{\tan 53^\circ}{\tan 33^\circ}$$

$$= \tan 53^\circ \cdot \cot 33^\circ = \cot 37^\circ \cdot \tan 57^\circ$$

## SSC CGL MAINS 2013

{29 Sept. 2013}

11. (d)  $\sin\theta + \cos\theta = \sqrt{2} \cos\theta$

square both sides:-

$$\sin^2\theta + \cos^2\theta + 2\sin\theta \cos\theta = 2\cos^2\theta$$

$$1 + 2\sin\theta \cos\theta = 2\cos^2\theta$$

$$2\sin\theta \cos\theta = 2\cos^2\theta - 1$$

$$\sin 2\theta = \cos 2\theta$$

Now,

$$\cos\theta - \sin\theta = \sqrt{(\cos\theta - \sin\theta)^2}$$

$$= \sqrt{\cos^2\theta + \sin^2\theta - 2\cos\theta \sin\theta}$$

$$= \sqrt{1 - \sin 2\theta}$$

$$= \sqrt{1 - \cos 2\theta}$$

$$= \sqrt{2\sin^2\theta} = \sqrt{2} \sin\theta$$

12. (a)  $x \sin 45^\circ = y \csc 30^\circ$

$$x \times \frac{1}{\sqrt{2}} = y \times 2$$

$$\frac{x}{y} = 2\sqrt{2}$$

$$\frac{x^4}{y^4} = (2\sqrt{2})^4 = 2^6 = 4^3$$

13. (a)  $\frac{\sin A}{1+\cos A} + \frac{\sin A}{1-\cos A}$   
Put,  $\theta = 30^\circ$

$$= \frac{1}{2+\sqrt{3}} + \frac{1}{2-\sqrt{3}}$$

$$= \frac{2-\sqrt{3}+2+\sqrt{3}}{(2+\sqrt{3})(2-\sqrt{3})}$$

$$= \frac{4}{4} = 1$$

only option (a) gives 4 at  $\theta = 30^\circ$

Alternatively:-

$$\frac{\sin A}{1+\cos A} + \frac{\sin A}{1-\cos A}$$

$$= \frac{\sin A(1-\cos A) + \sin A(1+\cos A)}{(1-\cos^2 A)}$$

$$= \frac{\sin A - \sin A \cos A + \sin A + \sin A \cos A}{\sin^2 A}$$

$$= \frac{2\sin A}{\sin^2 A}$$

$$= 2\operatorname{cosec} A$$

14. (d)  $\sin \theta = \frac{1}{r}$ ,

$$\cos \theta = \frac{\sqrt{3}}{r}$$

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{\sqrt{3}}, \tan \theta = \frac{1}{\sqrt{3}}$$

$$\therefore \sqrt{3} \tan \theta + 1$$

$$= \sqrt{3} \times \frac{1}{\sqrt{3}} + 1$$

$$= 2$$

### SSC CGL MAINS 2013

{21 Sept. 2014}

15. (b)  $\tan x = \sin 45^\circ \cdot \cos 45^\circ + \sin 30^\circ$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \Rightarrow \frac{1}{2} + \frac{1}{2} = 1$$

$$\tan x = 1$$

$$\tan x = \tan 45^\circ$$

$$\Rightarrow x = 45^\circ$$

16. (c) Put,  $\theta = 60^\circ$

$$\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} = \sqrt{\frac{2-1}{2+1}} = \frac{1}{\sqrt{3}}$$

$$\therefore \text{only option (c) gives } \frac{1}{\sqrt{3}}$$

Alternatively:-

$$\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}}$$

rationalize it-

$$\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} \times \frac{(\sec \theta - 1)}{(\sec \theta - 1)}$$

$$= \sqrt{\frac{(\sec \theta - 1)^2}{\sec^2 \theta - 1}} = \sqrt{\frac{(\sec \theta - 1)^2}{\tan^2 \theta}}$$

$$= \frac{\sec \theta - 1}{\tan \theta} = \frac{\sec \theta}{\tan \theta} - \frac{1}{\tan \theta}$$

$$\frac{1}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} - \frac{\cos \theta}{\sin \theta}$$

$$= \operatorname{cosec} \theta - \cot \theta$$

17. (a)  $\sin 3A = \cos(A - 26^\circ)$

$$\sin 3A = \sin[(90^\circ - (A - 26^\circ))]$$

$$3A + A - 26^\circ = 90^\circ$$

$$4A = 90^\circ + 26^\circ = 116^\circ$$

$$A = \frac{116^\circ}{4} = 29^\circ$$

18. (a)  $\sec^2 \theta - \frac{\sin^2 \theta - 2\sin^4 \theta}{2\cos^4 \theta - \cos^2 \theta}$

$$= \sec^2 \theta - \frac{\sin^2 \theta(1 - 2\sin^2 \theta)}{\cos^2 \theta(2\cos^2 \theta - 1)}$$

$$= \sec^2 \theta - \tan^2 \theta \times \frac{\cos 2\theta}{\cos 2\theta}$$

$$= \sec^2 \theta - \tan^2 \theta = 1$$

19. (c)  $x = a(\sin \theta + \cos \theta)$

$$\frac{x}{a} = (\sin \theta + \cos \theta)$$

square both side

$$\frac{x^2}{a^2} = 1 + 2\sin \theta \cos \theta$$

$$y = b(\sin \theta - \cos \theta)$$

$$\frac{y}{b} = \sin \theta - \cos \theta$$

square both side

$$\frac{y^2}{b^2} = 1 - 2\sin \theta \cos \theta \quad \dots(ii)$$

Add eq. (i) and (ii)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$$

20. (d)  $\sin 5\theta = \cos 2\theta$

$$\sin 5\theta = \sin(90^\circ - 2\theta)$$

$$5\theta = 70^\circ$$

$$\theta = 14^\circ$$

21. (c)  $2\sin^2 \theta = 3\cos \theta$

$$\text{Let, } \theta = 60^\circ$$

$$2\sin^2 60^\circ = 3\cos 60^\circ$$

$$2 \times \frac{3}{4} = 3 \times \frac{1}{2}$$

$$\frac{3}{2} = \frac{3}{2}$$

$\theta = 60^\circ$  satisfied

22. (d)  $a\sin \theta + b\cos \theta = c \quad \dots(i)$

$$a\cos \theta - b\sin \theta = x \quad \dots(ii)$$

Squaring and then add eq. (i) and eq. (ii) equation.

$$a^2\sin^2 \theta + b^2\cos^2 \theta + 2ab\sin \theta \cos \theta + a^2\cos^2 \theta + b^2\sin^2 \theta - 2ab\sin \theta \cos \theta$$

$$= c^2 + x^2$$

$$a^2[\sin^2 \theta + \cos^2 \theta] + b^2[\cos^2 \theta + \sin^2 \theta]$$

$$= c^2 + x^2$$

$$[\because \sin^2 \theta + \cos^2 \theta = 1]$$

$$a^2 + b^2 = c^2 + x^2$$

$$x = \pm \sqrt{a^2 + b^2 - c^2}$$

23. (d)  $\sin \theta + \cos \theta = \sqrt{2} \sin(90^\circ - \theta)$

$$\sin \theta + \cos \theta = \sqrt{2} \cos \theta$$

$$[\because \sin(90^\circ - \theta) = \cos \theta]$$

divide by  $(\cos \theta)$

$$\tan \theta + 1 = \sqrt{2}$$

$$\frac{1}{\cot \theta} = \sqrt{2} - 1$$

$$\cot \theta = \frac{1}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

$$\cot \theta = \sqrt{2} + 1$$

24. (c)  $3[\sec^2 \theta + \tan^2 \theta] = 5$

$$\sec^2 \theta + \tan^2 \theta = \frac{5}{3}$$

$$\text{Let, } \theta = 30^\circ$$

$$\frac{4}{3} + \frac{1}{3} = \frac{5}{3}$$

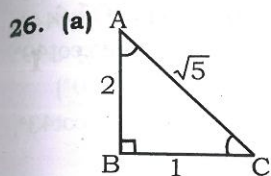
$$\frac{5}{3} = \frac{5}{3} \quad (\text{satisfied})$$

$$\cos 2\theta = \cos 60^\circ = \frac{1}{2}$$

25. (d)  $x \cos^2 30^\circ \cdot \sin 60^\circ$

$$= \frac{\tan^2 45^\circ \cdot \sec 60^\circ}{\operatorname{cosec} 60^\circ}$$

$$x \times \frac{3}{4} \times \frac{\sqrt{3}}{2} = \frac{1 \times 2}{2} \times \sqrt{3} = 2 \times \frac{2}{3}$$



$$\tan \alpha = 2$$

$$= \frac{\operatorname{cosec}^2 \alpha - \sec^2 \alpha}{\operatorname{cosec}^2 \alpha + \sec^2 \alpha}$$

$$= \frac{\frac{1}{\sin^2 \alpha} - \frac{1}{\cos^2 \alpha}}{\frac{1}{\sin^2 \alpha} + \frac{1}{\cos^2 \alpha}}$$

$$= \frac{\cos^2 \alpha - \sin^2 \alpha}{\cos^2 \alpha + \sin^2 \alpha}$$

$$= \cos^2 \alpha - \sin^2 \alpha$$

$$= \left( \frac{1}{\sqrt{5}} \right)^2 - \left( \frac{2}{\sqrt{5}} \right)^2$$

$$= \frac{1}{5} - \frac{4}{5} = -\frac{3}{5}$$

27. (d)  $\sin(\theta + 30^\circ) = \frac{3}{2\sqrt{3}}$

$$\sin(\theta + 30^\circ) = \frac{\sqrt{3}}{2} = \sin 60^\circ$$

$$\theta = 30^\circ$$

$$\cos^2 \theta = \cos^2 30^\circ = \frac{3}{4}$$

28. (c)  $4\cos^2 \theta - 4\sqrt{3} - \cos \theta + 3 = 0$

$$\text{Let, } \cos \theta = x$$

$$4x^2 - 4\sqrt{3}x + 3 = 0$$

$$4x^2 - (2\sqrt{3} + 2\sqrt{3})x + 3 = 0$$

$$4x^2 - 2\sqrt{3}x - 2\sqrt{3}x + 3 = 0$$

$$2x(2x - \sqrt{3}) - \sqrt{3}(2x - \sqrt{3}) = 0$$

$$x = \frac{\sqrt{3}}{2}$$

$$\cos \theta = \frac{\sqrt{3}}{2} = 30^\circ$$

Alternatively:-

$$4\cos^2 \theta - 4\sqrt{3} \cos \theta + 3 = 0$$

$$\text{Let, } \theta = 30^\circ$$

$$= 4\cos^2 30^\circ - 4\sqrt{3} \cos 30^\circ + 3$$

$$= 4 \times \frac{3}{4} - 4\sqrt{3} \times \frac{\sqrt{3}}{2} + 3$$

$$= 6 - 6 = 0$$

$$\text{Value for } \theta = 30^\circ \text{ is satisfied}$$

**SSC CGL MAINS 2014**

**{12 Apr. 2015} PAPER - II**

29. (a)  $\frac{H}{B} - \frac{B}{H} = \frac{3}{2}$

$$\frac{H^2 - B^2}{BH} = \frac{3}{2}$$

$$\frac{P^2}{BH} = \frac{3}{2}$$

$$\therefore \sec \theta = \frac{H}{B} = \frac{2}{1} = 2$$

Alternatively:-

$$\sec \theta - \cos \theta = \frac{3}{2}$$

$$\Rightarrow \frac{1}{\cos \theta} - \cos \theta = \frac{3}{2}$$

$$\frac{1 - \cos^2 \theta}{\cos \theta} = \frac{3}{2}$$

$$2 - 2\cos^2 \theta = 3 \cos \theta$$

$$\text{Let, } \cos \theta = x$$

$$2x^2 + 3x - 2 = 0$$

$$2x^2 + 4x - x - 2 = 0$$

$$2x(x + 2) - 1(x + 2) = 0$$

$$x = -2, \frac{1}{2} \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \sec \theta = 2$$

30. (d)  $\tan A = \cot B$  (If  $A + B = 90^\circ$ )

$$5x - 10 + 5y + 20 = 90^\circ$$

$$x + y = \frac{80^\circ}{5} = 16^\circ$$

Alternatively:-

$$\tan(5x - 10^\circ) = \cot(5y + 20^\circ)$$

$$\tan(5x - 10^\circ) = \tan(90^\circ - (5y + 20^\circ))$$

$$\therefore 5x - 10^\circ = 90^\circ - 5y - 20^\circ$$

$$5x + 5y = 80^\circ$$

$$x + y = 16^\circ$$

31. (b)  $\sin \theta + \sin^2 \theta = 1$

$$\sin \theta = 1 - \sin^2 \theta$$

$$\sin \theta = \cos^2 \theta$$

$$\cos^{12} \theta + 3\cos^{10} \theta + 3\cos^8 \theta + \cos^6 \theta - 1$$

$$(\cos^2 \theta + \cos^4 \theta)^3 - 1 = 1 - 1 = 0$$

32. (a)  $a(\tan \theta + \cot \theta) = 1$

$$\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{1}{a}$$

$$a = \sin \theta \cos \theta$$

$$b = \sin \theta + \cos \theta$$

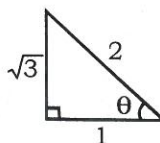
$$\text{Solving from option}$$

$$\therefore 2a = b^2 - 1$$

$$2\sin \theta \cos \theta = (\sin \theta + \cos \theta)^2 - 1$$

$$2\sin \theta \cos \theta = 1 + 2\sin \theta \cos \theta - 1$$

$$1 = 1 \Rightarrow \text{So, } 2a = b^2 - 1$$



33. (a)  $\tan A \cdot \tan B = 1$  (If  $A + B = 90^\circ$ )

$$\therefore \tan 11^\circ \cdot \tan 79^\circ = 1$$

$$\tan 17^\circ \cdot \tan 73^\circ = 1$$

$$\therefore 1 \times 1 = 1$$

Alternatively:-

$$\tan 11^\circ \tan 17^\circ \tan 79^\circ \tan 73^\circ$$

$$\Rightarrow \tan 11^\circ \tan 17^\circ \tan (90^\circ - 11^\circ)$$

$$\tan (90^\circ - 17^\circ)$$

$$\Rightarrow \tan 11^\circ \tan 17^\circ \cot 11^\circ \cot 17^\circ$$

$$\Rightarrow \tan 11^\circ \tan 17^\circ \times \frac{1}{\tan 11^\circ} \cdot \frac{1}{\tan 17^\circ}$$

$$\Rightarrow 1$$

34. (d)  $\sin A + \sin^2 A = 1$

$$\sin A = 1 - \sin^2 A$$

$$\sin A = \cos^2 A$$

$$\therefore \cos^2 A + \cos^4 A = \sin A + \sin^2 A = 1$$

35. (a)  $(1 + \sec 20^\circ + \cot 70^\circ)$

$$(1 - \operatorname{cosec} 20^\circ + \tan 70^\circ)$$

$$= [1 + \sec(90^\circ - 70^\circ) + \cot 70^\circ][1$$

$$+ \operatorname{cosec}(90^\circ - 70^\circ) + \tan 70^\circ]$$

$$= [1 + \operatorname{cosec} 70^\circ + \cot 70^\circ](1 - \sec 70^\circ + \tan 70^\circ)$$

$$= \left( 1 + \frac{1}{\sin 70^\circ} + \frac{\cos 70^\circ}{\sin 70^\circ} \right)$$

$$\left( 1 - \frac{1}{\cos 70^\circ} + \frac{\sin 70^\circ}{\cos 70^\circ} \right)$$

$$= \left( \frac{\sin 70^\circ + 1 + \cos 70^\circ}{\sin 70^\circ} \right)$$

$$\left( \frac{\cos 70^\circ - 1 + \sin 70^\circ}{\cos 70^\circ} \right)$$

$$= \left( \frac{(\cos 70^\circ + \sin 70^\circ)^2}{\cos 70^\circ \cdot \sin 70^\circ} - (1)^2 \right)$$

$$= \frac{\cos^2 70^\circ + \sin^2 70^\circ + 2\cos 70^\circ \cdot \sin 70^\circ - 1}{\cos 70^\circ \cdot \sin 70^\circ}$$

$$= \frac{2\cos 70^\circ \cdot \sin 70^\circ}{\cos 70^\circ \cdot \sin 70^\circ} = 2$$

36. (c) Put,  $A = 45^\circ$

$$\frac{\tan A - \sec A - 1}{\tan A + \sec A + 1}$$

$$= \frac{-\sqrt{2}}{2 + \sqrt{2}} = \frac{-\sqrt{2}}{\sqrt{2}(\sqrt{2} + 1)}$$

$$= \frac{-1}{(\sqrt{2} + 1)} \times \frac{(\sqrt{2} - 1)}{(\sqrt{2} - 1)} = 1 - \sqrt{2}$$

$$\text{option (c) will give } (1 - \sqrt{2})$$

**Alternatively:-**

$$\begin{aligned} & \frac{\tan A - \sec A - 1}{\tan A + \sec A + 1} \\ &= \frac{\frac{\sin A}{\cos A} - \frac{1}{\cos A} - 1}{\frac{\sin A}{\cos A} + \frac{1}{\cos A} + 1} \\ &= \frac{\sin^2 A + 1 + \cos^2 A - 2\sin A + 2\cos A - 2\sin A \cos A}{(\sin A)^2 - (1 + \cos A)^2} \\ &= \frac{2 - 2\sin A + 2\cos A - 2\sin A \cos A}{\sin^2 A - 1 - \cos^2 A - 2\cos A} \\ &= \frac{2(1 - \sin A) + 2\cos A(1 - \sin A)}{1 - \cos^2 - 1 - \cos^2 A - 2\cos A} \\ &= \frac{2(1 + \cos A)(1 - \sin A)}{-2\cos A(1 + \cos A)} \\ &= \frac{\sin A - 1}{\cos A} \end{aligned}$$

**SSC CGL MAINS 2015****{25 Aug. 2015}**

37. (a)  $\sec \theta - \tan \theta = \frac{1}{\sqrt{3}}$  ... (i)

$$\therefore \sec^2 \theta - \tan^2 \theta = 1$$

$$(\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1$$

$$\sec \theta + \tan \theta = \sqrt{3} \quad \dots (ii)$$

From eq. (i) &amp; (ii)

$$\sec \theta = \frac{2}{\sqrt{3}}, \tan \theta = \frac{1}{\sqrt{3}}$$

$$\therefore \sec \theta \cdot \tan \theta = \frac{2}{\sqrt{3}} \times \frac{1}{\sqrt{3}} = \frac{2}{3}$$

38. (b)  $x = a \sin \theta - b \cos \theta$  ... (i)

$$y = a \cos \theta + b \sin \theta \quad \dots (ii)$$

$$\text{Eq. (i)}^2 + \text{(ii)}^2$$

$$x^2 + y^2 = (a \sin \theta - b \cos \theta)^2 + (a \cos \theta + b \sin \theta)^2$$

$$x^2 + y^2 = a^2 \sin^2 \theta + b^2 \cos^2 \theta - 2ab \sin \theta \cos \theta + a^2 \cos^2 \theta + b^2 \sin^2 \theta + 2ab \sin \theta \cos \theta$$

$$x^2 + y^2 = a^2 \sin^2 \theta + a^2 \cos^2 \theta + b^2 \sin^2 \theta + b^2 \cos^2 \theta$$

$$x^2 + y^2 = a^2 + b^2$$

39. (d)  $\tan A = n \tan B$

$$\frac{\sin A}{\cos A} = n \frac{\sin B}{\cos B}$$

$$\sin A \cos B = n \sin B \cos A$$

$$\sin^2 A \cos^2 B = n^2 \sin^2 B \cos^2 A$$

$$\sin^2 A (1 - \sin^2 B) = n^2 \frac{\sin^2 A}{m^2} \cos^2 A$$

$$\left( \therefore \frac{\sin^2 A}{m^2} = \sin^2 B \right)$$

$$\left( 1 - \frac{\sin^2 A}{m^2} \right) = \frac{n^2 \cos^2 A}{m^2}$$

$$\left( 1 - \frac{1 - \cos^2 A}{m^2} \right) = \frac{n^2 \cos^2 A}{m^2}$$

$$\frac{m^2 - 1 + \cos^2 A}{m^2} = \frac{n^2 \cos^2 A}{m^2}$$

$$m^2 - 1 = \cos^2 A (n^2 - 1)$$

$$\cos^2 A = \frac{m^2 - 1}{n^2 - 1}$$

**Alternatively:-**

$$\text{Put, } A = 30^\circ, B = 60^\circ$$

$$\tan A = n \tan B$$

$$n = \frac{1}{3}$$

$$\sin A = m \sin B$$

$$m = \frac{1}{\sqrt{3}}$$

$$\cos^2 A = \cos^2 30^\circ = \left( \frac{\sqrt{3}}{2} \right)^2 = \frac{3}{4}$$

Put, the value in option (d)

$$\begin{aligned} & \left( \frac{1}{\sqrt{3}} \right)^2 - 1 = \frac{-2}{-3} = \frac{2}{3} \\ & \left( \frac{1}{3} \right)^2 - 1 = \frac{-8}{9} = \frac{3}{4} \end{aligned}$$

40. (b)  $\tan \theta - \cot \theta = 0$

$$\tan \theta = \cot \theta$$

$$\frac{\tan(\theta + 15^\circ)}{\tan(\theta - 15^\circ)}$$

$$\text{Put, } \theta = 45^\circ \quad [\because \tan \theta = \cot \theta]$$

$$\frac{\tan(45^\circ + 15^\circ)}{\tan(45^\circ - 15^\circ)}$$

$$\frac{\tan 60^\circ}{\tan 30^\circ} = \frac{\sqrt{3}}{1} = 3$$

41. (b)  $\sin A + \sin^2 A = 1$

$$\sin A = 1 - \sin^2 A$$

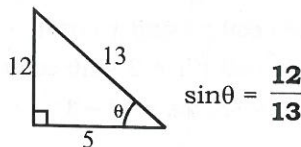
$$\sin A = \cos^2 A$$

$$\therefore \cos^2 A + \cos^4 A = \sin A + \sin^2 A = 1$$

42. (a)  $5 \cos \theta + 12 \sin \theta = 13$

Divide eq. by 13

$$\frac{5}{13} \cos \theta + \frac{12}{13} \sin \theta = 1$$



$$\sin \theta = \frac{12}{13}$$

43. (b)  $\cot 41^\circ \cdot \cot 42^\circ \cdot \cot 43^\circ \cdot \cot 44^\circ \cdot \cot 45^\circ \cdot \cot 46^\circ \cdot \cot 47^\circ \cdot \cot 48^\circ \cdot \cot 49^\circ$   
 $\cot A \cdot \cot B = 1$  (If  $A+B=90^\circ$ )  
 $\cot 41^\circ \cdot \cot 49^\circ \cdot \cot 42^\circ \cdot \cot 48^\circ \cdot \cot 43^\circ \cdot \cot 47^\circ \cdot \cot 44^\circ \cdot \cot 46^\circ \cdot \cot 45^\circ$   
 $1 \times 1 \times 1 \times 1 \times 1 = 1$

44. (d)  $7 \sin^2 \theta + 3 \cos^2 \theta = 4$   
 $7 \sin^2 \theta + 3(1 - \sin^2 \theta) = 4$   
 $4 \sin^2 \theta + 3 = 4$

$$\sin^2 \theta = \frac{1}{4}$$

$$\sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

$$\tan \theta = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

45. (a)  $(\operatorname{cosec} \alpha - \sin \alpha)(\sec \alpha - \cos \alpha)(\tan \alpha + \cot \alpha)$

$$\Rightarrow \left( \frac{1}{\sin \alpha} - \sin \alpha \right) \left( \frac{1}{\cos \alpha} - \cos \alpha \right)$$

$$\left( \frac{\sin \alpha}{\cos \alpha} + \frac{\cos \alpha}{\sin \alpha} \right)$$

$$= \left( \frac{1 - \sin^2 \alpha}{\sin \alpha} \right) \left( \frac{1 - \cos^2 \alpha}{\cos \alpha} \right)$$

$$\left( \frac{1}{\cos \alpha \sin \alpha} \right) = 1$$

**SSC CGL MAINS 2016****{30 Nov. 2016}**

46. (b)  $x \tan 60^\circ + \cos 45^\circ = \sec 45^\circ$

$$x\sqrt{3} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

$$\sqrt{3}x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x = \frac{1}{\sqrt{6}}$$

$$\therefore x^2 + 1 = \left( \frac{1}{\sqrt{6}} \right)^2 + 1$$

$$= \frac{1}{6} + 1 = \frac{7}{6}$$

47. (c)  $\sin(2x - 20^\circ) = \cos(2y + 20^\circ)$   
 $\sin(2x - 20^\circ) = \sin(90^\circ - 2y - 20^\circ)$   
 $2x - 20^\circ = 90^\circ - 2y - 20^\circ$

$$2x + 2y = 90^\circ$$

$$x + y = 45^\circ$$

$$\therefore \tan(x + y) = \tan 45^\circ = 1$$

48. (b)  $a^2 \sec^2 x - b^2 \tan^2 x = c^2$

$$\frac{a^2}{\cos^2 x} - \frac{b^2 \sin^2 x}{\cos^2 x} = c^2$$

$$\frac{a^2 - b^2(1 - \cos^2 x)}{\cos^2 x} = c^2$$

$$a^2 - b^2 + b^2 \cos^2 x = c^2 \cos^2 x$$

$$\frac{a^2 - b^2}{c^2 - b^2} = \cos^2 x$$

$$\sec^2 x = \frac{c^2 - b^2}{a^2 - b^2}$$

$$\tan^2 x = \sec^2 x - 1$$

$$= \frac{c^2 - b^2}{a^2 - b^2} - 1$$

$$= \frac{c^2 - b^2 - a^2 + b^2}{a^2 - b^2}$$

$$= \frac{c^2 - a^2}{a^2 - b^2}$$

$$\therefore \sec^2 x + \tan^2 x$$

$$= \frac{c^2 - b^2}{a^2 - b^2} + \frac{c^2 - a^2}{a^2 - b^2}$$

$$= \frac{c^2 - b^2 + c^2 - a^2}{a^2 - b^2}$$

$$= \frac{b^2 + a^2 - 2c^2}{b^2 - a^2}$$

49. (c)  $(1 + \sec 20^\circ + \cot 70^\circ)$

$$(1 - \operatorname{cosec} 20^\circ + \tan 70^\circ)$$

$$= (1 + \sec 20^\circ + \tan 20^\circ) (1 - \operatorname{cosec} 20^\circ + \cot 20^\circ)$$

$$= \left[ \frac{(\cos 20^\circ + \sin 20^\circ + 1)}{\cos 20^\circ} \right]$$

$$\left[ \frac{(\cos 20^\circ + \sin 20^\circ - 1)}{\sin 20^\circ} \right]$$

$$= \frac{(\cos 20^\circ + \sin 20^\circ)^2 - 1}{\sin 20^\circ \cos 20^\circ}$$

$$= \frac{\cos^2 20^\circ + \sin^2 20^\circ + 2 \sin 20^\circ \cos 20^\circ - 1}{\sin 20^\circ \cos 20^\circ}$$

$$= \frac{1 + 2 \sin 20^\circ \cos 20^\circ - 1}{\sin 20^\circ \cos 20^\circ}$$

$$= \frac{2 \sin 20^\circ \cos 20^\circ}{\sin 20^\circ \cos 20^\circ} = 2$$

50. (c)  $\tan^4 \theta + \tan^2 \theta = 1$

$$\frac{\sin^4 \theta}{\cos^4 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta} = 1$$

$$\frac{\sin^4 \theta + \sin^2 \theta \cos^2 \theta}{\cos^4 \theta} = 1$$

$$\frac{\sin^2 \theta (\sin^2 \theta + \cos^2 \theta)}{\cos^4 \theta} = 1$$

$$\sin^2 \theta = \cos^4 \theta$$

$$\tan^2 \theta = \cos^2 \theta$$

$$\therefore \tan^4 \theta + \tan^2 \theta = 1$$

$$(\cos^2 \theta)^2 + \cos^2 \theta = 1$$

$$\cos^4 \theta + \cos^2 \theta = 1$$

51. (c)  $8(\sin^6 \theta + \cos^6 \theta) - 12(\sin^4 \theta + \cos^4 \theta)$

$$= 8(1 - 3\sin^2 \theta \cos^2 \theta) - 12(1 - 2\sin^2 \theta \cos^2 \theta)$$

$$= 8 - 24 \sin^2 \theta \cos^2 \theta - 12 + 24 \sin^2 \theta \cos^2 \theta$$

$$= -4$$

### SSC CGL MAINS 2016

{01 Dec. 2016}

52. (d)  $(1 + \sec 22^\circ + \cot 68^\circ) (1 - \operatorname{cosec} 22^\circ + \tan 68^\circ)$

$$= [1 + \sec 22^\circ + \cot(90^\circ - 22^\circ)]$$

$$[1 - \operatorname{cosec} 22^\circ + \tan(90^\circ - 22^\circ)]$$

$$= [1 + \sec 22^\circ + \tan 22^\circ] [1 - \operatorname{cosec} 22^\circ + \cot 22^\circ]$$

$$= \left( \frac{\sin 22^\circ + \cos 22^\circ + 1}{\cos 22^\circ} \right)$$

$$\left( \frac{\sin 22^\circ + \cos 22^\circ - 1}{\sin 22^\circ} \right)$$

Now take common-

$$= \frac{(\sin 22^\circ + \cos 22^\circ)^2 - 1}{\sin 22^\circ \cos 22^\circ}$$

$$= \frac{\sin^2 22^\circ + \cos^2 22^\circ + 2 \sin 22^\circ \cos 22^\circ - 1}{\sin 22^\circ \cos 22^\circ}$$

$$= \frac{2 \sin 22^\circ \cos 22^\circ}{\sin 22^\circ \cos 22^\circ} = 2$$

53. (a)  $x \sin \theta - y \cos \theta = 0$

$$x \sin \theta = y \cos \theta \Rightarrow \tan \theta = y/x$$

$$x \sin^3 \theta = y \cos^3 \theta = \sin \theta \cos \theta$$

$$x \sin \theta \cdot \sin^2 \theta + y \cos \theta \cdot \cos^2 \theta$$

$$= \sin \theta \cos \theta$$

Put,  $y \cos \theta$  in place of  $x \sin \theta$

$$y \cos \theta (\sin^2 \theta + \cos^2 \theta) = \sin \theta \cos \theta$$

$$y = \sin \theta$$

Similarly,  $x = \cos \theta$

$$x^2 + y^2 = \cos^2 \theta + \sin^2 \theta = 1$$

54. (b)  $\sec \theta - \tan \theta = m \dots (i)$

$$\therefore \sec^2 \theta - \tan^2 \theta = 1$$

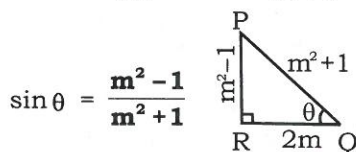
$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

$$\sec \theta + \tan \theta = \frac{1}{m} \dots (ii)$$

From equation (i) and (ii)

$$2 \sec \theta = m + \frac{1}{m}$$

$$\sec \theta = \frac{m^2 + 1}{2m}, \cos \theta = \frac{2m}{m^2 + 1}$$



$$\sin \theta = \frac{m^2 - 1}{m^2 + 1}$$

or,  $\sin \theta = \sqrt{1 - \cos^2 \theta}$

$$\sin \theta = \sqrt{1 - \left( \frac{2m}{m^2 + 1} \right)^2}$$

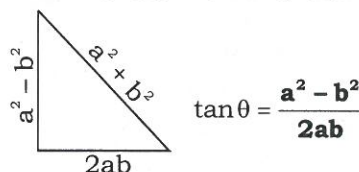
$$\sin \theta = \sqrt{\frac{(m^2 - 1)^2}{(m^2 + 1)^2}} = \frac{m^2 - 1}{m^2 + 1}$$

55. (b)

$$(a^2 - b^2) \sin \theta + 2ab \cos \theta = a^2 + b^2$$

$$\left( \frac{a^2 - b^2}{a^2 + b^2} \right) \sin \theta + \frac{2ab}{a^2 + b^2} \cos \theta = 1$$

$$\cos \theta = \frac{2ab}{a^2 + b^2}, \sin \theta = \frac{a^2 - b^2}{a^2 + b^2}$$



56. (d)  $2y \cos \theta = x \sin \theta$  [if  $\theta = 45^\circ$ ]

$$x = 2y \dots (i)$$

$$2x \sec \theta - y \operatorname{cosec} \theta = 3$$

$$2(2y) \sec 45^\circ - y \operatorname{cosec} 45^\circ = 3$$

$$4y\sqrt{2} - y\sqrt{2} = 3$$

$$\Rightarrow y = \frac{1}{\sqrt{2}}$$

Put, the value of  $y$  in equation (i)

$$x = 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$$

$$\therefore x^2 + 4y^2 = 2 + 4 \times \frac{1}{2} = 4$$

57. (b)  $\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta + \operatorname{cosec} \theta + 1}$

Put,  $\theta = 45^\circ$

$$\frac{1 + \sqrt{2} - 1}{1 + \sqrt{2} + 1} = \frac{\sqrt{2}}{2 + \sqrt{2}}$$

$$\frac{1}{\sqrt{2} + 1} = \sqrt{2} - 1$$

now go through option (b)

$$\frac{1 - \cos \theta}{\sin \theta} = \frac{1 - \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} = \sqrt{2} - 1$$

...satisfied

## SSC CGL MAINS 2016

{12 Jan. 2017}

58. (b)  $\sec A + \tan A = a$  ... (i)

$[\sec^2 A - \tan^2 A = 1]$

$\sec A - \tan A = \frac{1}{a}$  ... (ii)

Adding eq. (i) and (ii),

$2\sec A = a + \frac{1}{a}$

$\sec A = \frac{a^2 + 1}{2a}$

$\cos A = \frac{2a}{a^2 + 1}$

59. (b)  $\sin p + \operatorname{cosec} p = 2$

$\sin p + \frac{1}{\sin p} = 2$

$\sin^2 p + 1 = 2\sin p$

$\sin^2 p + 1 - 2\sin p = 0$

$(\sin^2 p - 1)^2 = 0 \Rightarrow \sin p = 1$

$\sin^7 p + \operatorname{cosec}^7 p = 1^7 + 1^7 = 2$

Alternatively:-

$\sin p + \operatorname{cosec} p = 2$

Put,  $p = 90^\circ$

$\sin^7 90^\circ + \operatorname{cosec}^7 90^\circ = 1^7 + 1^7 = 2$

60. (c)  $\cos x \cdot \cos y + \sin x \cdot \sin y = -1$

$\cos(x - y) = -1$

$\cos(x - y) = \cos 180^\circ$

$x - y = 180^\circ$

$\Rightarrow x = 180^\circ + y$

$\cos x + \cos y$

$\Rightarrow \cos(180^\circ + y) + \cos y$

$\Rightarrow -\cos y + \cos y = 0$

61. (b)  $2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta) + 1$

$[\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta]$

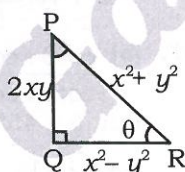
$[\sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta \cos^2 \theta]$

$2(1 - 3\sin^2 \theta \cos^2 \theta) - 3(1 - 2\sin^2 \theta \cos^2 \theta) + 1$

$2 - 6\sin^2 \theta \cos^2 \theta - 3 + 6\sin^2 \theta \cos^2 \theta + 1$

$= 0$

62. (d)  $\cos \theta = \frac{x^2 - y^2}{x^2 + y^2}$

 $2xy, x^2 - y^2, x^2 + y^2$  are triplets.

$\cot \theta = \frac{x^2 - y^2}{2xy}$

63. (b)  $x = \operatorname{cosec} \theta - \sin \theta$

$= \frac{1 - \sin^2 \theta}{\sin \theta} = \frac{\cos^2 \theta}{\sin \theta}$

$y = \sec \theta - \cos \theta$

$= \frac{1 - \cos^2 \theta}{\cos \theta} = \frac{\sin^2 \theta}{\cos \theta}$

Put,  $\theta = 45^\circ$

$x = \frac{1/2}{1/\sqrt{2}} = \frac{1}{\sqrt{2}}$

$y = \frac{1/2}{1/\sqrt{2}} = \frac{1}{\sqrt{2}}$

By option (b)

$x^2 y^2 (x^2 + y^2 + 3) = 1$

$\frac{1}{2} \times \frac{1}{2} \left( \frac{1}{2} + \frac{1}{2} + 3 \right) = 1$

$\frac{1}{4} \times (4) = 1$

$\therefore x^2 y^2 (x^2 + y^2 + 3) = 1$

## SSC CGL MAINS 2017

{17 Feb. 2018}

64. (b)  $\frac{[(\sin x + \sin y)(\sin x - \sin y)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$

Put,  $x = 90^\circ, y = 0$

$\frac{[(\sin 90^\circ + \sin 0^\circ)(\sin 90^\circ - \sin 0^\circ)]}{[(\cos 90^\circ + \cos 0^\circ)(\cos 0^\circ - \cos 90^\circ)]}$

$= \frac{[(1+0)(1-0)]}{[(0+1)(1-0)]} = 1$

65. (b)  $= \frac{(\tan 50^\circ + \tan 30^\circ)}{4 \cos 40^\circ (\tan 50^\circ - \tan 30^\circ)}$

Put,  $\theta = 60^\circ$

$= \frac{(\tan 50^\circ + \tan 180^\circ)}{4 \cos 240^\circ (\tan 50^\circ - \tan 180^\circ)}$

$\frac{\tan 50^\circ}{4 \cos 240^\circ \times \tan 50^\circ} \quad [\because \tan 180^\circ = 0]$

$= \frac{1}{4 \cos 240^\circ} = \frac{1}{4 \cos(180^\circ + 60^\circ)}$

$= \frac{1}{4 \left( -\frac{1}{2} \right)} = \frac{-1}{2}$

From option (b)

$\cos 2\theta = \cos 120^\circ$

$\cos(90^\circ + 30^\circ) \quad [\because \cos(90^\circ + \theta) = -\sin \theta]$

$= -\sin 30^\circ = \frac{-1}{2}$

66. (a)  $\frac{4}{3} \cot^2 \left( \frac{\pi}{6} \right) + 3 \cos^2(150^\circ)$

$= 4 \operatorname{cosec}^2 45^\circ + 8 \sin^2 \left( \frac{\pi}{2} \right)$

$= \frac{4}{3} \cot^2(30^\circ) + 3 \cos^2(90^\circ + 60^\circ)$

$-4 \operatorname{cosec}^2 45^\circ + 8 \sin^2 \left( \frac{\pi}{2} \right)$

$= \frac{4}{3} \times 3 + 3 \times \frac{3}{4} - 4 \times 2 + 8 \times 1$

$= 4 + \frac{9}{4} - 8 + 8 = \frac{25}{4}$

67. (d)  $\sin(B-C) \cdot \cos(A-D) + \sin(A-B) \cdot \cos(C+D) \cdot \sin(C-A) \cdot \cos(B-D)$

put,  $A = B = C = D = 0$

$\sin 0^\circ \cos 0^\circ + \sin 0^\circ \cdot \cos 0^\circ + \sin 0^\circ \cos 0^\circ$

$0 + 0 + 0 = 0$

68. (b)

$\frac{[4 \cos(90-A) \cdot \sin^3(90+A)] - 4 \sin(90+A) \cdot \cos^3(90-A)}{\cos \left( \frac{180+8A}{2} \right)}$

$= \frac{4 \sin A \cdot \cos^3 A - 4 \cos A \cdot \sin^3 A}{\cos(90+4A)}$

$= \frac{4 \sin A \cdot \cos^3 A - 4 \cos A \cdot \sin^3 A}{-\sin 4A}$

$= \frac{4 \sin A \cdot \cos A [\cos^2 A - \sin^2 A]}{-\sin 4A}$

$= \frac{4 \sin A \cdot \cos A \times \cos 2A}{2 \sin 2A \cdot \cos 2A}$

$= \frac{4 \sin A \cdot \cos A}{-2 \times 2 \sin A \cdot \cos A} = -1$

69. (b)

$\cos \left[ \frac{(180-\theta)}{2} \right] \cdot \cos \left[ \frac{(180-90)}{2} \right] +$

$\sin \left[ \frac{180-30}{3} \right] \cdot \sin \left[ \frac{(180-130)}{2} \right]$

$= \cos \left( 90 - \frac{\theta}{2} \right) \cdot \cos \left( 90 - \frac{90}{2} \right) +$

$\sin \left( 90 - \frac{30}{2} \right) \cdot \sin \left( 90 - \frac{130}{2} \right)$

$= \sin \frac{\theta}{2} \cdot \sin \frac{90}{2} + \cos \frac{30}{2} \cdot \cos \frac{130}{2}$

Multiply and divide by 2

$= \frac{1}{2} \left[ 2 \sin \frac{\theta}{2} \cdot \sin \frac{90}{2} + 2 \cos \frac{30}{2} \cdot \cos \frac{130}{2} \right]$

$[2 \sin A \sin B = \cos(A-B) - \cos(A+B)]$

$[2 \cos A \cos B = \cos(A-B) + \cos(A+B)]$

$= \frac{1}{2} [\cos 40 - \cos 50 + \cos 50 + \cos 80]$

$\frac{1}{2} [2 \cos 60 \cdot \cos 20] = \cos 20 \cdot \cos 60$

$$70. (b) [(\tan^2(90-\theta) - \sin^2(90-\theta))$$

$$\operatorname{cosec}^2(90-\theta)\cot^2(90-\theta) \\ \Rightarrow (\cot^2\theta - \cos^2\theta)\sec^2\theta.\tan^2\theta.$$

$$\Rightarrow \left[ \frac{1}{\tan^2\theta} - \frac{1}{\sec^2\theta} \right] \sec^2\theta.\tan^2\theta$$

$$\Rightarrow \left[ \frac{\sec^2\theta - \tan^2\theta}{\tan^2\theta \cdot \sec^2\theta} \right] \times \sec^2\theta.\tan^2\theta$$

$$= 1$$

Alternatively:-

$$[(\tan^2(90-\theta) - \sin^2(90-\theta))$$

$$\operatorname{cosec}^2(90-\theta)\cot^2(90-\theta)$$

$$\text{Put, } \theta = 45^\circ$$

$$= \left[ 1 - \frac{1}{2} \right] \times 2 \times 1$$

$$= 1$$

**SSC CGL MAINS 2017**

**{18 Feb. 2018}**

$$71. (c) \frac{\cos^3 2\theta + 3 \cos 2\theta}{\cos^6 \theta - \sin^6 \theta}$$

$$\text{Put, } \theta = 30^\circ$$

$$\frac{\cos^3 60^\circ + 3 \cos 60^\circ}{\cos^6 30^\circ - \sin^6 30^\circ}$$

$$= \frac{\left(\frac{1}{2}\right)^3 + 3 \times \frac{1}{2}}{\left(\frac{\sqrt{3}}{2}\right)^6 - \left(\frac{1}{2}\right)^6}$$

$$= \frac{\frac{1}{8} + \frac{3}{2}}{\frac{27}{64} - \frac{1}{64}} = \frac{1+12}{27-1}$$

$$= \frac{13}{8} \times \frac{64}{26}$$

$$= 4$$

Alternatively:-

$$\frac{\cos^3 2\theta + 3 \cos 2\theta}{\cos^6 \theta - \sin^6 \theta}$$

$$\text{Put, } \theta = 0^\circ$$

$$\frac{1+3(1)}{1-0} = 4$$

$$72. (d) \tan\left(\frac{\pi}{4} + A\right) \times \tan\left(\frac{3\pi}{4} + A\right) = ?$$

$$\therefore \tan\left(\frac{\pi}{4} + A\right) = \frac{1+\tan A}{1-\tan A}$$

$$\tan\left(\frac{3\pi}{4} + A\right) = \tan[90^\circ + (45^\circ + A)]$$

$$\tan(90^\circ + \theta) = -\cot \theta$$

$$\therefore \tan(90^\circ + (45^\circ + A)) = -\cot(45^\circ + A)$$

$$\cot(45^\circ + A) = \frac{\cot A - 1}{1 + \cot A}$$

$$-\frac{(1 + \tan A)}{(1 - \tan A)} \times \left( \frac{1}{\tan A} - 1 \right)$$

$$\Rightarrow \frac{(1 + \tan A)}{(1 - \tan A)} \times \frac{(-\tan A + 1)}{(1 + \tan A)} = -1$$

Alternatively:-

options are numerical

$$\tan\left(\frac{\pi}{4} + A\right) \times \tan\left(\frac{3\pi}{4} + A\right)$$

$$\text{put } A = 0^\circ$$

$$\tan 45^\circ \times \tan 135^\circ$$

$$1 \times -1 = -1$$

$$73. (b) [(\sec 2\theta + 1) \left[ \sqrt{\sec^2 \theta - 1} \right] \times \frac{1}{2}$$

$$[\cot \theta - \tan \theta] = ?$$

$$\left[ \frac{1}{\cos 2\theta} + 1 \right] \left[ \sqrt{\tan^2 \theta} \right] \times \frac{1}{2}$$

$$\left[ \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \right]$$

$$\left[ \frac{1 + \cos 2\theta}{\cos 2\theta} \right] \times \tan \theta \times \frac{1}{2} \left[ \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cdot \cos \theta} \right]$$

$$\therefore \cos^2 \theta - \sin^2 \theta = \cos 2\theta$$

$$\left[ \frac{1 + \cos 2\theta}{\cos 2\theta} \right] \times \tan \theta \times \frac{1}{2} \left[ \frac{\cos 2\theta}{\sin \theta \cdot \cos \theta} \right]$$

$$[1 + \cos 2\theta] \times \frac{\sin \theta}{\cos \theta} \times \frac{1}{2} \left[ \frac{1}{\sin \theta \cdot \cos \theta} \right]$$

$$\{\cos 2\theta = 2\cos^2 \theta - 1\}$$

$$[1 + 2\cos^2 \theta - 1] \times \frac{1}{2\cos^2 \theta}$$

$$= 2\cos^2 \theta \times \frac{1}{2\cos^2 \theta} = 1$$

Alternatively:-

$$\text{Put, } \theta = 30^\circ$$

$$(\sec 60^\circ + 1) \left[ \sqrt{\sec^2 30^\circ - 1} \right] \times \frac{1}{2} [\cot$$

$$30^\circ - \tan 30^\circ]$$

$$3 \times \sqrt{\left(\frac{4}{3} - 1\right)} \times \frac{1}{2} \left( \sqrt{3} - \frac{1}{\sqrt{3}} \right)$$

$$= 3 \times \frac{1}{\sqrt{3}} \times \frac{1}{2} \times \frac{2}{\sqrt{3}} = 1$$

$$74. (c) \sin(630^\circ + A) + \cos A = ?$$

$$\Rightarrow \sin[720^\circ - (90^\circ - A)] + \cos A$$

$$= -\sin(90^\circ - A) + \cos A$$

$$= -\cos A + \cos A$$

$$= 0$$

$$75. (d) [\sin 59^\circ \cos 31^\circ + \cos 59^\circ \sin 31^\circ]$$

$$+ [\cos 20^\circ \cos 25^\circ - \sin 20^\circ \sin 25^\circ]$$

$$\sin 59^\circ = \sin(90^\circ - 31^\circ) = \cos 31^\circ$$

$$\therefore [\sin(C+D) = \sin C \cdot \cos D + \cos C \sin D]$$

$$\{\cos(C+D) = \cos C \cdot \cos D - \sin C \cdot \sin D\}$$

$$\sin(59^\circ + 31^\circ) + \cos(20^\circ + 25^\circ)$$

$$= \frac{\sin 90^\circ}{\cos 45^\circ} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$$

$$76. (d) \cos(90^\circ - B) \sin(C - A) + \sin$$

$$(90^\circ + A) \cos(B + C) - \sin(90^\circ - C)$$

$$\cos(A + B)$$

$$= \sin B(\sin C \cos A - \cos C \sin A) +$$

$$\cos A(\cos B \cos C - \sin B \sin C) - \cos C$$

$$(\cos A \cos B - \sin A \sin B)$$

$$= \sin B \cdot \sin C \cdot \cos A - \sin B \cdot \cos C \cdot \sin A$$

$$+ \cos A \cdot \cos B \cdot \cos C - \cos A \cdot \sin B \cdot \sin C$$

$$- \cos C \cdot \cos A \cdot \cos B + \cos C \cdot \sin A \cdot \sin B$$

$$= 0$$

Alternatively:-

$$\text{Put } A = B = C = 45^\circ$$

$$\cos(90^\circ - B) \sin(C - A) + \sin(90^\circ + A)$$

$$\cos(B + C) - \sin(90^\circ - C) \cos(A + B)$$

$$= 0 + 0 + 0 = 0$$

$$\therefore \begin{cases} \sin(C - A) = 0 \\ \cos(B + C) = 0 \\ \cos(A + B) = 0 \end{cases}$$

**SSC CGL MAINS 2017**

**{19 Feb. 2018}**

$$77. (d) (\cos 7A + \cos 5A) \div (\sin 7A - \sin 5A)$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$= \frac{2 \cos \left( \frac{7A+5A}{2} \right) \cos \left( \frac{7A-5A}{2} \right)}{2 \cos \left( \frac{7A+5A}{2} \right) \sin \left( \frac{7A-5A}{2} \right)}$$

$$= \frac{\cos 6A \cdot \cos A}{\cos 6A \sin A}$$

$$= \cot A$$

$$78. (c) \frac{[1 - \sin(90 - 2A)]}{[1 + \sin(90 + 2A)]}$$

$$= \frac{1 - \cos 2A}{1 + \cos 2A} = \frac{1 - 1 + 2\sin^2 A}{1 + 2\cos^2 A - 1}$$

$$= \frac{2\sin^2 A}{2\cos^2 A} = \tan^2 A$$

$$[\therefore \cos^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1]$$

$$79. (c) \sin 75^\circ + \sin 15^\circ$$

$$\sin C + \sin D = 2\sin \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$= 2\cos 45^\circ \cdot \cos 30^\circ$$

$$= 2 \times \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{\sqrt{2}}$$

Alternatively:-

$$\sin 75^\circ + \sin 15^\circ$$

$$= \sin(45^\circ + 30^\circ) + \sin(45^\circ - 30^\circ)$$

$$= \sin 45^\circ \cdot \cos 30^\circ + \sin 30^\circ \cdot \cos 45^\circ$$

$$+ \sin 45^\circ \cdot \cos 30^\circ - \sin 30^\circ \cdot \cos 45^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2}$$

$$= \frac{2\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{3}}{\sqrt{2}}$$

Alternatively:-

$$\sin 75^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$\sin 15^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

$$\sin 75^\circ + \sin 15^\circ$$

$$\frac{\sqrt{3} + 1}{2\sqrt{2}} + \frac{\sqrt{3} - 1}{2\sqrt{2}} = \frac{\sqrt{3}}{\sqrt{2}}$$

$$80. (a) \frac{\cos 3\theta + 2\cos 5\theta + \cos 7\theta}{\cos \theta + 2\cos 3\theta + \cos 5\theta} + \sin 2\theta \cdot \tan 3\theta$$

$$\cos C + \cos D = 2\cos \frac{C+D}{2} \cos \frac{C-D}{2}$$

$$\Rightarrow \frac{2\cos 5\theta \times \cos 2\theta + 2\cos 5\theta}{2\cos 3\theta \times \cos 2\theta + 2\cos 3\theta} +$$

$$\sin 2\theta \times \tan 3\theta$$

$$\Rightarrow \frac{2\cos 5\theta (\cos 2\theta + 1)}{2\cos 3\theta (\cos 2\theta + 1)} +$$

$$\sin 2\theta \times \tan 3\theta$$

$$\Rightarrow \frac{\cos 5\theta}{\cos 3\theta} + \frac{\sin 2\theta \sin 3\theta}{\cos 3\theta}$$

$$\Rightarrow \left( \frac{\cos 5\theta + \sin 2\theta \cdot \sin 3\theta}{\cos 3\theta} \right) \times 2$$

$$\Rightarrow \frac{2\cos 5\theta + 2\sin 3\theta \times \sin 2\theta}{2\cos 3\theta}$$

$$2\sin A \cdot \sin B = \cos(A-B) - \cos(A+B)$$

$$\Rightarrow \frac{2\cos 5\theta + \cos \theta - \cos 5\theta}{2\cos 3\theta}$$

$$\Rightarrow \frac{\cos 5\theta + \cos \theta}{2\cos 3\theta}$$

$$\Rightarrow \frac{2\cos 3\theta \cdot \cos 2\theta}{2\cos 3\theta} \Rightarrow \cos 2\theta$$

$$81. (d) \frac{2\sin(45 + \theta)\sin(45 - \sin \theta)}{\cos 2\theta}$$

$$2\sin A \cdot \sin B = \cos(A-B) - \cos(A+B)$$

$$= \frac{\cos(2\theta) - \cos 90^\circ}{\cos 2\theta} = \frac{\cos(2\theta) - 0}{\cos 2\theta}$$

$$= \frac{\cos 2\theta}{\cos 2\theta} = 1$$

Alternatively:-

$$\frac{2\sin(45^\circ + \theta)\sin(45^\circ - \sin \theta)}{\cos 2\theta}$$

$$\text{put } \theta = 15^\circ$$

$$\frac{2\sin 60^\circ \times \sin 30^\circ}{\cos 30^\circ} =$$

$$2 \times \frac{\sqrt{3}}{2} \times \frac{1}{2} = \frac{\sqrt{3}}{2} = 1$$

only (d) option is Satisfied with these condition

Alternatively:-

$$= \frac{2\sin(45 + \theta)\sin(45 - \theta)}{\cos 2\theta}$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$= \frac{2[\sin 45 \cos \theta + \cos 45 \sin \theta]}{\cos 2\theta} \times$$

$$\frac{[\sin 45 \cos \theta - \cos 45 \sin \theta]}{\cos 2\theta}$$

$$= 2 \left[ \frac{1}{\sqrt{2}} [\cos \theta + \sin \theta] \frac{1}{\sqrt{2}} [\cos \theta - \sin \theta] \right]$$

$$= 2 \times \frac{1}{2} = 1$$

$$82. (d) \sin(90+2A) \cdot [4 - \cos^2(90-2A)]$$

$$= \cos 2A \cdot (4 - \sin^2 2A)$$

$$\text{Put, Value of } A = 90^\circ$$

$$= \cos 180^\circ \cdot (4 - \sin^2 180^\circ)$$

$$= (-1)(4 - 0)$$

$$= -4$$

using option,

only (d) option is Satisfied with these condition

$$4(\cos^6 A - \sin^6 A)$$

$$= 4(\cos^6 90^\circ - \sin^6 90^\circ)$$

$$= 4(0 - 1) = -4$$

Alternatively:-

$$\sin(90+2A) \cdot [4 - \cos^2(90-2A)]$$

$$\cos 2A \cdot (4 - \sin^2 2A)$$

$$= (\cos^2 A - \sin^2 A)[4 - (2\sin A \cos A)^2]$$

$$= (\cos^2 A - \sin^2 A) 4(1 - \sin^2 A \cos^2 A)$$

$$= 4(\cos^6 A - \sin^6 A)$$

$$[\therefore (a^3 - b^3) = (a-b)(a^2 + b^2 + ab)]$$

$$83. (d) \frac{\cos(90+A)}{\sec(270-A)} + \frac{(\sin 270+A)}{\operatorname{cosec}(630-A)}$$

$$= \frac{-\sin A}{-\operatorname{cosec} A} + \frac{-\cos A}{-\sec A}$$

$$= \sin^2 A + \cos^2 A = 1$$

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$$84. (b) \frac{\sin(y-z) + \sin(y+z) + 2\sin y}{\sin(x-z) + \sin(x+z) + 2\sin x}$$

$$[\sin(A+B) + \sin(A-B) = 2\sin A \cos B]$$

$$\frac{2\sin y \times \cos z + 2\sin y}{2\sin x \times \cos z + 2\sin x}$$

$$\frac{2\sin y(\cos z + 1)}{2\sin x(\cos z + 1)} = \frac{\sin y}{\sin x}$$

$$85. (b)$$

$$= \frac{[\sin(x+y) - 2\sin x + \sin(x-y)]}{[\cos(x-y) + \cos(x+y) - 2\cos x]}$$

$$\times \left[ \frac{\sin 10x - \sin 8x}{\cos 10x + \cos 8x} \right]$$

$$[\cos(A+B) + \cos(A-B) = 2\cos A \cos B]$$

$$= \left[ \frac{2\sin x \cos y - 2\sin x}{2\cos x \cos y - 2\cos x} \right]$$

$$\times \left[ \frac{2\sin 9x \cdot \sin x}{2\cos 9x \cos x} \right]$$

$$= \left[ \frac{2\sin x(\cos y - 1)}{2\cos x(\cos y - 1)} \right] \times \tan x$$

$$\Rightarrow \tan x \tan x = \tan^2 x$$

$$\begin{aligned}
 86. (b) & \frac{[\sin(90-10\theta) - \cos(\pi-6\theta)]}{\cos\left(\frac{\pi}{2}-10\theta\right) - \sin(\pi-6\theta)} \\
 &= \frac{[\sin(90-10\theta) - \cos(180-6\theta)]}{\cos(90-10\theta) - \sin(180-6\theta)} \\
 &= \frac{\cos 10\theta + \cos 6\theta}{\sin 10\theta - \sin 6\theta} \\
 &= \frac{2 \cos 8\theta \cos 2\theta}{2 \cos 8\theta \sin 2\theta} = \cot 2\theta
 \end{aligned}$$

$$\begin{aligned}
 87. (d) \sec\theta(\cos\theta + \sin\theta) &= \sqrt{2} \\
 \sec\theta \cdot \cos\theta + \sec\theta \cdot \sin\theta &= \sqrt{2} \\
 1 + \tan\theta &= \sqrt{2} \\
 \tan\theta &= \sqrt{2} - 1 \\
 \cot\theta &= \sqrt{2} + 1 \\
 \frac{(2\sin\theta)}{\cos\theta - \sin\theta} &= \frac{2\sin\theta}{\sin\theta(\cot\theta - 1)} \\
 &= \frac{2}{\cot\theta - 1} = \frac{2}{\sqrt{2} + 1 - 1} \\
 \frac{2\sin\theta}{\cos\theta - \sin\theta} &= \sqrt{2}
 \end{aligned}$$

**Alternatively:-**

$$\begin{aligned}
 \sec\theta(\cos\theta + \sin\theta) &= \sqrt{2} \\
 \sec\theta \cos\theta + \sec\theta \sin\theta &= \sqrt{2} \\
 \Rightarrow 1 + \tan\theta &= \sqrt{2} \quad \dots(i) \\
 \sec\theta(\cos\theta + \sin\theta)
 \end{aligned}$$

$$\Rightarrow \cos\theta + \sin\theta = \frac{\sqrt{2}}{\sec\theta}$$

squaring both sides,

$$\cos^2\theta + \sin^2\theta + 2\cos\theta\sin\theta = \frac{2}{\sec^2\theta}$$

$$\Rightarrow 1 + 2\cos\theta\sin\theta = 2\cos^2\theta$$

$$\Rightarrow 2\sin\theta\cos\theta = 2\cos^2\theta - 1$$

$$\Rightarrow 2\sin\theta = \frac{2\cos^2\theta - 1}{\cos\theta}$$

By putting this value in question

$$\frac{2\sin\theta}{\cos\theta - \sin\theta}$$

$$\Rightarrow \frac{2\cos^2\theta - 1}{\cos\theta - \sin\theta}$$

$$\Rightarrow \frac{2\cos^2\theta - 1}{\cos\theta(\cos\theta - \sin\theta)}$$

$$\begin{aligned}
 &\Rightarrow \frac{\cos^2\theta - \sin^2\theta}{\cos\theta(\cos\theta - \sin\theta)} \\
 &\Rightarrow \frac{(\cos\theta - \sin\theta)(\cos\theta + \sin\theta)}{\cos\theta(\cos\theta - \sin\theta)} \\
 &\Rightarrow \frac{\cos\theta + \sin\theta}{\cos\theta} \\
 \therefore 1 + \tan\theta & \\
 \text{By equation(i) value of} & \\
 \Rightarrow 1 + \tan\theta &= \sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 88. (a) & \frac{1}{\sin^4(90-\theta)} + \frac{1}{[\cos^2(90-\theta)] - 1} \\
 &\Rightarrow \frac{1}{\cos^4\theta} + \frac{1}{\sin^2 - 1} \\
 &\Rightarrow \frac{1}{\cos^4\theta} - \frac{1}{1 - \sin^2\theta} \\
 &\Rightarrow \frac{1}{\cos^4\theta} - \frac{1}{\cos^2\theta} \\
 &\Rightarrow \frac{1 - \cos^2\theta}{\cos^4\theta} \\
 &\Rightarrow \frac{\sin^2\theta}{\cos^4\theta} \\
 &\Rightarrow \frac{\sin^2\theta}{\cos^2\theta} \sec^2\theta \\
 &\Rightarrow \tan^2\theta \sec^2\theta
 \end{aligned}$$

$$89. (c) \frac{[\tan(90-A) + \cot(90-A)]^2}{2\sec^2(90-2A)}$$

By putting  $A = 45^\circ$

$$\Rightarrow \frac{[\tan 45^\circ + \cot 45^\circ]^2}{2\sec^2(90^\circ - 90^\circ)}$$

$$\Rightarrow \frac{[1+1]^2}{2\sec^2\theta}$$

$$\Rightarrow \frac{4}{2 \times 1}$$

$$\Rightarrow 2$$

**Alternatively:-**

$$\frac{(\cot A + \tan A)^2}{2\sec^2 2A}$$

$$\begin{aligned}
 &= \frac{\left(\frac{\cos A}{\sin A} + \frac{\sin A}{\cos A}\right)^2}{2} \\
 &= \frac{(\sec A + \csc A)^2}{2}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2\sin^2 A \cos^2 A} (2\sin A \cdot \cos A)^2 \\
 &= \frac{4\sin^2 A \cos^2 A}{2\sin^2 A \cos^2 A} = 2
 \end{aligned}$$

$$\begin{aligned}
 90. (a) & \{\sin(90^\circ - x) \cdot \cos[\pi - (x - y)]\} + \\
 & \{\cos(90^\circ - x) \cdot \sin(\pi - (y - x))\} \\
 &= -\cos x \cos(x - y) + \sin x \sin(y - x) \\
 &= -\cos x \cos(x - y) - \sin x \sin(x - y) \\
 &= -[\cos x \cos(x - y) + \sin x \sin(x - y)] \\
 &\{\cos(A - B) = \cos A \cos B + \sin A \sin B\} \\
 &= -[\cos(x - y + y)] \\
 &= -\cos y
 \end{aligned}$$

**Alternatively:-**

$$\begin{aligned}
 &\sin(90^\circ - x) \cos(\pi - (x - y)) + \cos(90^\circ - x) \\
 &\sin[\pi - (y - x)] \\
 &\text{putting } x = y = 0^\circ \\
 &\sin 90^\circ \cos 180^\circ + \cos 90^\circ \sin 180^\circ \\
 &1 \times (-1) + 0 = -1 \\
 &\text{By options : } -\cos y = -1
 \end{aligned}$$

**Alternatively:-**

$$\begin{aligned}
 &\sin(90^\circ - x) \cdot \cos(\pi - (x - y)) \\
 &+ \cos(90^\circ - x) \cdot \sin(\pi - (y - x)) \\
 &\Rightarrow \cos x (-\cos(x - y)) + \sin x \cdot \sin(y - x) \\
 &\Rightarrow -\cos x \cos(x - y) + \sin x \sin(y - x) \\
 &\Rightarrow -\cos x (\cos x \cos y + \sin x \sin y) \\
 &+ \sin x (\sin y \cos x - \cos y \sin x) \\
 &\Rightarrow -\cos^2 x \cos y - \cos x \sin x \sin y \\
 &+ \sin x \sin y \cos x - \cos y \sin^2 x \\
 &\Rightarrow -\cos y (\cos^2 x + \sin^2 x) \Rightarrow -\cos y
 \end{aligned}$$

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$$91. (d) \frac{(\sin 4x + \sin 4y)[\tan(2x - 2y)]}{\sin 4x - \sin 4y}$$

$$\sin A + \sin B = 2\sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\Rightarrow \frac{2\sin\left(\frac{4x+4y}{2}\right) \cos\left(\frac{4x-4y}{2}\right) \left[\frac{\sin(2x-2y)}{\cos(2x-2y)}\right]}{2\cos\left(\frac{4x+4y}{2}\right) \sin\left(\frac{4x-4y}{2}\right)}$$

$$\Rightarrow \frac{\sin(2x+2y) \sin(2x-2y)}{\cos(2x+2y) \sin(2x-2y)} \times$$

$$\left[\frac{\sin(2x-2y)}{\cos(2x-2y)}\right] \Rightarrow \tan(2x+2y)$$

$$\begin{aligned}
 92. (c) & \frac{32\cos^6 x - 48\cos^4 x + 18\cos^2 x - 1}{4\sin x \times \cos x \times \sin(60^\circ - x) \times} \\
 & \cos(60^\circ - x) \times \sin(60^\circ + x) \cos(60^\circ + x) \\
 & \text{adding } (+4 \text{ and } -4),
 \end{aligned}$$

$$\rightarrow 32\cos^6x - 4 - 48\cos^4x + 18\cos^2x - 3$$

$$\rightarrow 32\cos^6x - 4 - 48\cos^4x + 24\cos^2x - 6\cos^2x - 3$$

$$\rightarrow 32\cos^6x - 4 - 48\cos^4x + 24\cos^2x - 3\cos^2x$$

$$\rightarrow 4(2\cos^3x - 1)^3 - 3\cos^2x$$

$$\rightarrow \cos 6x$$

$$\therefore \cos^3x \rightarrow 4\cos^3x - 3\cos x$$

$$\therefore \cos 6x \rightarrow 4\cos^3 2x - 3\cos^2 x$$

$$\text{Now, } \therefore 4\sin x \cdot \cos x \cdot \sin(60-x) \cdot \cos(60-x) \cdot \sin(60+x) \cdot \cos(60+x)$$

$$\cos(60-x) \cdot \sin(60+x) \cdot \cos(60+x)$$

$$[\therefore \sin x \cdot \sin(60^\circ - x) \cdot \sin(60^\circ + x) = \frac{1}{4} \sin 3x]$$

$$[\therefore \cos x \cdot \cos(60^\circ - x) \cdot \cos(60^\circ + x) = \frac{1}{4} \cos 3x]$$

$$= \frac{\cos 6x}{\frac{1}{4} \sin 3x \cdot \cos 3x}$$

$$= \frac{2}{2} \times \frac{4\cos 6x}{\sin 3x \cdot \cos 3x}$$

$$= \frac{8\cos 6x}{2\sin 3x \cdot \cos 3x}$$

$$= \frac{8\cos 6x}{\sin 6x}$$

$$= 8\cot 6x$$

$$93. (c) \left[ \frac{2\cot\left(\frac{\pi-A}{2}\right)}{1+\tan^2\left(\frac{2\pi-A}{2}\right)} \right]$$

$$\Rightarrow \frac{2\cot\left(\frac{\pi}{2} - \frac{A}{2}\right)}{1+\tan^2\left(\frac{\pi}{2} - \frac{A}{2}\right)}$$

$$\Rightarrow \frac{2\tan\frac{A}{2}}{1+\tan^2\frac{A}{2}}$$

$$\Rightarrow \sin A \left[ \therefore \sin A = \frac{2\tan\frac{A}{2}}{1+\tan^2\frac{A}{2}} \right]$$

$$94. (c) \tan\theta + \sec\theta = \frac{(x-2)}{x+2} \quad \dots(i)$$

$$\frac{(\sec\theta + \tan\theta)}{1} \times \frac{(\sec\theta - \tan\theta)}{(\sec\theta - \tan\theta)}$$

$$= \frac{x-2}{x+2}$$

$$\frac{1}{\sec\theta - \tan\theta} = \frac{x-2}{x+2}$$

$$\sec\theta - \tan\theta = \frac{x+2}{x-2} \quad \dots(ii)$$

$$\text{add eq. (ii) + (i)}$$

$$2\sec\theta = \frac{x+2}{x-2} + \frac{x-2}{x+2}$$

$$2\sec\theta = \frac{(x+2)^2 + (x-2)^2}{(x^2 - 2^2)}$$

$$\text{use } (a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$$

$$2\sec\theta = \frac{2(x^2 + 4)}{(x^2 - 4)}$$

$$\sec\theta = \frac{x^2 + 4}{x^2 - 4}$$

$$\frac{1}{\cos\theta} = \frac{x^2 + 4}{x^2 - 4}$$

$$\Rightarrow \cos\theta = \frac{x^2 - 4}{x^2 + 4}$$

$$95. (b) (\cos 40^\circ - \cos 140^\circ) / (\sin 80^\circ + \sin 20^\circ)$$

$$\sin A + \sin B = 2\sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2\sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$= \frac{-2\sin\left(\frac{40^\circ + 140^\circ}{2}\right) \sin\left(\frac{40^\circ - 140^\circ}{2}\right)}{2\sin\left(\frac{80^\circ + 20^\circ}{2}\right) \cos\left(\frac{80^\circ - 20^\circ}{2}\right)}$$

$$= \frac{-2\sin\left(\frac{180^\circ}{2}\right) \sin\left(-\frac{100^\circ}{2}\right)}{2\sin\left(\frac{100^\circ}{2}\right) \cos\left(\frac{60^\circ}{2}\right)}$$

$$= \frac{-2\sin 90^\circ \times \sin(-50^\circ)}{2\sin 50^\circ \times \cos 30^\circ}$$

$$= \frac{-2\sin 90^\circ \times \sin(-\sin 50^\circ)}{2\sin 50^\circ \times \cos 30^\circ}$$

$$= \frac{2\sin 90^\circ \times \sin 50^\circ}{2\sin 50^\circ \times \cos 30^\circ}$$

$$= \frac{\sin 90^\circ}{\cos 30^\circ}$$

$$= \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$$

$$96. (b) \text{ Put, } \theta = 60^\circ$$

$$\left[ \frac{1 - \tan(90^\circ - \theta) + \sec(90^\circ - \theta)}{\tan(90^\circ - \theta) + \sec(90^\circ - \theta) + 1} \right]$$

$$= \left[ \frac{1 - \tan 30^\circ + \sec 30^\circ}{\tan 30^\circ + \sec 30^\circ + 1} \right]$$

$$= \frac{1 - \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}}$$

$$= \frac{1 + \frac{1}{\sqrt{3}}}{1 + \frac{3}{\sqrt{3}}} = \frac{\sqrt{3} + 1}{3 + \sqrt{3}}$$

$$= \frac{1}{\sqrt{3}} \times \left( \frac{\sqrt{3} + 1}{\sqrt{3} + 1} \right) = \frac{1}{\sqrt{3}}$$

$$\frac{1}{\sqrt{3}} = \tan 30^\circ = \tan \frac{\theta}{2}$$

Alternatively:-

$$\left[ \frac{1 - \tan(90^\circ - \theta) + \sec(90^\circ - \theta)}{\tan(90^\circ - \theta) + \sec(90^\circ - \theta) + 1} \right]$$

$$\Rightarrow \left[ \frac{1 - \cot\theta + \operatorname{cosec}\theta}{\cot\theta + \operatorname{cosec}\theta + 1} \right]$$

$$\Rightarrow \left[ \frac{1 - \frac{\cos\theta}{\sin\theta} + \frac{1}{\sin\theta}}{\frac{\cos\theta}{\sin\theta} + \frac{1}{\sin\theta} + 1} \right]$$

$$\Rightarrow \left[ \frac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos\theta + 1} \right]$$

$$\left[ \begin{aligned} \therefore \sin\theta &= 2\sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2} \\ \cos\theta &= 1 - 2\sin^2 \frac{\theta}{2} \\ \cos \frac{\theta}{2} &= 2\cos^2 \frac{\theta}{2} - 1 \end{aligned} \right]$$

$$\Rightarrow \frac{2\sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2} + 1 - 1 + 2\sin^2 \frac{\theta}{2}}{2\sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2} + 1 + 2\cos^2 \frac{\theta}{2} - 1}$$

$$\Rightarrow \frac{2\sin \frac{\theta}{2} \left( \sin \frac{\theta}{2} + \cos \frac{\theta}{2} \right)}{2\cos \frac{\theta}{2} \left( \sin \frac{\theta}{2} + \cos \frac{\theta}{2} \right)}$$

$$\Rightarrow \tan \left( \frac{\theta}{2} \right)$$

$$97. (c) \text{ Put, } A = 60^\circ$$

$$\frac{\sin 30^\circ + \cos 60^\circ}{\cos(-30^\circ) + \sin 120^\circ}$$

$$= \frac{\frac{1}{2} + \frac{1}{2}}{\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$$

$$\frac{1}{\sqrt{3}} = \tan 30^\circ = \tan \frac{A}{2}$$

Alternatively:-

$$\frac{[\sin(90-A) + \cos(180-2A)]}{[\cos(90-2A) + \sin(180-A)]}$$

$$\Rightarrow \frac{\cos A + (-\cos 2A)}{\sin 2A + \sin A}$$

$$\Rightarrow \frac{\cos A - \cos 2A}{\sin 2A + \sin A}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\Rightarrow \frac{-2 \sin \left( \frac{A+2A}{2} \right) \times \sin \left( \frac{A-2A}{2} \right)}{2 \sin \left( \frac{A+2A}{2} \right) \times \cos \left( \frac{2A-A}{2} \right)}$$

$$\Rightarrow \frac{-\sin \left( \frac{A-2A}{2} \right)}{\cos \left( \frac{2A-A}{2} \right)} \Rightarrow \frac{\sin \left( \frac{2A-A}{2} \right)}{\cos \left( \frac{2A-A}{2} \right)}$$

$$\Rightarrow \frac{\sin(A/2)}{\cos(A/2)}$$

$$\Rightarrow \tan(A/2)$$

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98. (d)  $\frac{2(1 - \sin^2 \theta) \operatorname{cosec}^2 \theta}{\cot^2 \theta (1 + \tan^2 \theta)} - 1$   
 $(\because \sec^2 \theta = 1 + \tan^2 \theta)$

$$= \frac{2 \cos^2 \theta \operatorname{cosec}^2 \theta}{\cot^2 \theta \sec^2 \theta} - 1$$

$$= \frac{2 \cos^2 \theta \sin^2 \theta}{\sin^2 \theta \cos^2 \theta \sec^2 \theta} - 1$$

$$= 2 \cos^2 \theta - 1$$

$$= \cos 2\theta$$

99. (\*) Wrong question

100. (c)  $\cos 15^\circ - \cos 165^\circ$ 

$$\cos 15^\circ - \cos (180^\circ - 15^\circ)$$

$$\cos 15^\circ + \cos 15^\circ$$

$$= 2 \cos 15^\circ$$

$$= 2 \times \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3} + 1}{\sqrt{2}}$$

$$\therefore \cos 15^\circ = \sin 75^\circ = \frac{\sqrt{3} + 1}{2}$$

Alternatively:-

$$\cos 15^\circ - \cos 165^\circ$$

Use formula

$$\cos C - \cos D =$$

$$2 \sin \left( \frac{C+D}{2} \right) \cdot \sin \left( \frac{D-C}{2} \right)$$

$$= 2 \sin \left( \frac{180^\circ}{2} \right) \cdot \sin \left( \frac{150^\circ}{2} \right)$$

$$= 2 \sin (90^\circ) \cdot \sin (75^\circ)$$

$$= 2 \times \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3} + 1}{\sqrt{2}}$$

101. (a)  $\therefore P+Q+R=60^\circ$ By putting  $P=0^\circ$ ,  $Q=0^\circ$  and  $R=60^\circ$ 

$$\Rightarrow \cos Q \cdot \cos R \cdot (\cos P - \sin P) + \sin Q$$

$$\sin R \cdot (\sin P - \cos P)$$

$$\Rightarrow 1 \times \cos 60^\circ \cdot (\cos 0^\circ - \sin 0^\circ) + \sin$$

$$0^\circ \cdot \sin 60^\circ \cdot (\sin 0^\circ - \cos 0^\circ)$$

$$\Rightarrow \frac{1}{2} (1-0) + 0 \Rightarrow \frac{1}{2}$$

102. (\*) Wrong question

103. (d)

$$\frac{1 + 2 \cot^2(90-x) - 2 \operatorname{cosec}(90-x) \cot(90-x)}{\operatorname{cosec}(90-x) - \cot(90-x)}$$

$$\Rightarrow \frac{1 + 2 \tan^2 x - 2 \sec x \tan x}{\sec x - \tan x}$$

$$(\because \tan^2 x = 1 + \sec^2 x - 1)$$

$$\Rightarrow \frac{1 + 2(\sec^2 x - 1) - 2 \sec x \tan x}{\sec x - \tan x}$$

$$\Rightarrow \frac{2 \sec^2 x - 1 - 2 \sec x \tan x}{\sec x - \tan x}$$

$$\Rightarrow \frac{2 \sec x \times (\sec x - \tan x)}{(\sec x - \tan x)}$$

$$\frac{1}{(\sec x - \tan x)}$$

$$\Rightarrow 2 \sec x - \frac{1}{\sec x - \tan x}$$

$$\Rightarrow 2 \sec x - \frac{1}{\sec x - \tan x} \times$$

$$\frac{\sec x + \tan x}{\sec x + \tan x}$$

$$\Rightarrow 2 \sec x - \frac{(\sec x + \tan x)}{\sec^2 x - \tan^2 x}$$

$$\Rightarrow 2 \sec x - \sec x - \tan x$$

$$(\because \sec^2 x = 1 + \tan^2 x)$$

$$\Rightarrow \sec x - \tan x$$

Alternatively:-

By putting  $\theta = 45^\circ$  in equation

$$\Rightarrow \frac{1 + 2 \tan^2 45 - 2 \sec 45 \tan 45}{\sec 45 - \tan 45}$$

$$\Rightarrow \frac{1 + 2 - 2\sqrt{2}}{\sqrt{2} - 1}$$

$$\Rightarrow \frac{3 - 2\sqrt{2}}{\sqrt{2} - 1} \times \frac{(\sqrt{2} + 1)}{(\sqrt{2} + 1)}$$

$$\Rightarrow 3\sqrt{2} - 2\sqrt{2} - 1 \Rightarrow \sqrt{2} - 1$$

By options (option d satisfied)

$$\Rightarrow \sec x - \tan x \Rightarrow \sqrt{2} - 1$$

104. (d)  $\sin(180-\theta) \cdot \sin(90-\theta)$ 

$$+ \frac{\cot(90-\theta)}{1 + \tan^2 \theta}$$

$$= \sin \theta \cdot \cos \theta + \frac{\tan \theta}{\sec^2 \theta}$$

$$= \sin \theta \cdot \cos \theta + \frac{\sin \theta}{\cos \theta} \times \cos^2 \theta$$

$$= \sin \theta \cdot \cos \theta + \sin \theta \cdot \cos \theta$$

$$= 2 \cos \theta \cdot \sin \theta$$

**SSC CGL MAINS 2017****{09 Mar. 2018} PAPER - II**105. (b)  $\cos 15^\circ + \cos 105^\circ$ 

$$\cos C + \cos D = 2 \cos \left( \frac{C+D}{2} \right) \cos \left( \frac{C-D}{2} \right)$$

$$= 2 \cos 60^\circ \cdot \cos 45^\circ$$

$$= 2 \times \frac{1}{2} \times \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

106. (c)  $A + B + C = 90^\circ$ Let,  $B = C = 0^\circ$ 

$$\Rightarrow A = 90^\circ$$

$$\therefore \sin \left( \frac{A}{2} \right) \sin \left[ \frac{(180 - B - C)}{2} \right] +$$

$$\cos \left( \frac{A}{2} \right) \sin \left( \frac{B+C}{2} \right)$$

$$= \sin 45^\circ \cdot \sin 90^\circ + \cos 45^\circ \cdot \sin 0^\circ$$

$$= \sin 45^\circ = \frac{1}{\sqrt{2}}$$

Alternatively:-

$$A + B + C = 90^\circ$$

$$= \sin \left( \frac{A}{2} \right) \cdot \sin \left( \frac{180 - B - C}{2} \right) +$$

$$\cos \left( \frac{A}{2} \right) \cdot \sin \left( \frac{B+C}{2} \right)$$

$$= \sin \left( \frac{A}{2} \right) \cdot \sin \left( 90 - \frac{(B+C)}{2} \right)$$

$$+\cos\left(\frac{A}{2}\right) \cdot \sin\left(\frac{B+C}{2}\right)$$

$$= \sin\left(\frac{A}{2}\right) \cdot \cos\left(\frac{B+C}{2}\right) +$$

$$\cos\left(\frac{A}{2}\right) \cdot \sin\left(\frac{B+C}{2}\right)$$

$$[\sin(a+b) = \sin a \cdot \cos b + \cos a \cdot \sin b]$$

$$= \sin\left(\frac{A}{2} + \frac{B+C}{2}\right) = \sin\left(\frac{90}{2}\right)$$

$$= \sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$107.(d) \cot(90-x) \cdot \sin^4(90-x) + \cot(180-x) \cdot \sin^4(180-x)$$

$$= \tan x \cdot \cos^4 x - \cot x \cdot \sin^4 x$$

$$= \frac{\sin x}{\cos x} \cdot \cos^4 x - \frac{\cos x}{\sin x} \cdot \sin^4 x$$

$$= \sin x \cdot \cos^3 x - \cos x \cdot \sin^3 x$$

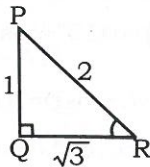
$$= \sin x \cdot \cos x (\cos^2 x - \sin^2 x)$$

$$= \frac{(2 \sin x \cos x)(\cos 2x)}{2}$$

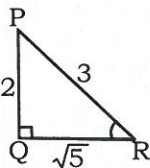
$$= \frac{2 \sin 2x \cdot \cos 2x}{2 \times 2}$$

$$= \frac{\sin 4x}{4}$$

$$108.(a) \sin x = \frac{1}{2}$$



$$\cos x = \frac{\sqrt{3}}{2}$$



$$\sin y = \frac{2}{3}$$

$$\cos y = \frac{\sqrt{5}}{3}$$

$$\frac{6 \cos^2 x - 4 \cos^4 x}{18 \cos^2 y - 27 \cos^4 y} =$$

$$\frac{\left(6 \times \frac{3}{4} - 4 \times \frac{9}{16}\right)}{\left(18 \times \frac{5}{9} - 27 \times \frac{25}{81}\right)}$$

$$= \frac{\left(\frac{72-36}{16}\right)}{\left(\frac{810-675}{81}\right)} = \frac{\frac{36}{16}}{\frac{135}{81}} = \frac{27}{20}$$

$$109.(b) \cos(A+B) = \frac{1}{2} = \cos 60^\circ$$

$$\sin(A-B) = \frac{1}{2} = \sin 30^\circ$$

$$A+B = 60^\circ$$

$$A-B = 30^\circ$$

$$\Rightarrow A = 45^\circ$$

$$B = 15^\circ$$

$$\sin A \cdot \cos A + \sin^2 A \cdot \sin B \cdot \cos B + \cos^3 A \cdot \cos B \cdot \tan A$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} + \left(\frac{1}{\sqrt{2}}\right)^2 \frac{(2 \sin B \cos B)}{2}$$

$$+ \cos^3 A \cdot \cos B \times \frac{\sin A}{\cos A}$$

$$= \frac{1}{2} + \frac{1}{2} \frac{\sin 2B}{2} + \cos^2 A \cdot \cos B \cdot \sin A$$

$$= \frac{1}{2} + \frac{1}{8} + \left(\frac{1}{\sqrt{2}}\right)^2 \times \frac{1}{\sqrt{2}} \cdot \cos B$$

$$[\because \cos 15^\circ = \sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}]$$

$$= \frac{1}{2} + \frac{1}{8} + \frac{1}{2\sqrt{2}} \left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)$$

$$= \frac{4+1+\sqrt{3}+1}{8}$$

$$= \frac{6+\sqrt{3}}{8}$$

$$110.(b) \cot(90^\circ+75^\circ)$$

$$= -\tan 75^\circ = -\tan(45^\circ+30^\circ)$$

$$= -\left[\frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}\right]$$

$$= -\left[\frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}\right] = \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$$= -\frac{4+2\sqrt{3}}{2}$$

$$= -(2+\sqrt{3})$$

$$111.(c) 2 \sin 15^\circ \cos 15^\circ - 4 \sin^3 15^\circ \cos 15^\circ$$

$$[\because 2 \sin A \cos A = \sin 2A]$$

$$= \sin 2 \times (15^\circ) - 2 \cdot \sin 15^\circ \cdot \cos 15^\circ \cdot 2 \sin^2 15^\circ$$

$$= \sin 30^\circ (1 - 2 \sin^2 15^\circ)$$

$$[\because 1 - 2 \sin^2 A = \cos 2A]$$

$$= \sin 30^\circ \times \cos 30^\circ$$

$$= \frac{1}{2} \times \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{3}}{4}$$

$$112.(c) \frac{(\cos 9^\circ + \sin 8^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\sin 56^\circ \cdot \sec 34^\circ + \cos 25^\circ \cdot \operatorname{cosec} 65^\circ}$$

$$= \frac{2 \sin 81^\circ \cdot 2 \operatorname{cosec} 81^\circ}{\sin 56^\circ \times \frac{1}{\sin 56^\circ} + \cos 25^\circ \times \frac{1}{\cos 25^\circ}}$$

$$\Rightarrow \frac{4}{2} = 2$$

$$113.(b) \operatorname{cosec}(65^\circ+\theta) - \sec(25^\circ-\theta) + \tan^2 20^\circ - \operatorname{cosec}^2 70^\circ$$

$$= \operatorname{cosec}[90^\circ - (25^\circ - \theta)] - \sec(25^\circ - \theta) + \tan^2 20^\circ - \cot^2 70^\circ - 1$$

$$= \sec(25^\circ - \theta) - \sec(25^\circ - \theta) + \tan^2 20^\circ - \tan^2 20^\circ - 1$$

$$= -1$$

$$114.(c) \sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$$

$$\text{Put } \theta = 45^\circ$$

$$= \sqrt{\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}} = \sqrt{\frac{\sqrt{2}+1}{\sqrt{2}-1}}$$

$$= \sqrt{\frac{\sqrt{2}+1}{\sqrt{2}-1}} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$= \sqrt{(\sqrt{2}+1)^2} = \sqrt{2}+1$$

$$\text{only option (c) gives at } (\sqrt{2}+1) \text{ at } \theta = 45^\circ$$

Alternatively:-

$$\sqrt{\frac{\cot \theta + \cos \theta}{\cot \theta - \cos \theta}}$$

$$= \sqrt{\frac{\frac{\cos \theta}{\sin \theta} + \cos \theta}{\frac{\cos \theta}{\sin \theta} - \cos \theta}}$$

$$= \sqrt{\frac{\cos \theta \left(\frac{1 + \sin \theta}{\sin \theta}\right)}{\cos \theta \left(\frac{1 - \sin \theta}{\sin \theta}\right)}}$$

Multiply and divide by  $(1 + \sin \theta)$

$$= \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}} \times \frac{(1 + \sin \theta)}{(1 + \sin \theta)}$$

$$= \frac{1 + \sin \theta}{\cos \theta}$$

$$= \sec \theta + \tan \theta$$

115.(a)  $\theta$  is in first quadrant

$$\cos^2 \theta - \sin^2 \theta = \frac{1}{2}$$

$$\therefore \cos^2 \theta = 1 - \sin^2 \theta$$

$$1 - \sin^2 \theta - \sin^2 \theta = \frac{1}{2}$$

$$\sin^2 \theta = \frac{1}{4} \Rightarrow \sin \theta = \frac{1}{2}$$

$$\therefore \sin \theta = 30^\circ$$

$$\tan^2 2\theta + \sin^2 3\theta$$

$$= \tan^2 60^\circ + \sin^2 90^\circ$$

$$= (\sqrt{3})^2 + 1 = 4$$

116.(a)  $5\sin\theta - 4\cos\theta = 0$ ,  $0 < \theta < 90$ 

$$5\sin\theta = 4\cos\theta$$

$$\tan\theta = \frac{4}{5}$$

$$\frac{5\sin\theta - 2\cos\theta}{5\sin\theta + 3\cos\theta}$$

Divide by  $\cos\theta$ 

$$= \frac{5\tan\theta - 2}{5\tan\theta + 3}$$

$$= \frac{5 \times \frac{4}{5} - 2}{5 \times \frac{4}{5} + 3} = \frac{2}{7}$$

$$117. (b) \frac{(1 + \cos\theta)^2 + \sin^2\theta}{(\csc^2\theta - 1)\sin^2\theta}$$

$$= \frac{1 + \cos^2\theta + 2\cos\theta + \sin^2\theta}{1 - \sin^2\theta}$$

$$= \frac{2 + 2\cos\theta}{\cos^2\theta}$$

$$= 2\sec^2\theta + 2\sec\theta$$

$$= 2\sec\theta(1 + \sec\theta)$$

$$118.(d) \frac{\sin\theta}{\cos\theta} = \sqrt{3}$$

$$\tan 60^\circ = \sqrt{3}, \theta = 60$$

$$2\sin^2\theta + \sec^2\theta + \sin\theta \cdot \sec\theta + \csc\theta$$

$$= 2 \times \left(\frac{\sqrt{3}}{2}\right)^2 + (2)^2 + \frac{\sqrt{3}}{2} \times 2 + \frac{2}{\sqrt{3}}$$

$$= \frac{3}{2} + 4 + \sqrt{3} + \frac{2}{\sqrt{3}}$$

$$= \frac{3\sqrt{3} + 8\sqrt{3} + 6 + 4}{2\sqrt{3}}$$

$$= \frac{10 + 11\sqrt{3}}{2\sqrt{3}}$$

divide and multiply by  $\sqrt{3}$ 

$$\frac{33 + 10\sqrt{3}}{6}$$

$$119.(a) = \left(\frac{1 - \tan\theta}{1 - \cot\theta}\right)^2 + 1$$

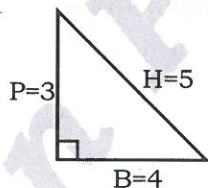
$$= \left(\frac{1 - \frac{\sin\theta}{\cos\theta}}{1 - \frac{\cos\theta}{\sin\theta}}\right)^2 + 1$$

$$= \left(\frac{\frac{\cos\theta - \sin\theta}{\cos\theta}}{\frac{\sin\theta - \cos\theta}{\sin\theta}}\right)^2 + 1$$

$$= \left(\frac{\sin\theta}{\cos\theta}\right)^2 + 1$$

$$= \tan^2\theta + 1 = \sec^2\theta$$

$$120.(d) \frac{(2\sin A)(1 + \sin A)}{1 + \sin A + \cos A}$$



$$= \frac{2 \times \frac{3}{5} \times (1 + \frac{3}{5})}{1 + \frac{3}{5} + \frac{4}{5}} = \frac{\frac{6}{5} \times \frac{8}{5}}{\frac{12}{5}} = \frac{4}{5}$$

by option  $(1 + \sin A - \cos A)$ 

$$1 + \frac{3}{5} - \frac{4}{5} = \frac{4}{5}$$

so  $(1 + \sin A - \cos A)$  is ans.

$$121.(a) (\cos^6\theta + \sin^6\theta - 1) \times (\tan^2\theta + \cot^2\theta + 2)$$

put,  $\theta = 45^\circ$ 

$$= \left[\left(\frac{1}{\sqrt{2}}\right)^6 + \left(\frac{1}{\sqrt{2}}\right)^6 - 1\right][1 + 1 + 2]$$

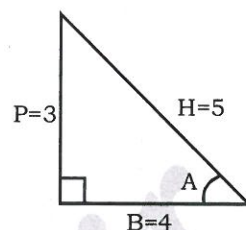
$$= \left(\frac{1}{8} + \frac{1}{8} - 1\right)(4)$$

$$= -3$$

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$$122.(d) \left(\frac{\sin A}{1 - \cos A} + \frac{1 - \cos A}{\sin A}\right) \div \left(\frac{\cot^2 A}{1 + \csc A} + 1\right)$$



$$\left(\frac{\frac{3}{5}}{1 - \frac{4}{5}} + \frac{1 - \frac{4}{5}}{\frac{3}{5}}\right) \div \left(\frac{\left(\frac{4}{3}\right)^2}{1 + \frac{5}{3}} + 1\right)$$

$$\left(3 + \frac{1}{3}\right) \div \left(\frac{16}{24} + 1\right)$$

$$\frac{10}{3} \div \left(\frac{5}{3}\right) \Rightarrow \frac{10}{3} \times \frac{3}{5} = 2$$

$$123.(b) \frac{1 + \sin\theta}{1 - \sin\theta} = \frac{p^2}{q^2}$$

$$p = 2, q = 1$$

$$\frac{1 + \sin\theta}{1 - \sin\theta} = \frac{4}{1}$$

$$1 + \sin\theta = 4 - 4\sin\theta$$

$$5\sin\theta = 3$$

$$\sin\theta = \frac{3}{5} \begin{matrix} \nearrow P \\ \searrow B=4 \\ \nearrow H \end{matrix}$$

$$\sec\theta = \frac{H}{B} = \frac{5}{4}$$

by option (b)

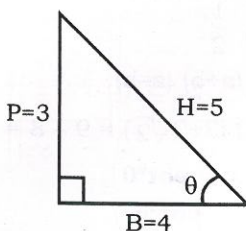
$$= \frac{1}{2} \left(\frac{q}{p} + \frac{p}{q}\right) \Rightarrow = \frac{1}{2} \left(\frac{1}{2} + \frac{2}{1}\right)$$

$$= \frac{1}{2} \times \frac{5}{2} = \frac{5}{4}$$

$$\Rightarrow \text{then, } \frac{1}{2} \left(\frac{q}{p} + \frac{p}{q}\right)$$

124.(d)

$$\frac{\sec\theta(1 - \sin\theta)(\sin\theta + \cos\theta)(\sec\theta + \tan\theta)}{\sin\theta(1 + \tan\theta) + \cos\theta(1 + \cot\theta)}$$



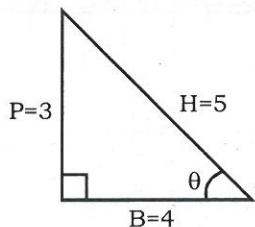
$$\begin{aligned} &= \frac{5}{4} \times \left(1 - \frac{3}{5}\right) \left(\frac{3}{5} + \frac{4}{5}\right) \left(\frac{5}{4} + \frac{3}{4}\right) \\ &= \frac{3}{5} \left(1 + \frac{3}{4}\right) + \left(\frac{4}{5} \times \left(1 + \frac{4}{3}\right)\right) \\ &= \frac{5}{4} \times \frac{2}{5} \times \frac{7}{5} \times \frac{8}{4} \\ &= \frac{3}{5} \times \frac{7}{4} + \frac{4}{5} \times \frac{7}{3} = \frac{12}{25} \\ &\frac{5}{4} \times \frac{2}{5} \times \frac{7}{5} \times \frac{8}{4} = \frac{12}{25} \end{aligned}$$

by option =  $\sin\theta \cdot \cos\theta$

$$\frac{3}{5} \times \frac{4}{5} = \frac{12}{25}$$

Then,  $\sin\theta \cdot \cos\theta$

**125.(a)**  $\tan^2\theta + \cot^2\theta - \sec^2\theta \operatorname{cosec}\theta$



$$\frac{9}{16} + \frac{16}{9} - \frac{25}{16} \times \frac{25}{9}$$

$$\frac{81 + 256 - 25 \times 25}{16 \times 9}$$

$$= \frac{337 - 625}{144}$$

$$= \frac{-288}{144} = -2$$

**Alternatively,**

$$\theta = 45^\circ$$

$$\tan^2\theta + \cot^2\theta - \sec^2\theta \operatorname{cosec}\theta$$

$$1 + 1 - (\sqrt{2})^2 \cdot (\sqrt{2})^2$$

$$2 - 4 = -2$$

**126.(b)**  $\frac{(\sec\theta - \tan\theta)^2 (1 + \sin\theta)^2}{\sin^2\theta}$

Put,  $\theta = 45^\circ$

$$\frac{(\sqrt{2} - 1)^2 \left(1 + \frac{1}{\sqrt{2}}\right)^2}{\left(\frac{1}{\sqrt{2}}\right)^2}$$

$$\frac{(3 - 2\sqrt{2})(3 + 2\sqrt{2})}{2} \cdot \frac{1}{2}$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$(3 - 2\sqrt{2})(3 + 2\sqrt{2}) = 9 - 8 = 1$$

by option =  $\cot^2\theta$

$$= 1^2 = 1 \Rightarrow \cot^2\theta$$

**127.(b)**  $3(\cot^2\theta - \cos^2\theta) = \cos^2\theta$

$$3\cot^2\theta - 3\cos^2\theta = \cos^2\theta$$

$$3\cot^2\theta = 4\cos^2\theta$$

$$3 \times \frac{\cos^2\theta}{\sin^2\theta} = 4 \times \cos^2\theta$$

$$\sin^2\theta = \frac{3}{4}, \quad \sin\theta = \frac{\sqrt{3}}{2}$$

$$\theta = 60^\circ$$

Then,

$$\tan^2\theta + \operatorname{cosec}^2\theta + \sin^2\theta$$

$$\tan^2 60^\circ + \operatorname{cosec}^2 60^\circ + \sin^2 60^\circ$$

$$3 + \frac{4}{3} + \frac{3}{4} = \frac{61}{12}$$

**128.(c)**  $\frac{\sin^2\theta - 3\sin\theta + 2}{\cos^2\theta} = 1$

$$\sin^2\theta - 3\sin\theta + 2 = \cos^2\theta$$

$$\sin^2\theta - 3\sin\theta + 2 = 1 - \sin^2\theta$$

$$2\sin^2\theta - 3\sin\theta + 1 = 0$$

$$2\sin^2\theta - 2\sin\theta - \sin\theta + 1 = 0$$

$$(2\sin\theta - 1)(\sin\theta - 1) = 0$$

$$2\sin\theta - 1 = 0$$

$$\sin\theta = \frac{1}{2}, \quad \theta = 30^\circ$$

$$\sin\theta - 1 = 0, \quad \sin\theta = 1$$

$$\theta = 90^\circ, \quad 0^\circ < \theta < 90^\circ$$

Then,

$$\cos 2\theta + \sin 3\theta + \operatorname{cosec} 2\theta$$

$$\cos 60^\circ + \sin 90^\circ + \operatorname{cosec} 60^\circ$$

$$\frac{1}{2} + 1 + \frac{2}{\sqrt{3}}$$

$$\frac{\sqrt{3} + 2\sqrt{3} + 4}{2\sqrt{3}}$$

$$\frac{(4 + 3\sqrt{3}) \times \sqrt{3}}{(2\sqrt{3}) \times \sqrt{3}} = \frac{9 + 4\sqrt{3}}{6}$$

**129.(b)**  $\frac{\sin(78^\circ + \theta) - \cos(12^\circ - \theta)}{\tan^2 70^\circ - \operatorname{cosec}^2 20^\circ}$

Put, the value of  $\theta = 0^\circ$

$$\sin 78^\circ - \cos(90^\circ - 78^\circ) +$$

$$= \frac{(\tan^2 70^\circ - \operatorname{cosec}^2(90^\circ - 70^\circ))}{\sin 25^\circ \cdot \cos(90^\circ - 25^\circ)}$$

$$+ \cos 25^\circ \cdot \sin(90^\circ - 25^\circ)$$

$$= \frac{\sin 78^\circ - \sin 78^\circ + (\tan^2 70^\circ - \sec^2 70^\circ)}{\sin 25^\circ \cdot \sin 25^\circ + \cos 25^\circ \cdot \cos 25^\circ}$$

$$= \frac{0 + \tan^2 70^\circ - \sec^2 70^\circ}{\sin 25^\circ + \cos^2 25^\circ}$$

$$= \frac{0 + \tan^2 70^\circ - \sec^2 70^\circ}{\sin 25^\circ + \cos^2 25^\circ}$$

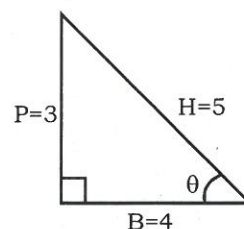
$$[\because \sec^2\theta - \tan^2\theta = 1]$$

$$= \frac{-1}{1} = -1$$

$$= \frac{-1}{1} = -1$$

**Alternatively:-** Put,  $\theta = 12^\circ$

**130.(d)**  $\sqrt{\frac{\operatorname{cosec}\theta - \cot\theta}{\operatorname{cosec}\theta + \cot\theta}} \div \frac{\sin\theta}{1 + \cos\theta}$



$$= \sqrt{\frac{\frac{5}{3} - \frac{4}{3}}{\frac{5}{3} + \frac{4}{3}}} \div \frac{\frac{3}{5}}{1 + \frac{4}{5}}$$

$$= \sqrt{\frac{1}{9}} \div \frac{3}{9}$$

$$\Rightarrow \frac{1}{3} \div \frac{1}{3} = 1$$

**131.(a)**  $(\tan 29^\circ \cdot \cot 61^\circ - \operatorname{cosec}^2 61^\circ)$   
 $+ \cot^2 54^\circ - \sec^2 36^\circ + (\sin^2 1^\circ + \sin^2 3^\circ + \dots + \sin^2 89^\circ)$   
 $= (\tan(90^\circ - 61^\circ) \cot 61^\circ - \operatorname{cosec}^2 61^\circ)$   
 $+ \cot^2 54^\circ - \sec^2(90^\circ - 54^\circ) + \sin^2 1^\circ$   
 $+ \sin^2 3^\circ + \dots + \sin^2 89^\circ$   
 $= (\cot^2 61^\circ - \operatorname{cosec}^2 61^\circ + (\cot^2 54^\circ - \operatorname{cosec}^2 54^\circ) + \sin^2 1^\circ + \sin^2 3^\circ + \dots + \sin^2 89^\circ + \cos^2 3^\circ + \cos^2 1^\circ)$   
 $[\because \operatorname{cosec}^2\theta - \cot^2\theta = 1]$   
 $= -1 - 1 + \sin^2 1^\circ + \sin^2 3^\circ + \dots + \sin^2 44^\circ + \sin^2 45^\circ$   
 $= -1 - 1 + 22 + \sin^2 45^\circ$   
 $= 20 \frac{1}{2}$

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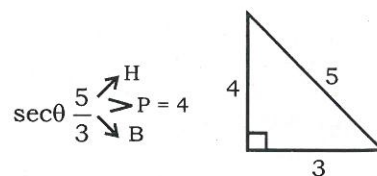
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**132.(b)** Put,  $P = 3$

$$\sec\theta + \tan\theta = 3$$

$$\sec\theta - \tan\theta = \frac{1}{3}$$

$$2\sec\theta = \frac{10}{3}$$



$$\frac{\operatorname{cosec}\theta + 1}{\operatorname{cosec}\theta - 1} = \frac{\frac{5}{4} + 1}{\frac{5}{4} - 1} = \frac{\frac{9}{4}}{\frac{1}{4}} = 9$$

By option  $P^2 = 3^2 = 9$

$$133.(d) \operatorname{cosec}(67^\circ + \theta) - \sec(23^\circ - \theta) + \cos 15^\circ \cos 35^\circ \operatorname{cosec} 55^\circ \cos 60^\circ \operatorname{cosec} 75^\circ$$

$$\text{Put } \theta = 0^\circ$$

$$= \operatorname{cosec} 67^\circ - \sec(90^\circ - 67^\circ) + \cos 15^\circ$$

$$\cos 35^\circ \times \frac{1}{\sin 55^\circ} \times \cos 60^\circ \times \frac{1}{\sin 75^\circ}$$

$$= \operatorname{cosec} 67^\circ - \operatorname{cosec} 67^\circ + \cos 15^\circ$$

$$\cos 35^\circ \times \frac{1}{\cos 35^\circ} \times \frac{1}{2} \times \frac{1}{\cos 15^\circ} = \frac{1}{2}$$

$$134.(c) 2\cos^2\theta + 3\sin\theta = 3$$

$$\text{Put, } \theta = 30^\circ$$

$$2 \times \left(\frac{\sqrt{3}}{2}\right)^2 + 3 \times \frac{1}{2} = \frac{3}{2} + \frac{3}{2} = 3$$

$$\text{then, } \theta = 30^\circ$$

$$\sin^2 2\theta + \cos^2 \theta + \tan^2 2\theta + \operatorname{cosec}^2 \theta = ?$$

$$= (\sin 60^\circ)^2 + (\cos 30^\circ)^2 + (\tan 60^\circ)^2 + (\operatorname{cosec} 60^\circ)^2$$

$$= \frac{3}{4} + \frac{3}{4} + 3 + \frac{4}{3}$$

$$= \frac{9+9+36+16}{12} = \frac{70}{12} = \frac{35}{6}$$

Alternatively:-

$$2(1 - \sin^2 \theta) + 3\sin \theta = 3$$

$$2 - 2\sin^2 \theta + 3\sin \theta = 3$$

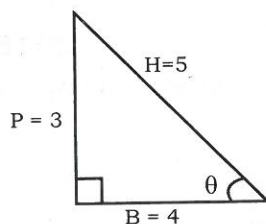
$$2\sin^2 \theta - 2\sin \theta - \sin \theta + 1 = 0$$

$$2\sin \theta (\sin \theta - 1) - 1(\sin \theta - 1) = 0$$

$$\sin \theta = 1, \sin \theta = \frac{1}{2}$$

$$\theta = 90^\circ, \theta = 30^\circ \text{ then } \theta = 30^\circ$$

$$135.(b) (1 + \cot \theta - \operatorname{cosec} \theta)(1 + \cos \theta + \sin \theta) \operatorname{sec} \theta$$



Put, value

$$\left(1 + \frac{4}{3} - \frac{5}{3}\right) \left(1 + \frac{4}{5} + \frac{3}{5}\right) \times \frac{5}{4}$$

$$\frac{2}{3} \times \frac{12}{5} \times \frac{5}{4} = 2$$

$$136.(b) \frac{\sec^2 \theta}{\operatorname{cosec}^2 \theta} + \frac{\operatorname{cosec}^2 \theta}{\sec^2 \theta} - (\sec^2 \theta + \operatorname{cosec}^2 \theta)$$

$$\theta = 45^\circ$$

$$\frac{(\sqrt{2})^2}{(\sqrt{2})^2} + \frac{(\sqrt{2})^2}{(\sqrt{2})^2} - ((\sqrt{2})^2 + (\sqrt{2})^2)$$

$$= -2$$

$$137.(d)$$

$$\frac{2(\sin^6 \theta + \cos^6 \theta) - 3(\sin^4 \theta + \cos^4 \theta)}{\cos^4 \theta - \sin^4 \theta - 2\cos^2 \theta}$$

$$\text{Put } \theta = 90^\circ$$

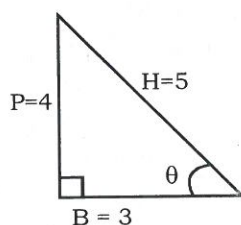
$$\frac{2(1+0) - 3(1+0)}{0-1-0} = \frac{2-3}{-1} = 1$$

$$138.(d) \sin^2 64^\circ + \cos 64^\circ (\sin(90^\circ - 64^\circ) + 2 \cos 43^\circ \operatorname{cosec}(90^\circ - 43^\circ))$$

$$\sin^2 64^\circ + \cos^2 64^\circ + 2 \cos 43^\circ \sec 43^\circ$$

$$1 + 2 = 3$$

$$139.(a) \frac{(\sin \theta - \cos \theta)(1 + \tan \theta + \cot \theta)}{1 + \sin \theta \cos \theta}$$



$$\left(\frac{4}{5} - \frac{3}{5}\right) \left(1 + \frac{4}{3} + \frac{3}{4}\right)$$

$$1 + \frac{4}{5} \times \frac{3}{5}$$

$$\frac{\frac{1}{5} \times (12+16+9)}{\frac{25+12}{25}} = \frac{\frac{1}{5} \times \frac{37}{12}}{\frac{37}{25}} = \frac{5}{12}$$

by option (a)

**sec $\theta$  - cosec $\theta$**

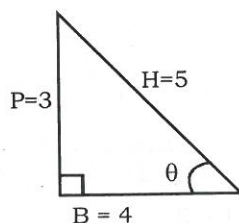
$$\frac{5}{3} - \frac{5}{4} = \frac{20-15}{12} = \frac{5}{12}$$

then option (a)

(sec $\theta$  - cosec $\theta$ ) is ans.

$$140.(c)$$

$$\frac{\sin \theta + \cos \theta - 1}{\sin \theta - \cos \theta + 1} \times \frac{\tan^2 \theta (\operatorname{cosec}^2 \theta - 1)}{\sec \theta - \tan \theta}$$



$$\frac{\frac{3}{5} + \frac{4}{5} - 1}{\frac{3}{5} - \frac{4}{5} + 1} \times \frac{\frac{9}{4} \times \left(\frac{25}{9} - 1\right)}{\frac{5}{4} - \frac{3}{4}} = 1$$

$$141.(a) \frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = \frac{4}{\sqrt{3}}$$

$$\frac{\sin^2 \theta + (1 + \cos \theta)^2}{(1 + \cos \theta)(\sin \theta)} = \frac{4}{\sqrt{3}}$$

$$\frac{\sin^2 \theta + \cos^2 \theta + 1 + 2 \cos \theta}{(1 + \cos \theta)(\sin \theta)} = \frac{4}{\sqrt{3}}$$

$$\frac{2(1 + \cos \theta)}{(1 + \cos \theta) \sin \theta} = \frac{4}{\sqrt{3}}$$

$$\sin \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 60^\circ$$

Then,

$$(\tan \theta + \sec \theta)^{-1} = (\tan 60^\circ + \sec 60^\circ)^{-1}$$

$$= (\sqrt{3} + 2)^{-1}$$

$$= \frac{1}{(2 + \sqrt{3})} \times \frac{(2 - \sqrt{3})}{2 - \sqrt{3}} = 2 - \sqrt{3}$$

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$$142.(d) \frac{\cos^2 30^\circ \sin^2 45^\circ + \sec^2 60^\circ}{\tan 60^\circ \cos^2 45^\circ - \sec^2 60^\circ \tan 45^\circ}$$

$$= \frac{(2)^2 \left(\frac{1}{\sqrt{2}}\right)^2 + (2)^2}{(\sqrt{3})(\sqrt{2})^2 - (2)^2(1)}$$

$$= \frac{6}{2\sqrt{3} - 4} = \frac{3}{\sqrt{3} - 2} = -3(2 + \sqrt{3})$$

$$143.(a) \frac{\sin \theta [(1 - \tan \theta) \tan \theta + \sec^2 \theta]}{(1 - \sin \theta) \tan \theta (1 + \tan \theta) (\sec \theta + \tan \theta)}$$

$$\Rightarrow \text{put, } \theta = 45^\circ$$

$$\Rightarrow \frac{\frac{1}{\sqrt{2}} [(1-1)1 + (\sqrt{2})^2]}{\left(1 - \frac{1}{\sqrt{2}}\right)(1)(1+1)(\sqrt{2}+1)}$$

$$\frac{\sqrt{2}}{(\sqrt{2}-1)(2)(\sqrt{2}+1)}$$

$$\Rightarrow \frac{(\sqrt{2}-1)}{\sqrt{2}} (2)(\sqrt{2}+1)$$

$$\Rightarrow \frac{1}{(\sqrt{2}-1)(\sqrt{2}+1)} = 1$$

$$144.(d) \cos A (\sec A - \cos A) (\cot A + \tan A)$$

When we put  $A = 45^\circ$  then two option gives equal value

Now put  $A = 60^\circ$

$$= \frac{1}{2} \left(2 - \frac{1}{2}\right) \left(\frac{1}{\sqrt{3}} + \sqrt{3}\right)$$

$$= \frac{3}{4} \left(\frac{4}{\sqrt{3}}\right) = \sqrt{3}$$

In option (d)  $\tan A = \tan 60^\circ = \sqrt{3}$

**145.(b)**  $\sin 3A = \cos(A+10)$

if  $\sin A = \cos B$

then,  $A + B = 90^\circ$

$3A + A + 10^\circ = 90^\circ$

$A = 20^\circ$

Now come to the question

$$2\operatorname{Cosec} \frac{3A}{2} + 6\sin^2 3A - \frac{3}{2} \tan^2 3A$$

$$= 2\operatorname{Cosec} 30^\circ + 6\sin^2 60^\circ - \frac{3}{2} \tan^2 60^\circ$$

$$= 2(2) + 6\left(\frac{\sqrt{3}}{2}\right)^2 - \frac{3}{2}(\sqrt{3})^2$$

$$= 4 + \frac{9}{2} - \frac{9}{2} = 4$$

**146.(a)**  $\left(\frac{1}{\cos \theta} - \frac{1}{\sin \theta}\right) + \frac{1}{\operatorname{cosec} \theta - \cot \theta}$

$$+ \frac{1}{\sec \theta + \tan \theta}$$

Put,  $\theta = 45^\circ$

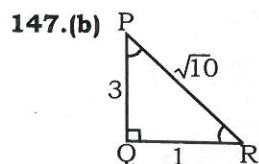
$$\Rightarrow \left(\frac{1}{\frac{1}{\sqrt{2}}} - \frac{1}{\frac{1}{\sqrt{2}}}\right) + \frac{1}{\sqrt{2}-1} - \frac{1}{\sqrt{2}+1}$$

$$\Rightarrow 0 + \frac{\sqrt{2}+1-\sqrt{2}+1}{(\sqrt{2}-1)(\sqrt{2}+1)} = 2$$

In option (a)  $\sec \theta \cdot \operatorname{cosec} \theta$

$$= \sec 45^\circ \cdot \operatorname{cosec} 45^\circ$$

$$= (\sqrt{2})(\sqrt{2}) = 2 \quad \dots \text{satisfied}$$

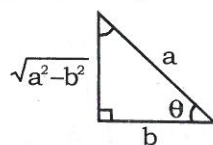


$$\Rightarrow \frac{\sec P(\cos R + \sin P)}{\operatorname{cosec} R(\sin R - \operatorname{cosec} P)}$$

$$\Rightarrow \frac{\left(\frac{\sqrt{10}}{3}\right)\left(\frac{1}{\sqrt{10}} + \frac{1}{\sqrt{10}}\right)}{\left(\frac{\sqrt{10}}{3}\right)\left(\frac{3}{\sqrt{10}} - \sqrt{10}\right)} = \frac{2}{-7}$$

$$= -\frac{2}{7}$$

**148.(c)**  $\sec \theta = \frac{a}{b}, b \neq 0$



$$\Rightarrow \frac{1 - \tan^2 \theta}{2 - \sin^2 \theta}$$

$$\Rightarrow \frac{1 - \left(\frac{a^2 - b^2}{b^2}\right)}{2 - \left(\frac{a^2 - b^2}{a^2}\right)}$$

$$\Rightarrow \frac{\frac{2b^2 - a^2}{b^2}}{\frac{a^2 + b^2}{a^2}} = \frac{a^2(2b^2 - a^2)}{b^2(a^2 + b^2)}$$

**149.(a)**

$$\sec^2 \theta (2 + \tan^2 \theta + \cot^2 \theta) + (\sin^2 \theta - \tan^2 \theta)$$

$$(\operatorname{cosec}^2 \theta + \sec^2 \theta)(1 + \cot^2 \theta)^2$$

put,  $\theta = 45^\circ$

$$\Rightarrow \frac{(\sqrt{2})^2(2+1+1) + \left[\left(\frac{1}{\sqrt{2}}\right)^2 - (1)\right]}{\left[(\sqrt{2})^2 + (\sqrt{2})^2\right](1+1)}$$

$$= \frac{8 \times (-2)}{16} = -1$$

**150.(d)** If  $A + B = 90^\circ$

then,  $\sin A \cdot \sec B = 1$

&  $\sin A = \cos B$

$$\Rightarrow \frac{2\sin^2 38^\circ \sec^2 52^\circ + \cos 64^\circ \sin 26^\circ + \sin^2 64^\circ}{\tan^2 23^\circ + \cot^2 23^\circ - \sec^2 67^\circ - \operatorname{cosec}^2 67^\circ}$$

$$\Rightarrow \frac{2 + \cos^2 64^\circ + \sin^2 64^\circ}{\cot^2 67^\circ + \tan^2 67^\circ - \sec^2 67^\circ - \operatorname{cosec}^2 67^\circ}$$

$$\Rightarrow \frac{2+1}{-1-1} = -\frac{3}{2}$$

**151.(c)**

$$\frac{\cos^6 \theta + \sin^6 + 3\sin^2 \theta \cdot \cos^2 \theta}{\operatorname{cosec} \theta \cdot \sec \theta (\sin \theta + \cos - 1)(\sin \theta + \cos \theta + 1)}$$

Put,  $\theta = 45^\circ$

$$\Rightarrow \frac{\left(\frac{1}{\sqrt{2}}\right)^6 + \left(\frac{1}{\sqrt{2}}\right)^6 + 3\left(\frac{1}{\sqrt{2}}\right)^2 \left(\frac{1}{\sqrt{2}}\right)^2}{(\sqrt{2})(\sqrt{2})\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1\right)\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 1\right)}$$

$$\Rightarrow \frac{\frac{1}{8} + \frac{1}{8} + \frac{3}{4}}{2(\sqrt{2}-1)(\sqrt{2}+1)}$$

$$\Rightarrow \frac{1+1+6}{8} = \frac{1}{2}$$

**Alternatively:-**

$$\frac{\cos^6 \theta + \sin^6 + 3\sin^2 \theta \cdot \cos^2 \theta}{\operatorname{cosec} \theta \cdot \sec \theta (\sin \theta + \cos - 1)(\sin \theta + \cos \theta + 1)}$$

$$\Rightarrow \frac{(\cos^2 \theta + \sin^2 \theta)^3}{\frac{1}{\sin \theta \cdot \cos \theta} [(\sin \theta + \cos \theta)^2 - (1)^2]}$$

$$\Rightarrow \frac{1}{\frac{1}{\sin \theta \cdot \cos \theta} [\sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta - 1]}$$

$$\Rightarrow \frac{1}{\frac{1}{\sin \theta \cdot \cos \theta} [1 + 2\sin \theta \cos \theta - 1]} = \frac{1}{2}$$

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**152.(a)** If  $A + B = 90^\circ$

Then,  $\sin^2 A + \sin^2 B = 1$

$$\cos^2 A + \cos^2 B = 1$$

$$\sin A \cdot \sec B = 1$$

$$\cos A \cdot \sec B = 1$$

$$\tan A \cdot \tan B = 1$$

$$\cot A \cdot \cot B = 1$$

Now come to the question-

$$x = \sec 57^\circ$$

$$x^2 = \sec^2 57^\circ$$

$$\sec^2 57^\circ - \tan^2 57^\circ = 1$$

$$\tan^2 57^\circ = x^2 - 1$$

$$\Rightarrow \cot^2 33^\circ + \sin^2 57^\circ + \sin^2 33^\circ + \operatorname{cosec}^2 57^\circ \cdot \cos^2 33^\circ + \sec^2 33^\circ \cdot \sin^2 57^\circ$$

$$\Rightarrow \cot^2 33^\circ + 1 + 1 + 1$$

$$= 3 + \cot^2 33^\circ$$

$$= 3 + \tan^2 57^\circ$$

$$= 3 + x^2 - 1$$

$$= x^2 + 2$$

(if  $A + B = 90$ , then  $\tan A = \cot B$ )

**153.(a)**  $(\tan^2 A + \cot^2 A - 2) - \sec^2 A \operatorname{cosec}^2 A$

$\Rightarrow$  Put,  $\theta = 45^\circ$

$$= (1 + 1 - 2) - (\sqrt{2})^2 (\sqrt{2})^2$$

$$= -4$$

**154.(b)**  $\frac{5\cos^2 60 + 4\sec^2 30 - \tan^2 45}{\tan^2 60 - \sin^2 30^\circ - \cos^2 45}$

$$= \frac{5\left(\frac{1}{2}\right)^2 + 4\left(\frac{2}{\sqrt{3}}\right)^2 - 1}{(\sqrt{3})^2 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{\sqrt{2}}\right)^2}$$

$$\Rightarrow \frac{\frac{5}{4} + \frac{16}{3} - 1}{3 - \frac{1}{4} - \frac{1}{2}}$$

$$\Rightarrow \frac{15+64-12}{\frac{12}{12-1-2}} = \frac{12}{4}$$

$$\Rightarrow \frac{67}{\frac{12}{9}} \Rightarrow \frac{67}{12} \times \frac{4}{9} = \frac{67}{27}$$

155.(d)  $\cos(2\theta + 54^\circ) = \sin\theta$   
 $\sin[90^\circ - (2\theta + 54^\circ)] = \sin\theta$   
 $\sin(36^\circ - 2\theta) = \sin\theta$   
 $3\theta = 36^\circ$   
 $\theta = 12^\circ$

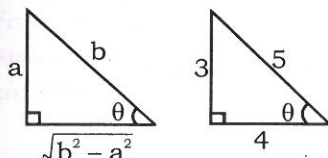
$$\frac{1}{\tan 5\theta + \operatorname{cosec} \frac{5\theta}{2}}$$

$$\frac{1}{\tan 60^\circ + \operatorname{cosec} 30^\circ}$$

$$\frac{1}{\sqrt{3}+2}$$

$$\frac{1}{\sqrt{3}+2} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = 2 - \sqrt{3}$$

156.(a)  $\operatorname{cosec} \theta = \frac{b}{a}$   
 Let,  $a = 3, b = 5$



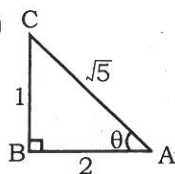
$$\frac{\sqrt{3}(\cot \theta + 1)}{\tan \theta + \sqrt{3}} = \frac{\sqrt{3}\left(\frac{4}{3} + 1\right)}{\frac{3}{4} + \sqrt{3}}$$

$$\Rightarrow \frac{\sqrt{3}(4 + \sqrt{3})}{\sqrt{3}(\sqrt{3} + 4)} = \frac{4}{3}$$

Now check the option (a)

$$\sqrt{\frac{b^2 - a^2}{a}} = \frac{4}{3}$$

157.(c)  $\frac{\sin^2 \theta}{\cos \theta(1 + \cos \theta)} + \frac{1 + \cos \theta}{\cos \theta}$   
 $= \frac{\sin^2 \theta + (1 + \cos \theta)^2}{\cos \theta(1 + \cos \theta)}$   
 $= \frac{\sin^2 \theta + 1 + \cos^2 \theta + 2 \cos \theta}{\cos \theta(1 + \cos \theta)}$   
 $= \frac{2(1 + \cos \theta)}{\cos \theta(1 + \cos \theta)} = 2 \sec \theta$

158.(b) 

$$= \frac{\sin A(\cos C + \cos A)}{\cos C(\sin C - \sin A)}$$

$$= \frac{\frac{1}{\sqrt{5}}\left(\frac{1}{\sqrt{5}} + \frac{2}{\sqrt{5}}\right)}{\frac{1}{\sqrt{5}}\left(\frac{2}{\sqrt{5}} - \frac{1}{\sqrt{5}}\right)} = \frac{3}{1} = 3$$

159.(a)  $\frac{\sin \theta + \cos \theta - 1}{\sin \theta - \cos \theta + 1} \times \sqrt{\frac{1 + \sin \theta}{1 - \sin \theta}}$

Put,  $\theta = 45^\circ$

$$\Rightarrow \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1}{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 1} \times \sqrt{\frac{1 + \frac{1}{\sqrt{2}}}{1 - \frac{1}{\sqrt{2}}}}$$

$$\Rightarrow \frac{1 + 1 - \sqrt{2}}{\sqrt{2}} \times \sqrt{2} + 1$$

$$\Rightarrow \frac{\sqrt{2}(\sqrt{2} - 1)}{\sqrt{2}} \times (\sqrt{2} + 1) = 1$$

160.(a)  $\frac{(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta)}{(\sec \theta + \tan \theta)(1 - \sin \theta)}$

$$= \frac{\left(1 + \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta}\right)\left(1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta}\right)}{\left(\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}\right)(1 - \sin \theta)}$$

$$= \frac{\left(\frac{\cos \theta + \sin \theta + 1}{\cos \theta}\right)\left(\frac{\sin \theta + \cos \theta - 1}{\sin \theta}\right)}{\left(\frac{1 + \sin \theta}{\cos \theta}\right)(1 - \sin \theta)}$$

$$= \frac{(\cos \theta + \sin \theta)^2 - (1)^2}{\frac{\sin \theta \cdot \cos \theta}{1 - \sin^2 \theta} \cos \theta}$$

$$= \frac{1 + 2 \sin \theta \cdot \cos \theta - 1}{\sin \theta \cdot \cos^2 \theta}$$

$$= \frac{2}{\cos \theta}$$

$$= 2 \sec \theta$$

161.(a)  $\frac{\sec A(\sec A + \tan A)(1 - \sin A)}{(\operatorname{cosec}^2 A - 1)\sin^2 A}$

Put,  $A = 45^\circ$

$$= \frac{\sqrt{2}(\sqrt{2} + 1)\left(1 - \frac{1}{\sqrt{2}}\right)}{(\sqrt{2})^2 - 1\left(\frac{1}{\sqrt{2}}\right)^2} = 2$$

Put,  $A = 45^\circ$  in options and the option (a)

$$\sec^2 A = \sec^2 45 = 2$$

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162.(c)  $\sec \theta + \tan \theta = 3$

$$\sec \theta - \tan \theta = \frac{1}{3}$$

$$2 \sec \theta = 3 + \frac{1}{3}$$

$$2 \sec \theta = \frac{10}{3}$$

$$\sec \theta = \frac{5}{3}$$

[If,  $\sec \theta + \tan \theta = P$

then,  $\sec \theta - \tan \theta = \frac{1}{P}$ ]

163.(c)  $\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} = 2 \frac{51}{79} = \frac{209}{79}$

using componendo & dividendo

$$\frac{(\sec \theta + \tan \theta) + (\sec \theta - \tan \theta)}{(\sec \theta - \tan \theta) - (\sec \theta - \tan \theta)}$$

$$= \frac{209 + 79}{209 - 79} = \frac{288}{130}$$

$$\frac{2 \sec \theta}{2 \tan \theta} = \frac{144}{65}$$

$$\frac{1}{\cos \theta} \cdot \frac{\sin \theta}{\cos \theta} = \frac{144}{65}$$

$$\sin \theta = \frac{65}{144}$$

164.(d)  $3 \sin x + 4 \cos x = 2 \dots (i)$

$3 \cos x - 4 \sin x = k \dots (ii)$

$(i)^2 + (ii)^2$

$$9(\sin^2 x + \cos^2 x) + 16(\sin^2 x + \cos^2 x) = 4 + k^2$$

$$25 = 4 + k^2$$

$$k^2 = 21$$

$$k = \sqrt{21}$$

{29 Jan. 2022}

165.(c)  $\sin\theta + \sin^2\theta = 1$

$\sin\theta = 1 - \sin^2\theta$

$\Rightarrow \sin\theta = \cos^2\theta$

$\cos^2\theta + \cos^4\theta$

$= \cos^2\theta + \sin^2\theta$

$= 1$  [using eq. (i)]

166.(a)  $\frac{\cos^2\theta}{\cot^2\theta - \cos^2\theta} = 3$

$$\frac{\cos^2\theta}{\cos^2\theta \left( \frac{1}{\sin^2\theta} - 1 \right)} = 3$$

$$\frac{\sin^2\theta}{\cos^2\theta} = 3$$

$\tan^2\theta = 3$

$\tan\theta = \sqrt{3}$

$\theta = 60^\circ$

167.(c)  $\cos\theta = \frac{5 \rightarrow B}{13 \rightarrow H}, P = 12$

$\tan^2\theta + \sec^2\theta$

$= \tan^2\theta + 1 + \tan^2\theta$

$= 1 + 2\tan^2\theta$

$= 1 + 2 \times \left( \frac{12}{5} \right)^2$

$= 1 + \frac{2 \times 144}{25}$

$= 1 + \frac{288}{25}$

$= \frac{313}{25}$

168.(d)  $\alpha + \beta = 90^\circ$

$\frac{\alpha}{\beta} = \frac{2}{1}$

$3 \text{ unit} = 90^\circ$

$\beta = 1 \text{ unit} = 30^\circ$

$\alpha = 2 \text{ unit} \times 30^\circ = 60^\circ$

$3\cos^2\alpha - 2\sin^2\beta$

$3\cos^2 60^\circ - 2\sin^2 30^\circ$

$= 3 \times \left( \frac{1}{2} \right)^2 - 2 \left( \frac{1}{2} \right)^2$

$= \frac{3-2}{4} = \frac{1}{4}$

169.(a)  $\sin(x+y) = \cos(x-y)$

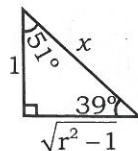
$(x+y) + (x-y) = 90^\circ$

$2x = 90^\circ$

$x = 45^\circ$

$\cos^2 45^\circ = \left( \frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2}$

170.(c)  $\operatorname{cosec} 39^\circ = \frac{x}{1} = \frac{H}{P}$



$$= \frac{1}{\operatorname{cosec}^2 51^\circ} + \sin^2 39^\circ + \tan^2 51^\circ -$$

$$\frac{1}{\sin^2 51^\circ \cdot \sec^2 39^\circ}$$

$$= \sin^2 51^\circ + \sin^2 39^\circ + \tan^2 51^\circ - 1$$

$$\boxed{\begin{matrix} \sin^2 A + \sin^2 B = 1 \\ A + B = 90^\circ \end{matrix}}$$

$$\boxed{\begin{matrix} \sec A \cdot \sec B = 1 \\ \text{when, } A + B = 90^\circ \end{matrix}}$$

$$= 1 + \tan^2 51^\circ - 1$$

$$= \tan^2 51^\circ$$

$$= \frac{(x^2 - 1)}{1} = x^2 - 1$$

171.(a) Put  $\theta = 45^\circ$

$$\Rightarrow (\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 =$$

$$k + \tan^2\theta + \cot^2\theta$$

$$\Rightarrow (\sin 45^\circ + \operatorname{cosec} 45^\circ)^2 + (\cos 45^\circ +$$

$$\sec 45^\circ)^2 = k + \tan^2 45^\circ + \cot^2 45^\circ$$

$$\Rightarrow \left( \frac{1}{\sqrt{2}} + \sqrt{2} \right)^2 + \left( \frac{1}{\sqrt{2}} + \sqrt{2} \right)^2$$

$$= k + (1)^2 + (1)^2$$

$$\Rightarrow \left( \frac{3}{\sqrt{2}} \right)^2 + \left( \frac{3}{\sqrt{2}} \right)^2 = k + 2$$

$$2 \times \frac{9}{2} = k + 2$$

$$k = 7$$

172.(b)  $3\tan\theta = 2\sqrt{3} \sin\theta$

$$\frac{3\sin\theta}{\cos\theta} = 2\sqrt{3} \sin\theta$$

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\cos\theta = \cos 30^\circ$$

$$\theta = 30$$

$$\Rightarrow \frac{\operatorname{cosec}^2 2\theta + \cot^2 2\theta}{\sin^2 \theta + \tan^2 \theta}$$

$$\text{Put, } \theta = 30$$

$$\frac{\operatorname{cosec}^2 60^\circ + \cot^2 60^\circ}{\sin^2 30^\circ + \tan^2 60^\circ}$$

$$= \frac{\frac{4}{3} + \frac{1}{3}}{\frac{1}{4} + 3} = \frac{\frac{5}{3}}{\frac{13}{4}} = \frac{20}{39}$$

173.(d)

$$\frac{3(\cot^2 47^\circ - \sec^2 43^\circ) - 2(\tan^2 23^\circ - \operatorname{cosec}^2 67^\circ)}{\operatorname{cosec}^2 (68^\circ + \theta) - \tan(\theta + 61^\circ) - \tan^2 (22^\circ - \theta) + \cot(29^\circ - \theta)}$$

$$\sec^2\theta = \tan^2\theta + 1$$

$$\operatorname{cosec}^2\theta = \cot^2\theta + 1$$

$$\Rightarrow \cot 47^\circ = \cot(90^\circ - 47^\circ)$$

$$= \tan 43^\circ \quad \dots(i)$$

$$\Rightarrow \tan 23^\circ = \tan(90^\circ - 23^\circ)$$

$$= \cot 67^\circ \quad \dots(ii)$$

$$\Rightarrow \tan(22^\circ - \theta) = \tan(90^\circ - 22^\circ + \theta)$$

$$= \cot(68^\circ + \theta) \quad \dots(iii)$$

$$\Rightarrow \tan(61^\circ + \theta) = \tan(90^\circ - 61^\circ - \theta)$$

$$= \cot(29^\circ - \theta) \quad \dots(iv)$$

Putting value of equ(i), eq(ii) and equ(iii) and equ(iv)

$$\frac{3(\tan^2 43^\circ - \sec^2 43^\circ) - 2(\cot^2 67^\circ - \operatorname{cosec}^2 67^\circ)}{\operatorname{cosec}^2 (68^\circ + \theta) - \cot^2 (68^\circ + \theta) - \cot(29^\circ - \theta) + \cot(29^\circ - \theta)}$$

$$\Rightarrow \frac{-3+2}{1} = -1$$

174.(c)  $1 + 2\tan^2\theta + 2\sin\theta\sec\theta = \frac{a}{b}$

$$1 + 2 \frac{\sin^2\theta}{\cos^2\theta} + \frac{2\sin\theta}{\cos^2\theta} = \frac{a}{b}$$

$$\frac{\cos^2\theta + 2\sin^2\theta + 2\sin\theta}{\cos^2\theta} = \frac{a}{b}$$

$$\frac{1 - \sin^2\theta + 2\sin^2\theta + 2\sin\theta}{1 - \sin^2\theta} = \frac{a}{b}$$

$$\frac{1 + \sin^2 \theta + 2 \sin \theta}{1 - \sin^2 \theta} = \frac{a}{b}$$

$$\frac{(1 + \sin \theta)^2}{(1 + \sin \theta)(1 - \sin \theta)} = \frac{a}{b}$$

$$\frac{1 + \sin \theta}{1 - \sin \theta} = \frac{a}{b}$$

Apply componendo and dividendo:-

$$\Rightarrow \frac{a+b}{a-b} = \frac{1}{\sin \theta} = \text{cosec} \theta$$

175.(a)

$$\left( \frac{\tan^3 \theta}{\sec^2 \theta} + \frac{\cot^3 \theta}{\text{cosec}^2 \theta} + 2 \sin \theta \cdot \cos \theta \right)$$

$$\div (1 + \text{cosec}^2 \theta + \tan^2 \theta)$$

$$= \left( \frac{\sin^3 \theta}{\cos^2 \theta} + \frac{\cos^3 \theta}{\sin^2 \theta} + 2 \sin \theta \cdot \cos \theta \right)$$

$$\div (1 + \text{cosec}^2 \theta + \sec^2 \theta - 1)$$

$$= \left( \frac{\sin^3 \theta}{\cos \theta} + \frac{\cos^3 \theta}{\sin \theta} + 2 \sin \theta \cdot \cos \theta \right) \div$$

$$\left( \frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta} \right)$$

$$= \left[ \frac{\sin^4 \theta + \cos^4 \theta + 2 \sin^2 \theta \cdot \cos^2 \theta}{\sin \theta \cdot \cos \theta} \right] \div$$

$$\left( \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cdot \cos^2 \theta} \right)$$

$$= \frac{(\sin^2 \theta + \cos^2 \theta)^2}{\sin \theta \cdot \cos \theta} \div \frac{1}{\sin^2 \theta \cdot \cos^2 \theta}$$

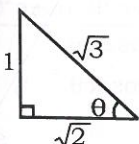
$$= \frac{\sin^2 \theta \cdot \cos^2 \theta}{\sin \theta \cdot \cos \theta} = \sin \theta \cdot \cos \theta$$

176.(a)  $7 \sin^2 \theta + 4 \cos^2 \theta = 5$

$$7 \sin^2 \theta + 4(1 - \sin^2 \theta) = 5$$

$$3 \sin^2 \theta = 1$$

$$\sin^2 \theta = \frac{1}{3}, \sin \theta = \frac{1}{\sqrt{3}}$$



After putting value in question

$$\Rightarrow \frac{\frac{\sqrt{3}}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{\sqrt{2} \times \sqrt{2} - \sqrt{3} \times \frac{\sqrt{2}}{\sqrt{3}}} = \frac{4}{2 - \sqrt{2}}$$

$$= \frac{2\sqrt{2}}{2 - \sqrt{2}} = \frac{2}{\sqrt{2} - 1} = 2(1 + \sqrt{2})$$

177.(a)  $(1 + \cot^2 \theta)(1 + \tan^2 \theta)(\sin \theta - \text{cosec} \theta)(\cos \theta - \sec \theta)$

Put,  $\theta = 45^\circ$

$$= (1+1)(1+1) \left( \frac{1}{\sqrt{2}} - \sqrt{2} \right) \left( \frac{1}{\sqrt{2}} - \sqrt{2} \right)$$

$$= (2)(2) \left( -\frac{1}{\sqrt{2}} \right) \times \left( -\frac{1}{\sqrt{2}} \right)$$

$$= 2$$

Also put  $\theta = 45^\circ$  in all options

In optin (c)

$$\Rightarrow \sec \theta \cdot \text{cosec} \theta = \sqrt{2} \times \sqrt{2} = 2$$

So, option (c) is correct.

Alternatively:-

$$(1 + \cot^2 \theta)(1 + \tan^2 \theta)(\sin \theta - \text{cosec} \theta)(\cos \theta - \sec \theta)$$

$$= \text{Cosec}^2 \theta \times \text{Sec}^2 \theta \times \left( \frac{\sin^2 \theta - 1}{\sin \theta} \right)$$

$$\times \left( \frac{\cos^2 \theta - 1}{\cos \theta} \right)$$

$$=$$

$$\text{Cosec}^2 \theta \times \text{Sec}^2 \theta \times \frac{-\cos^2 \theta}{\sin \theta} \times \frac{-\sin^2 \theta}{\cos \theta}$$

$$= \text{Sec} \theta \cdot \text{Cosec} \theta$$

178.(b)

$$\frac{(1 + \sec \theta \text{cosec} \theta)^2 (\sec \theta - \tan \theta)^2 (1 + \sin \theta)}{(\sin \theta + \sec \theta)^2 + (\cos \theta + \text{cosec} \theta)^2}$$

$$= \frac{\left( 1 + \frac{1}{\sin \theta \cos \theta} \right)^2 \left( \frac{1 - \sin \theta}{\cos \theta} \right)^2 (1 + \sin \theta)}{\left( \sin \theta + \frac{1}{\cos \theta} \right)^2 + \left( \cos \theta + \frac{1}{\sin \theta} \right)^2}$$

$$= \frac{(\sin \theta \cos \theta + 1)^2 (1 - \sin \theta)(1 + \sin \theta)(1 - \sin \theta)}{\sin^2 \theta \cdot \cos^2 \theta \times \cos^2 \theta}$$

$$(\sin \theta \cdot \cos \theta + 1)^2 \left[ \frac{1}{\sin^2 \theta \cos^2 \theta} \right]$$

$$= (1 - \sin \theta)$$

Alternatively:-

$$= \frac{\left( 1 + \frac{2}{\sqrt{3}} \times 2 \right)^2 \left( \frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}} \right)^2 \left( 1 + \frac{1}{2} \right)}{\left( \frac{1}{2} + \frac{2}{\sqrt{3}} \right)^2 + \left( \frac{\sqrt{3}}{2} + 2 \right)^2}$$

$$= \frac{\left( \frac{4 + \sqrt{3}}{\sqrt{3}} \right)^2 \times \frac{1}{3} \times \frac{3}{2}}{\left( \frac{4 + \sqrt{3}}{2\sqrt{3}} \right)^2 + \left( \frac{4 + \sqrt{3}}{2} \right)^2}$$

$$= \frac{(4 + \sqrt{3})^2 \times \frac{1}{3} \times \frac{1}{2}}{(4 + \sqrt{3})^2 + \left( \frac{1}{12} + \frac{1}{4} \right)}$$

$$= \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{1}{3}} = \frac{1}{2}$$

Only option (b) gives  $\frac{1}{2}$  at  $\theta = 30^\circ$

179.(d)

$$\frac{(1 + \cos \theta - \sin^2 \theta)}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\sec^2 \theta \text{cosec}^2 \theta}}{\tan \theta + \cot \theta}$$

$$= \frac{(1 + \cos \theta - 1 + \cos^2 \theta)}{\sin \theta (1 + \cos \theta)} \times \frac{\sqrt{\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cdot \cos^2 \theta}}}{\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cdot \cos \theta}}$$

$$= \frac{\cos \theta (1 + \cos \theta)}{\sin \theta (1 + \cos \theta)}$$

$$= \cot \theta$$

180.(b)  $\sin A = \frac{5}{13}$

$$\Rightarrow \cos A = \frac{12}{13}$$

$$\cot B = \frac{24}{7} \Rightarrow \tan B = \frac{7}{24}$$

$$\Rightarrow (\sec A \cdot \cos B)(\text{cosec} B \tan A)$$

$$= \left( \frac{13}{12} \times \frac{24}{25} \right) \times \left( \frac{25}{7} \times \frac{5}{12} \right) = \frac{65}{42}$$

181.(b)

$$\frac{4 \tan^2 30^\circ + \sin^2 30^\circ \cdot \cos^2 45^\circ + \sec^2 48^\circ - \cot^2 42^\circ}{\cos 37^\circ \cdot \sin 53^\circ + \sin 37^\circ \cdot \cos 53^\circ + \tan 18^\circ \cdot \tan 72^\circ}$$

$$\cot 42^\circ = \cot (90^\circ - 48^\circ) = \tan 48^\circ$$

$$\cos 37^\circ = \cos (90^\circ - 53^\circ) = \sin 53^\circ$$

$$\sin 37^\circ = \sin (90^\circ - 53^\circ) = \cos 53^\circ$$

$$= \frac{4 \times \frac{1}{3} + \frac{1}{4} \times \frac{1}{2} + 1}{1 + 1} = \frac{59}{48}$$

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182.(d)

$$\frac{3(\cos^2 26^\circ - \tan^2 64^\circ) + (\cot^2 42^\circ - \sec^2 48^\circ)}{\cot(22^\circ - \theta) - \cos^2(62^\circ + \theta) - \tan(\theta + 68^\circ) + \tan^2(28^\circ - \theta)}$$

$$\operatorname{cosec}^2 26^\circ = \sec^2 64^\circ$$

$$\cot^2 42^\circ = \tan^2 48^\circ$$

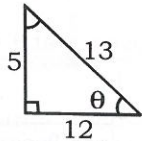
$$\cot(22^\circ - \theta) = \tan(68^\circ + \theta)$$

$$\operatorname{cosec}^2(62^\circ + \theta) = \sec^2(28^\circ - \theta)$$

$$\frac{3(\sec^2 64^\circ - \tan^2 64^\circ) + \left( \frac{\tan^2 48^\circ}{-\sec^2 48^\circ} \right)}{\tan(68^\circ + \theta) - \tan(68^\circ + \theta) - \sec^2(28^\circ - \theta) + \tan^2(28^\circ - \theta)}$$

$$= \frac{3(1) + (-1)}{-1} = \frac{3-1}{-1} = -2$$

$$183.(b) \cos \theta = \frac{12}{13} = \frac{B}{H}$$

use triplets  $\rightarrow 5, 12, 13$ 

$$\Rightarrow \frac{\sin \theta(1 - \tan \theta)}{\tan \theta(1 + \operatorname{cosec} \theta)}$$

$$= \frac{\frac{5}{13} \left( 1 - \frac{5}{12} \right)}{\frac{5}{12} \left( 1 + \frac{13}{5} \right)} = \frac{\frac{5}{13} \times \frac{7}{12}}{\frac{5}{12} \times \frac{18}{5}}$$

$$= \frac{35}{13 \times 12} \times \frac{12}{18} = \frac{35}{234}$$

$$184.(a) \frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta} = \frac{1}{7}$$

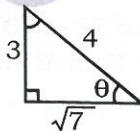
Apply componendo and dividendo:-

$$\frac{\sec \theta}{\tan \theta} = \frac{8}{6} = \frac{4}{3}$$

$$\frac{1}{\cos \theta} \times \frac{\cos \theta}{\sin \theta} = \frac{4}{3}$$

$$\frac{1}{\sin \theta} = \frac{4}{3}$$

$$\sin \theta = \frac{3}{4}$$



$$\Rightarrow \frac{\operatorname{cosec} \theta + \cot^2 \theta}{\operatorname{cosec} \theta - \cot^2 \theta}$$

$$= \frac{\frac{4}{3} + \frac{7}{9}}{\frac{4}{3} - \frac{7}{9}} = \frac{36 + 21}{36 - 21} = \frac{57}{15} = \frac{19}{5}$$

$$185.(b) \theta = 30^\circ, 45^\circ, 60^\circ$$

 $\theta = 45^\circ$  not suitable because two option will be sameso we can put  $\theta = 30^\circ$  or  $60^\circ$ Put,  $\theta = 30^\circ$ 

$$\Rightarrow (\tan \theta + \cot \theta)(\sec \theta + \tan \theta)(1 - \sin \theta)$$

$$= \left( \frac{1}{\sqrt{3}} + \sqrt{3} \right) \left( \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} \right) \left( 1 - \frac{1}{2} \right)$$

$$= \frac{4}{\sqrt{3}} \times \frac{3}{\sqrt{3}} \times \frac{1}{2} = 2$$

Option (b) gives value 2 after putting  $\theta = 30^\circ$ **cosec****Alternatively:-**

$$\Rightarrow (\tan \theta + \cot \theta)(\sec \theta + \tan \theta)(1 - \sin \theta)$$

$$\left( \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right) \left( \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \right) (1 - \sin \theta)$$

$$= \left( \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} \right) \left( \frac{1 + \sin \theta}{\cos \theta} \right) (1 - \sin \theta)$$

$$= \frac{1}{\sin \theta \cos \theta} \times \frac{(1 - \sin^2 \theta)}{\cos \theta}$$

$$= \frac{1}{\sin \theta \cos \theta} \times \frac{\cos^2 \theta}{\cos \theta}$$

$$= \frac{1}{\sin \theta} = \operatorname{cosec} \theta$$

$$186.(d) \frac{\tan^6 \theta - \sec^6 \theta + 3 \sec^2 \theta \tan^2 \theta}{\tan^2 \theta + \cot^2 \theta + 1}$$

$$\frac{(\tan^2 \theta)^3 - (\sec^2 \theta)^3 + 3 \sec^2 \theta \tan^2 \theta}{\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} + 1}$$

$$\frac{(\tan^2 \theta - \sec^2 \theta)^3 [\tan^4 \theta + \sec^4 \theta + \tan^2 \theta \sec^2 \theta] + 3 \sec^2 \theta \tan^2 \theta}{\frac{\sin^4 \theta + \cos^4 \theta + 2 \sin^2 \theta \cos^2 \theta}{\sin^2 \theta \cos^2 \theta}}$$

$$= \frac{(-1)^3 (\tan^4 \theta + \sec^4 \theta + \tan^2 \theta \sec^2 \theta) + 3 \sec^2 \theta \tan^2 \theta}{(\sin^2 \theta + \cos^2 \theta)^2}$$

$$= \frac{-\tan^4 \theta - \sec^4 \theta - \tan^2 \theta \sec^2 \theta + 3 \sec^2 \theta \tan^2 \theta}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{-[\sec^4 \theta + \tan^4 \theta - 2 \sec^2 \theta \tan^2 \theta]}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{-(\sec^2 \theta - \tan^2 \theta)^2}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{-1}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{-1}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{-1}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{-1}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{-1}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{-1}{\sin^2 \theta \cos^2 \theta}$$

$$= \frac{-1}{\sin^2 \theta \cos^2 \theta}$$

$$= -1 \times \frac{\sin^2 \theta \cos^2 \theta}{1}$$

$$= -\sin^2 \theta \cos^2 \theta$$

187.(b)

$$\frac{\cos^4 \theta - \sin^4 \theta + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + \cot \theta + 1)(\operatorname{cosec} \theta - \cot \theta + 1) - 2}$$

$$= \frac{(\cos^2 \theta)^2 - (\sin^2 \theta)^2 + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + 1 + \cot \theta)(\operatorname{cosec} \theta + 1 - \cot \theta) - 2}$$

$$= \frac{(\cos^2 \theta - \sin^2 \theta)(1) + 2 \sin^2 \theta + 3}{(\operatorname{cosec} \theta + 1)^2 - (\cot \theta)^2 - 2}$$

$$= \frac{\cos^2 \theta + \sin^2 \theta + 3}{\operatorname{cosec}^2 \theta + 1 + 2 \operatorname{cosec} \theta - \cot^2 \theta - 2}$$

$$= \frac{1 + 3}{1 + 1 + 2 \operatorname{cosec} \theta - 2}$$

$$= \frac{4}{2 \operatorname{cosec} \theta} = 2 \sin \theta$$

188.(d)

$$\frac{(1 - \sin \theta + \cos \theta)^2 (1 - \cos \theta) \sec^3 \theta \operatorname{cosec}^2 \theta}{(\sec \theta - \tan \theta)(\tan \theta + \cot \theta)}$$

Put,  $\theta = 45^\circ$ 

$$\frac{\left( 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right)^2 \left( 1 - \frac{1}{\sqrt{2}} \right) (\sqrt{2})^3 (\sqrt{2})^2}{(\sqrt{2} - 1)(1 + 1)}$$

$$= \frac{1 \left( \frac{\sqrt{2} - 1}{\sqrt{2}} \right) 2\sqrt{2} \times 2}{(\sqrt{2} - 1) \times 2} = 2$$

Only 2  $\tan \theta$  gives 2 after putting  $\theta = 45^\circ$ Hence, **2 tan****Alternatively:-**

$$\frac{(1 - \sin \theta + \cos \theta)^2 (1 - \cos \theta) \sec^3 \theta \operatorname{cosec}^2 \theta}{(\sec \theta - \tan \theta)(\tan \theta + \cot \theta)}$$

$$(1 - \sin \theta + \cos \theta)^2 = 2(1 + \cos \theta)(1 - \sin \theta)$$

(Result)  $(a+b+c)^2$ 

$$= \frac{2(1 + \cos \theta)(1 - \sin \theta) \times (1 - \cos \theta) \sec^3 \theta \operatorname{cosec}^2 \theta}{\frac{(1 - \sin \theta)}{\cos \theta} \times \frac{1}{\cos \theta \sin \theta}}$$

$$= \frac{2(1 - \cos^2 \theta) \times \sec^3 \theta \operatorname{cosec}^2 \theta}{\sec^2 \theta \operatorname{cosec} \theta}$$

$$= 2 \sin^2 \theta \sec^2 \theta \operatorname{cosec} \theta$$

$$= 2 \frac{\sin \theta}{\cos \theta} \Rightarrow 2 \tan \theta$$

189.(d)

$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\operatorname{cosec}^2 71^\circ + \tan^2(90^\circ - 71^\circ) + 1}$$

$$\cos 9^\circ = \sin 81^\circ$$

$$\sec 9^\circ = \operatorname{cosec} 81^\circ$$

$$\frac{(\sin 81^\circ + \sin 81^\circ)(\operatorname{cosec} 81^\circ + \operatorname{cosec} 81^\circ)}{\operatorname{cosec}^2 71^\circ - \tan^2(90^\circ - 71^\circ) + 1}$$

$$\frac{2 \sin 81^\circ \times 2 \operatorname{cosec} 81^\circ}{\operatorname{cosec}^2 71^\circ - \cot^2 71^\circ + 1}$$

$$= \frac{4}{2} = 2$$

$$190.(d) \frac{\sin^2 \theta}{\cos^2 \theta - 3 \cos \theta + 2}$$

$$\text{Put, } \theta = 60^\circ$$

$$\frac{\frac{3}{4}}{\frac{1}{4} - \frac{3}{2} + 2} = \frac{\frac{3}{4}}{\frac{1}{4}} = 1$$

Satisfies, Hence  $\theta = 60^\circ$ 

$$\Rightarrow \frac{\tan^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2}}{\tan \theta + \sin \theta}$$

$$= \frac{\left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{1}{2}\right)^2}{\sqrt{3} + \frac{\sqrt{3}}{2}} = \frac{\frac{1}{3} + \frac{1}{4}}{\frac{2\sqrt{3} + \sqrt{3}}{2}}$$

$$= \frac{7}{12} \times \frac{2}{3\sqrt{3}} = \frac{7}{18\sqrt{3}}$$

$$= \frac{7}{18\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{7\sqrt{3}}{54}$$

$$191.(b) \frac{(1 - 2 \sin^2 \theta \cos^2 \theta)(\cot \theta + 1) \cos \theta}{(\sin^4 \theta + \cos^4 \theta)(1 + \tan \theta) \operatorname{cosec} \theta}$$

$$\text{Put, } \theta = 30^\circ$$

$$= \frac{\left(1 - 2 \times \frac{1}{4} \times \frac{3}{4}\right)(\sqrt{3} + 1) \times \frac{\sqrt{3}}{2}}{\left(\frac{1}{16} + \frac{9}{16}\right)\left(1 + \frac{1}{\sqrt{3}}\right) \times 2} - 1$$

$$= \frac{\left(1 - \frac{3}{8}\right)\left(\frac{\sqrt{3} + 1}{2}\right) \times \sqrt{3}}{\frac{10}{16}\left(\frac{\sqrt{3} + 1}{\sqrt{3}}\right) \times 2} - 1$$

$$= \frac{\frac{5}{8}\left(\frac{\sqrt{3} + 1}{2}\right) \cdot \sqrt{3}}{\frac{5}{4}\left(\frac{\sqrt{3} + 1}{\sqrt{3}}\right)} - 1$$

$$= \frac{1}{2} \times \frac{\sqrt{3} \times \sqrt{3}}{2} - 1$$

$$= \frac{3}{4} - 1 \rightarrow -\frac{1}{4}$$

Go through option:-(b)

$$\Rightarrow -\sin^2 \theta \text{ gives } -\frac{1}{4} \text{ at } \theta = 30^\circ$$

$$\therefore -\sin^2 \theta$$

Alternatively:-

$$\frac{(1 - 2 \sin^2 \theta \cos^2 \theta) \left(\frac{\cos \theta}{\sin \theta} + 1\right) \cos \theta}{(\sin^2 \theta + \cos^2 \theta)^2 - (2 \sin^2 \theta \cos^2 \theta) \left(1 + \frac{\sin \theta}{\cos \theta}\right) \frac{1}{\sin \theta}} - 1$$

$$\frac{(1 - 2 \sin^2 \theta \cos^2 \theta) \left(\frac{\cos \theta + \sin \theta}{\sin \theta}\right) \cos \theta}{(1 - 2 \sin^2 \theta \cos^2 \theta) \left(\frac{\cos \theta + \sin \theta}{\cos \theta}\right) \frac{1}{\sin \theta}} - 1$$

$$= \frac{\cos \theta}{\sin \theta} \times \sin \theta \cos \theta - 1$$

$$= \cos^2 \theta - 1$$

$$= -\sin^2 \theta$$